

Exam:-Physiology
Year:-2013 E.C

Dr. Doc

[24/09/2026 8:23 AM] Imaanaa Harmee: **1. Identify the one that is a typical example of a positive feedback.**

- A. Glucose homeostasis**
- B. Calcium homeostasis**
- C. Gonadal hormones secretion**
- D. Milk ejection**
- E. Milk production**

Answer D. Milk ejection

Milk ejection (let-down reflex) is a classic example of positive feedback, where the response amplifies the original stimulus. During breastfeeding, suckling stimulates nerve receptors in the nipple, leading to oxytocin release from the posterior pituitary. Oxytocin causes contraction of myoepithelial cells, resulting in milk ejection. Continued suckling increases oxytocin release until the stimulus stops.

Why Not Other Options?

A. Glucose homeostasis ✗

Maintained by negative feedback, where insulin and glucagon act to reverse changes in blood glucose levels and keep them within a normal range.

B. Calcium homeostasis ✗

Maintained by negative feedback involving PTH and calcitonin to stabilize blood calcium concentration.

C. Gonadal hormones secretion ✗

Regulated mainly by negative feedback through the hypothalamic-pituitary-gonadal axis, where sex hormones inhibit further hormone release.

E. Milk production ✗

Controlled mainly by prolactin and is not the classic example of positive feedback. Milk ejection, mediated by oxytocin, is the typical example.

[24/09/2026 8:23 AM] Imaanaa Harmee: **2. General properties of negative feedback include all of the following EXCEPT:**

- A. It is unwavering of the set point**
- B. Ideal value of a controlled variable is defined as its "set point"**
- C. It is a closed-loop control system in which the output accelerates the control system**
- D. The term negative is used because the effector response opposes the departure from the set point**
- E. It often works towards homeostasis**

Answer C. It is a closed-loop control system in which the output accelerates the control system

Negative feedback is a regulatory mechanism that maintains homeostasis by reducing or opposing changes from the normal range. It is a closed-loop control system, meaning the output is monitored and used to adjust the response. However, the output in negative feedback does not accelerate or amplify the control system; it reduces the original disturbance and brings the variable closer to its set point.

Why Not Other Options?

A. It is unwavering of the set point ✗

Negative feedback helps maintain a controlled variable near its set point. Although minor fluctuations occur, the system continuously corrects deviations to preserve stability.

B. Ideal value of a controlled variable is defined as its "set point" ✗

The set point is the desired normal value of a physiological variable, such as body temperature or blood glucose level, around which the body regulates.

D. The term negative is used because the effector response opposes the departure from the set point ✗

Correct. The response acts in the opposite direction of the initial change. For example, increased blood glucose stimulates insulin release, which lowers glucose levels.

E. It often works towards homeostasis ✗

Correct. Negative feedback is the most common mechanism used by the body to maintain a stable internal environment, including regulation of temperature, glucose, blood pressure, and hormone levels.

[24/09/2026 8:23 AM] Imaanaa Harmee: **3.A structure not importantly involved in the regulation of blood glucose is:**

A. Liver

B. Lung

C. Kidney

D. Small intestine

E. Brain

Answer B. Lung

The lung is not importantly involved in blood glucose regulation. Blood glucose homeostasis mainly depends on organs that store, produce, use, or regulate glucose availability, such as the liver, kidneys, small intestine, and brain.

Why Not Other Options?

A. Liver ✗

The liver is a major regulator of blood glucose. It stores glucose as glycogen after meals and releases glucose through glycogenolysis and gluconeogenesis during fasting.

C. Kidney ✗

The kidney contributes to glucose regulation through gluconeogenesis and by reabsorbing glucose from the filtrate to prevent glucose loss in urine.

D. Small intestine ✗

The small intestine plays an important role by absorbing dietary carbohydrates and releasing glucose into the bloodstream after meals.

E. Brain ✗

The brain depends heavily on glucose as an energy source and participates in glucose regulation through the hypothalamus, which helps coordinate hormonal responses.

[24/09/2026 8:23 AM] Imaanaa Harmee: **4. Wrong about hormonal and nervous system regulation is:**

A. Both involve chemical messengers

B. Nervous regulation is widely diffused

C. Hormonal regulation is of short duration

D. Both act via interaction with receptors

E. Both act by altering their target organs

Answer C. Hormonal regulation is of short duration

The endocrine (hormonal) and nervous systems are the two major systems that regulate body functions. Although both use chemical messengers and receptors, they differ in the speed and duration of their effects. Hormonal regulation is usually slower in onset but produces long-lasting effects because hormones travel through the bloodstream and remain active for a longer time. In contrast, nervous regulation is rapid and short-lived, as nerve impulses and neurotransmitters act quickly and their effects stop soon after the stimulus is removed. Therefore, stating that hormonal regulation is of short duration is incorrect.

Why Not Other Options?

A. Both involve chemical messengers ✗

Correct. The nervous system communicates using neurotransmitters, while the endocrine system uses hormones. Both are chemical messengers that transmit signals to target cells.

B. Nervous regulation is widely diffused ✗

Nervous regulation is generally localized and specific, affecting only cells supplied by particular nerve fibers. In contrast, hormonal regulation is usually more widespread because hormones circulate in the bloodstream.

D. Both act via interaction with receptors ✗

Correct. Both neurotransmitters and hormones produce their effects by binding to specific receptors on target cells. Without receptors, no response occurs.

E. Both act by altering their target organs ✗

Correct. Both systems regulate body functions by changing the activity of target cells or organs, helping maintain normal physiological function.

[24/09/2026 8:23 AM] Imaanaa Harmee: **5. Which one of the following does not belong to the cell membrane proteins?**

- A. Ion channels**
- B. Na⁺-K⁺ ATPase**
- C. Carrier proteins**
- D. Membrane receptor**
- E. Enzymes**

Answer: None of the above (All are cell membrane proteins).

All the listed structures are membrane proteins. Cell membrane proteins perform essential functions such as transport, signaling, enzymatic activity, and maintenance of ion gradients. Therefore, there is no incorrect option in this question. If this is an exam question, it is likely misprinted.

Why Not Other Options?

A. Ion channels ✗

Ion channels are integral membrane proteins that allow specific ions (e.g., Na⁺, K⁺, Ca²⁺, Cl⁻) to pass through the cell membrane by facilitated diffusion.

B. Na⁺-K⁺ ATPase ✗

Na⁺-K⁺ ATPase is an integral membrane transport protein (pump). It uses ATP to actively transport 3 Na⁺ out of the cell and 2 K⁺ into the cell, maintaining the resting membrane potential.

C. Carrier proteins ✗

Carrier proteins are membrane proteins that transport molecules such as glucose and amino acids across the membrane by facilitated diffusion or active transport.

D. Membrane receptor ✗

Membrane receptors are proteins that bind hormones, neurotransmitters, or other signaling molecules and initiate cellular responses.

E. Enzymes ✗

Some enzymes are attached to the cell membrane and catalyze biochemical reactions on the membrane surface, such as adenylyl cyclase and ATPases.

[24/09/2026 8:23 AM] Imaanaa Harmee: **6. The statement that is correct about structure of cell membrane is:**

- A. Portions of the lipid bilayer that face the ECF and ICF are hydrophobic**
- B. Polar molecules can easily pass through the lipid bilayer**
- C. Protein channels within the lipid bilayer allow the passage of ions and water**
- D. The polar heads of the lipid bilayer face each other in the interior of the cell**
- E. None of the above**

Answer C. Protein channels within the lipid bilayer allow the passage of ions and water

The cell membrane is composed of a phospholipid bilayer with embedded proteins (fluid mosaic model). It is selectively permeable, allowing small nonpolar molecules to diffuse directly, while ions and water cross through membrane proteins such as ion channels and aquaporins.

Why Not Other Options?

A. Portions of the lipid bilayer that face the ECF and ICF are hydrophobic ✗

Incorrect. The outer and inner surfaces of the membrane are formed by hydrophilic polar heads, while the hydrophobic tails face inward.

B. Polar molecules can easily pass through the lipid bilayer ✗

Incorrect. Polar molecules and ions cannot easily cross the hydrophobic core and usually require transport proteins.

D. The polar heads of the lipid bilayer face each other in the interior of the cell ✗

Incorrect. The hydrophobic tails face each other in the center of the membrane, while the polar heads face the aqueous environments.

E. None of the above ✗

Incorrect because option C is correct.

[24/09/2026 8:23 AM] Imaanaa Harmee: **7. Cellular organelle that provides protection to the cell is:**

A. Lysosome

B. Endoplasmic reticulum

C. Golgi complex

D. Mitochondrion

E. Ribosome

Answer A. Lysosome

Lysosomes are membrane-bound organelles that contain hydrolytic (digestive) enzymes. They protect the cell by digesting bacteria, viruses, worn-out organelles, and other foreign or damaged materials. This process, known as intracellular digestion, helps maintain cell health and prevents the accumulation of harmful substances.

Why Not Other Options?

B. Endoplasmic reticulum ✗

The endoplasmic reticulum (ER) is involved in protein synthesis (rough ER) and lipid synthesis, detoxification, and calcium storage (smooth ER). It does not primarily protect the cell.

C. Golgi complex ✗

The Golgi apparatus modifies, sorts, packages, and secretes proteins and lipids. It also forms lysosomes but is not itself the main protective organelle.

D. Mitochondrion ✗

Mitochondria are the powerhouses of the cell, producing ATP through aerobic respiration. Their main role is energy production.

E. Ribosome ✗

Ribosomes are the sites of protein synthesis. They translate mRNA into proteins but have no direct protective function.

[24/09/2026 8:23 AM] Imaanaa Harmee: **8. Insensible water loss is water lost via ___.**

A. Skin evaporation and from the lungs

B. Urine

C. Excessive sweating

D. Vomiting or diarrhea

E. None of the above

Answer A. Skin evaporation and from the lungs

Insensible water loss refers to the continuous, unnoticeable loss of water from the body that occurs without conscious awareness or regulation. It mainly occurs through evaporation from the skin (not sweat) and water vapor lost during respiration from the lungs. In a healthy adult, insensible water loss is approximately 600–800 mL/day, depending on environmental conditions and body temperature.

Why Not Other Options?

B. Urine ✗

Urine loss is a sensible water loss because it can be measured and is regulated by the kidneys.

C. Excessive sweating ✗

Sweating is considered sensible water loss because it is noticeable, measurable, and varies with temperature and physical activity.

D. Vomiting or diarrhea ✗

These are forms of abnormal sensible water loss and can lead to significant dehydration.

E. None of the above ✗

Incorrect because option A correctly describes insensible water loss.

[24/09/2026 8:23 AM] Imaanaa Harmee: **9. One that does not have direct relation with simple diffusion is:**

A. Temperature

B. Concentration

C. Solubility

D. Membrane thickness

E. Surface area

Answer A. Temperature

Simple diffusion is the passive movement of molecules from an area of higher concentration to lower concentration without the use of energy or carrier proteins.

The rate of simple diffusion is directly determined by Fick's law, which depends mainly on the concentration gradient, membrane surface area, membrane thickness, and the solubility (permeability) of the substance. Although temperature can influence the speed of molecular movement, it is not considered a direct determinant of simple diffusion in physiological membranes.

Why Not Other Options?

B. Concentration ✗

The greater the concentration gradient, the faster the rate of diffusion.

C. Solubility ✗

Lipid-soluble substances diffuse more readily through the phospholipid bilayer than water-soluble substances.

D. Membrane thickness ✗

Diffusion is inversely proportional to membrane thickness. A thicker membrane slows diffusion.

E. Surface area ✗

Diffusion is directly proportional to the membrane surface area. A larger surface area increases the rate of diffusion.

[24/09/2026 8:23 AM] Imaanaa Harmee: **10. Which of the following is true about the Na⁺-K⁺ ATPase?**

A. It is made up of five subunits

B. Its alpha subunit is facing the ECF and has binding site for Na⁺ ions

C. Its ICF domain contains four binding sites

D. The ICF domain has K⁺ ion binding sites

E. ATPase binding site is on its ECF domain

Answer C. Its ICF domain contains four binding sites

The Na⁺-K⁺ ATPase (sodium-potassium pump) is an integral membrane protein responsible for actively transporting 3 Na⁺ ions out of the cell and 2 K⁺ ions into the cell using energy from ATP. The intracellular (ICF) domain of the α -subunit contains four important binding sites: three binding sites for Na⁺ ions and one ATP-binding site. ATP hydrolysis provides the energy needed for the transport cycle.

Why Not Other Options?

A. It is made up of five subunits ✗

Incorrect. The pump consists mainly of two subunits: an α (alpha) catalytic subunit and a β (beta) subunit. Some tissues also have a small regulatory FXYD (γ) protein.

B. Its alpha subunit is facing the ECF and has binding site for Na^+ ions ✗

Incorrect. The α -subunit spans the membrane, and Na^+ binds from the intracellular side, not the extracellular side.

D. The ICF domain has K^+ ion binding sites ✗

Incorrect. K^+ binds to the extracellular side of the pump before being transported into the cell.

E. ATPase binding site is on its ECF domain ✗

Incorrect. The ATP-binding (ATPase) site is located on the intracellular domain, where ATP is hydrolyzed to provide energy.

[24/09/2026 8:23 AM] Imaanaa Harmee: **11. The ion that is lowest in the blood is:**

A. Sodium

B. Potassium

C. Chloride

D. Bicarbonate

E. Calcium

Answer E. Calcium

Among the listed ions, calcium (Ca^{2+}) has the lowest concentration in the blood. Total plasma calcium is about 2.2–2.6 mmol/L, whereas sodium, chloride, bicarbonate, and potassium are present in much higher concentrations. Calcium is essential for muscle contraction, nerve impulse transmission, blood clotting, enzyme activity, and bone formation.

Why Not Other Options?

A. Sodium ✗

Sodium is the major extracellular cation, with a plasma concentration of 135–145 mmol/L.

B. Potassium ✗

Potassium is the major intracellular cation, but its plasma concentration (3.5–5.0 mmol/L) is still higher than calcium.

C. Chloride ✗

Chloride is the major extracellular anion, with a plasma concentration of about 98–106 mmol/L.

D. Bicarbonate ✗

Bicarbonate has a normal plasma concentration of about 22–28 mmol/L and plays an important role in acid-base balance.

[24/09/2026 8:23 AM] Imaanaa Harmee: **12. Wrong about the body fluid and electrolyte**

environment of the body is:

- A. ICF and ECF is isotonic**
- B. ICF contains less proteins**
- C. Magnesium is more in the ICF**
- D. Chloride is more in the ECF**
- E. Calcium is less in the ICF**

Answer B. ICF contains less proteins

The intracellular fluid (ICF) and extracellular fluid (ECF) have different electrolyte compositions. While they have the same osmolarity (are isotonic), the ICF contains more proteins and potassium, whereas the ECF contains more sodium, chloride, and calcium. Therefore, the statement that the ICF contains less proteins is incorrect.

Why Not Other Options?

A. ICF and ECF is isotonic ✗

Correct. Both compartments have approximately the same osmolarity ($\approx 285\text{--}295$ mOsm/L), allowing water to move freely without causing net fluid shifts.

C. Magnesium is more in the ICF ✗

Correct. Magnesium (Mg^{2+}) is predominantly an intracellular cation and is present in higher concentrations inside cells.

D. Chloride is more in the ECF ✗

Correct. Chloride (Cl^-) is the major extracellular anion and is much more concentrated in the ECF.

E. Calcium is less in the ICF ✗

Correct. Free intracellular calcium concentration is very low, while calcium is much more abundant in the ECF.

[24/09/2026 8:23 AM] Imaanaa Harmee: **13.FALSE about facilitated diffusion is:**

- A. It requires carrier proteins to transport substances**
- B. It needs ATP to transport substances**
- C. It transports substances down their concentration gradient**
- D. Carrier proteins are involved**
- E. It is a saturable process**

Answer B. It needs ATP to transport substances

Facilitated diffusion is a form of passive transport in which substances move down their concentration or electrochemical gradient with the help of carrier proteins or channel proteins. Since the movement is down the gradient, no ATP is required. The process is specific and saturable because there is a limited number of transport proteins available.

Why Not Other Options?

A. It requires carrier proteins to transport substances ✗

Correct. Many substances, such as glucose, require carrier proteins to cross the cell membrane by facilitated diffusion.

C. It transports substances down their concentration gradient ✗

Correct. Substances move from an area of higher concentration to lower concentration without the use of cellular energy.

D. Carrier proteins are involved ✗

Correct. Carrier proteins (and, in some cases, channel proteins) facilitate the movement of molecules that cannot diffuse directly through the lipid bilayer.

E. It is a saturable process ✗

Correct. Because the number of carrier proteins is limited, increasing the concentration of a substance eventually reaches a transport maximum (T_m), beyond which the rate cannot increase.

[24/09/2026 8:23 AM] Imaanaa Harmee: **14. Which one of the following represents an antiport carrier?**

A. Transport of glucose by facilitated diffusion

B. Na^+ - K^+ pump

C. Na^+ -glucose co-transporter

D. Transport of O_2 and CO_2

E. Osmosis

Answer B. Na^+ - K^+ pump

An antiport carrier (counter-transporter) transports two different substances in opposite directions across the cell membrane. The Na^+ - K^+ ATPase is the classic example of an antiport transporter. It uses ATP to pump 3 Na^+ ions out of the cell while simultaneously transporting 2 K^+ ions into the cell, helping maintain the resting membrane potential and normal cell volume.

Why Not Other Options?

A. Transport of glucose by facilitated diffusion ✗

Glucose moves through a uniport carrier (GLUT transporter). Only one substance is transported in one direction.

C. Na^+ -glucose co-transporter ✗

This is a symport (co-transport) mechanism. Both Na^+ and glucose move in the same direction across the membrane.

D. Transport of O_2 and CO_2 ✗

O_2 and CO_2 cross the membrane by simple diffusion, not by carrier proteins.

E. Osmosis ✗

Osmosis is the movement of water across a semipermeable membrane. It is not an antiport mechanism.

[24/09/2026 8:23 AM] Imaanaa Harmee: **15. Bipolar neurons are located in the:**

- A. Anterior horn of the gray matter of spinal cord**
- B. Retina**
- C. Lateral horn of the gray matter of the spinal cord**
- D. Posterior horn of the gray matter of the spinal cord**
- E. Peripheral effector organs**

Answer B. Retina

Bipolar neurons are special sensory neurons with one axon and one dendrite extending from opposite ends of the cell body. They are relatively uncommon and are mainly found in the retina of the eye, where they transmit visual signals from photoreceptor cells (rods and cones) to ganglion cells. Bipolar neurons are also present in the olfactory epithelium and vestibulocochlear pathways.

Why Not Other Options?

A. Anterior horn of the gray matter of spinal cord ✗

The anterior horn contains multipolar motor neurons, which supply skeletal muscles.

C. Lateral horn of the gray matter of the spinal cord ✗

The lateral horn contains multipolar autonomic (preganglionic sympathetic) neurons, not bipolar neurons.

D. Posterior horn of the gray matter of the spinal cord ✗

The posterior horn contains mainly interneurons and sensory relay neurons, which are predominantly multipolar.

E. Peripheral effector organs ✗

Peripheral effector organs (muscles and glands) are targets of neurons, not locations where bipolar neurons are found.

[24/09/2026 8:23 AM] Imaanaa Harmee: **16. Identify the correct statement from the following.**

- A. Motor neurons transmit impulses from periphery to the CNS**
- B. Microglia provides nutrients to neurons in the CNS**
- C. Neuroglia are excitable cells**
- D. Nerve insulators for central and peripheral NS are of the same attributes**
- E. Satellite cells are located in CNS**

Answer: None of the above (The question appears to be flawed.)

After evaluating each option, none is completely correct based on standard physiology and anatomy.

Why Not Other Options?

A. Motor neurons transmit impulses from periphery to the CNS ✗

Incorrect. Motor (efferent) neurons carry impulses from the CNS to muscles and glands. Sensory (afferent) neurons carry impulses from the periphery to the CNS.

B. Microglia provides nutrients to neurons in the CNS ✗

Incorrect. Microglia are the immune cells (macrophages) of the CNS. Astrocytes, not microglia, provide metabolic support and nutrients to neurons.

C. Neuroglia are excitable cells ✗

Incorrect. Neuroglial cells are non-excitable. They support, protect, nourish, and insulate neurons but do not generate action potentials.

D. Nerve insulators for central and peripheral NS are of the same attributes ✗

Incorrect. The insulating cells differ:

☞ CNS: Oligodendrocytes

☞ PNS: Schwann cells

Although both produce myelin, they differ in structure and function. One oligodendrocyte myelinates multiple axons, whereas one Schwann cell myelinates only one segment of a single axon.

E. Satellite cells are located in CNS ✗

Incorrect. Satellite cells are found in peripheral ganglia (PNS), where they surround neuronal cell bodies. They are not present in the CNS.

[24/09/2026 8:23 AM] Imaanaa Harmee: **17. During the rising phase of action potential:**

A. Voltage-gated Na^+ channels are opened

B. Voltage-gated K^+ channels opened

C. Voltage-gated K^+ channels are closed

D. Ligand-gated K^+ channels are closed

E. Ligand-gated Na^+ channels do open

Answer A. Voltage-gated Na^+ channels are opened

During the rising phase (depolarization) of the action potential, the membrane reaches the threshold potential, causing voltage-gated Na^+ channels to open rapidly. This results in a large influx of Na^+ ions into the cell, making the inside of the membrane progressively more positive. This rapid sodium entry is responsible for the steep upward (rising) phase of the action potential. Voltage-gated K^+ channels begin to open more slowly and mainly contribute to the repolarization phase.

Why Not Other Options?

B. Voltage-gated K^+ channels opened ✗

Incorrect. Although these channels begin to open with a delay, they are primarily responsible

for repolarization, not the rising phase.

C. Voltage-gated K^+ channels are closed ✗

Incorrect. They are not completely closed during the rising phase; they start opening slowly but have little effect until repolarization.

D. Ligand-gated K^+ channels are closed ✗

Incorrect. Ligand-gated K^+ channels are not involved in generating the action potential along the axon.

E. Ligand-gated Na^+ channels do open ✗

Incorrect. Ligand-gated Na^+ channels generate local (graded) potentials at synapses, whereas the action potential rising phase is produced by voltage-gated Na^+ channels.

[24/09/2026 8:23 AM] Imaanaa Harmee: **18. Which one of the following is wrong about the action potential of the nerve fiber?**

A. It is caused by a transient membrane depolarization

B. It needs a threshold stimulus to be generated

C. It has a definite RMP

D. It is all-or-none in its nature

E. Amplitude of action potential is stimulus dependent

Answer E. Amplitude of action potential is stimulus dependent

An action potential is a rapid, transient reversal of the resting membrane potential that occurs when a stimulus reaches the threshold. Once initiated, the action potential has a constant amplitude regardless of the strength of the stimulus, provided the stimulus is above threshold. Stronger stimuli increase the frequency (number) of action potentials, not their amplitude. This is known as the all-or-none principle.

Why Not Other Options?

A. It is caused by a transient membrane depolarization ✗

Correct. An action potential begins with a rapid depolarization due to opening of voltage-gated Na^+ channels, followed by repolarization.

B. It needs a threshold stimulus to be generated ✗

Correct. A stimulus must reach the threshold potential to trigger an action potential. Subthreshold stimuli do not produce an action potential.

C. It has a definite RMP ✗

Correct. Nerve fibers have a resting membrane potential (RMP) of about -70 to -90 mV, depending on the type of neuron.

D. It is all-or-none in its nature ✗

Correct. Once the threshold is reached, a full-sized action potential is generated. Increasing stimulus strength does not increase its size.

[24/09/2026 8:23 AM] Imaanaa Harmee: **19. The most sluggish nerve fibers are:**

- A. Type A nerve fibers**
- B. Type B nerve fibers**
- C. Type C nerve fibers**
- D. Type A beta nerve fibers**
- E. Type C nerve fibers**

Answer C. Type C nerve fibers

Type C nerve fibers are the slowest (most sluggish) conducting nerve fibers because they are small in diameter and lack myelin sheath. Their conduction velocity is very low compared with myelinated fibers. They transmit slow, dull pain, temperature sensations, and postganglionic autonomic signals.

Why Not Other Options?

A. Type A nerve fibers ✗

Type A fibers are large, myelinated fibers and have the fastest conduction velocities. They carry motor impulses and sensory information such as touch and proprioception.

B. Type B nerve fibers ✗

Type B fibers are myelinated and conduct impulses faster than Type C fibers. They mainly transmit preganglionic autonomic signals.

D. Type A beta nerve fibers ✗

Type A β fibers are large, heavily myelinated fibers with rapid conduction. They transmit touch, pressure, and vibration sensations.

E. Type C nerve fibers ✗

This option is identical to option C and appears to be a duplicate.

[24/09/2026 8:23 AM] Imaanaa Harmee: **20. An action potential arriving at the neuromuscular junction causes release of:**

- A. Acetylcholine**
- B. Na⁺ ions**
- C. Ca²⁺ ions**
- D. Noradrenalin**
- E. Dopamine**

Answer A. Acetylcholine

When an action potential reaches the neuromuscular junction, it depolarizes the presynaptic motor nerve terminal, causing voltage-gated Ca²⁺ channels to open. Calcium enters the nerve terminal and triggers synaptic vesicles to release acetylcholine (ACh) into the synaptic cleft. Acetylcholine then binds to receptors on the muscle fiber, producing muscle depolarization and contraction.

Why Not Other Options?

B. Na⁺ ions ✗

Incorrect. Na⁺ influx occurs in the muscle membrane after acetylcholine binds to its receptors, but Na⁺ is not the neurotransmitter released from the nerve terminal.

C. Ca²⁺ ions ✗

Incorrect. Ca²⁺ enters the presynaptic terminal and triggers neurotransmitter release, but it is not the substance released into the synaptic cleft.

D. Noradrenalin ✗

Incorrect. Noradrenaline is mainly released by sympathetic postganglionic neurons, not at the neuromuscular junction.

E. Dopamine ✗

Incorrect. Dopamine is a neurotransmitter involved mainly in the central nervous system and specific pathways, not skeletal muscle neuromuscular junctions.

[24/09/2026 8:28 AM] Imaanaa Harmee: **21. Regulatory proteins in the skeletal muscle are:**

A. Actin and myosin

B. Troponin tropomyosin

C. G-proteins

D. Actin and troponin

E. Myosin and tropomyosin

Answer B. Troponin and tropomyosin

In skeletal muscle, troponin and tropomyosin are regulatory proteins that control muscle contraction. At rest, tropomyosin covers the myosin-binding sites on actin, preventing contraction. When Ca²⁺ binds to troponin, it causes a conformational change that moves tropomyosin away, allowing actin and myosin to interact and produce contraction.

Why Not Other Options?

A. Actin and myosin ✗

Actin and myosin are contractile proteins, not regulatory proteins. They directly interact to produce muscle contraction through cross-bridge formation.

C. G-proteins ✗

G-proteins are involved in cell signaling pathways, especially in receptor-mediated signaling, but they are not involved in skeletal muscle contraction regulation.

D. Actin and troponin ✗

Actin is a contractile protein, while troponin is a regulatory protein. They do not both belong to the regulatory protein group.

E. Myosin and tropomyosin ✕

Myosin is a contractile protein, whereas tropomyosin is a regulatory protein.

[24/09/2026 8:28 AM] Imaanaa Harmee: **22. Which drug of the following blocks neuromuscular junction?**

A. D-tubocurarine

B. Methacholine

C. Nicotine

D. Carbacholine

E. Acetylcholine

F. None

Answer A. D-tubocurarine

D-tubocurarine is a competitive antagonist of nicotinic acetylcholine receptors at the neuromuscular junction. It blocks the action of acetylcholine on skeletal muscle fibers, preventing depolarization and causing skeletal muscle paralysis.

Why Not Other Options?

B. Methacholine ✕

Methacholine is a muscarinic receptor agonist used mainly in bronchial challenge testing. It does not block the neuromuscular junction.

C. Nicotine ✕

Nicotine is a nicotinic acetylcholine receptor agonist. It stimulates nicotinic receptors rather than blocking them.

D. Carbacholine ✕

Carbachol is a cholinergic agonist that stimulates both muscarinic and nicotinic receptors; it does not block neuromuscular transmission.

E. Acetylcholine ✕

Acetylcholine is the normal neurotransmitter released at the neuromuscular junction and activates nicotinic receptors to cause muscle contraction.

F. None ✕

Incorrect because D-tubocurarine is a classic neuromuscular junction blocker.

Exam:-Physiology

Year:-2014 E.C

Dr. Doc

[24/09/2026 8:28 AM] Imaanaa Harmee: **1. Which of the following statement is not correct about human physiology?**

- A. It deals with the normal functions of human body.**
- B. It studies how human body organs and organ systems are controlled, how they interact, and how they adapt to changing conditions.**
- C. It is the qualitative science by which the physiological parameters are expressed in figures/numbers and units.**
- D. It has a strong link with other disciplines like Anatomy, Biochemistry, Pathology, Pharmacology, Physics and Anesthesia.**

Answer: C. It is the qualitative science by which the physiological parameters are expressed in figures/numbers and units.

Explanation:

Human physiology is the scientific study of the normal functions of the human body, including how cells, tissues, organs, and organ systems function and interact. Physiology is primarily considered a quantitative science, not a qualitative science, because many physiological functions and parameters can be measured and expressed numerically with appropriate units—for example, blood pressure in mmHg, heart rate in beats/minute, and body temperature in °C. Therefore, option C is incorrect because it describes physiology as qualitative rather than quantitative. The other statements correctly describe the scope and interdisciplinary nature of human physiology. Physiology is closely related to anatomy because structure and function are interconnected, and it also has important relationships with biochemistry, pathology, pharmacology, physics, and other medical sciences.

2. The scholar who stated that the cell is bathed within internal environment or ECF was:

- A. William Harvey**
- B. Claude Bernard**
- C. Walter Cannon**
- D. John Snow**

Answer: B. Claude Bernard

Explanation:

The correct answer is Claude Bernard, a prominent French physiologist who

introduced the concept of the internal environment (*milieu intérieur*). Bernard recognized that the cells of the body are surrounded by and depend upon the extracellular fluid (ECF) for their survival. He emphasized that maintaining a relatively stable internal environment is essential for normal cellular function. This idea became a fundamental basis for the modern concept of homeostasis. Walter Cannon later developed and popularized the term homeostasis, describing the mechanisms by which the body maintains stability of its internal environment. William Harvey is famous for describing the circulation of blood, while John Snow is well known for his contributions to epidemiology, particularly his investigation of cholera.

3. Which of the following statement does not characterize homeostasis?

- A. It controls the deviation of certain physiological parameters.**
- B. It is a physiological adjustment by responds to stimuli.**
- C. It works at physiological limit/narrow range of value change.**
- D. It stabilizes the deranged substances in intracellular fluid.**

Answer: D. It stabilizes the deranged substances in intracellular fluid.

Explanation:

Homeostasis refers to the body's ability to maintain a relatively stable internal environment, particularly the conditions of the extracellular fluid (ECF) surrounding cells. It does not mean that the body keeps every intracellular substance directly stable. Instead, many homeostatic mechanisms regulate variables such as temperature, blood glucose, blood pressure, pH, oxygen, carbon dioxide, and electrolyte concentrations within appropriate physiological ranges. Homeostasis involves detecting changes or deviations from a normal range and producing appropriate responses to counteract those changes. Therefore, A, B, and C describe important characteristics of homeostatic regulation. Option D is incorrect because homeostatic control is primarily concerned with maintaining the stability of the internal environment/ECF, which in turn provides suitable conditions for cells and helps maintain their intracellular environment.

[24/09/2026 8:28 AM] Imaanaa Harmee: **4. Which of the following is false about homeostasis regulatory processes?**

- A. Compensated disturbance brings disorder in homeostasis wing.**
- B. All systems are equally involved in our homeostasis.**
- C. Homeostasis control system returns the deviated parameter to optimal range.**
- D. Majority of the parameters are maintained by negative feedback mechanism.**

Answer: B. All systems are equally involved in our homeostasis.

Explanation:

The statement that all body systems are equally involved in homeostasis is false. Different organ systems have different contributions to maintaining the internal

environment, and they are not equally involved in every homeostatic process. For example, the nervous and endocrine systems are major regulatory systems that coordinate many homeostatic responses, while the kidneys are especially important in regulating fluid volume, electrolytes, and acid-base balance. The lungs are important in regulating oxygen and carbon dioxide, while the cardiovascular system transports substances throughout the body. Most homeostatic mechanisms operate through negative feedback, in which a change away from the desired range triggers responses that oppose or reduce the original change. A compensated disturbance can also mean that regulatory mechanisms have acted to reduce the effect of a disturbance and restore the variable toward its normal range. Thus, B is the false statement.

5. Removal of one of the organ systems doesn't threaten the survival of an individual:

- A. Respiratory system**
- B. Renal system**
- C. Reproductive system**
- D. Integumentary system**

Answer: C. Reproductive system

Explanation:

The reproductive system is the correct answer because it is not essential for the immediate survival of an individual, although it is essential for the continuation of the species. Its primary function is reproduction, including the production of gametes and reproductive hormones and, in females, supporting pregnancy and childbirth. In contrast, the respiratory system is essential for obtaining oxygen and eliminating carbon dioxide; the renal system is crucial for removing metabolic waste and maintaining fluid, electrolyte, and acid-base balance; and the integumentary system provides an important protective barrier and contributes to temperature regulation, prevention of fluid loss, and other vital functions. Therefore, an individual can survive without a functioning reproductive system, but loss of essential functions of the respiratory or renal systems would rapidly threaten life.

[24/09/2026 8:28 AM] Imaanaa Harmee: **6. Which of the following body processes is not regulated by the positive feedback homeostasis mechanism?**

- A. Plasma blood glucose regulation**
- B. The process of child delivery**
- C. Blood clotting**
- D. Milk letdown process**
- E. Ovulation**

Answer: A. Plasma blood glucose regulation

Explanation:

Plasma blood glucose regulation is an example of negative feedback, not positive feedback. When blood glucose rises, pancreatic β -cells release insulin, which increases glucose uptake and storage and consequently lowers blood glucose toward its normal range. When blood glucose falls, glucagon helps increase it. The response therefore opposes the original disturbance, which is the defining feature of negative feedback. In contrast, childbirth, blood clotting, milk letdown, and the hormonal events leading to ovulation involve positive-feedback mechanisms in which the response reinforces the initial stimulus until the physiological event is completed. During childbirth, for example, uterine contractions promote oxytocin release, and oxytocin further strengthens uterine contractions until delivery. Thus, A is correct.

7. Which one is not true about the regulatory mechanism of human body?

- A. The human body has dynamic mechanisms that detect and respond to deviations.**
- B. The human body has the capacity to maintain a steady state in response to changes in the intracellular fluid.**
- C. Negative feedback initiates responses that counteract the original stimuli.**
- D. Positive feedback promotes the occurrence of the underlying stimuli.**

Answer: B. The human body has the capacity to maintain a steady state in response to changes in the intracellular fluid.

Explanation:

Homeostatic regulation primarily maintains a relatively stable internal environment, especially the extracellular fluid (ECF) surrounding the cells. The intracellular fluid is the fluid inside the cells and is influenced by the conditions of the ECF, but the body's major homeostatic systems regulate the ECF so that cells can function properly. The body constantly detects deviations and produces corrective responses. In negative feedback, the response opposes the initial change, helping return a variable toward its normal range. In positive feedback, the response reinforces the original stimulus and continues to amplify the process until a particular event is completed. Therefore, B is the incorrect statement.

8. The hormone that doesn't involve in regulation of calcium is:

- A. Parathyroid hormone**
- B. Cortisol**
- C. Calcitonin**
- D. Calcitriol**

Answer: B. Cortisol

Explanation:

The principal hormones involved in calcium regulation include parathyroid hormone (PTH), calcitriol, and calcitonin. PTH is released when blood calcium decreases and acts to increase calcium concentration in the blood through effects on bone and the kidneys and by promoting activation of vitamin D. Calcitriol, the active form of vitamin D, increases intestinal absorption of calcium and phosphate. Calcitonin generally acts to decrease blood calcium, mainly by inhibiting bone resorption, although its contribution to day-to-day calcium regulation is less important than that of PTH. Cortisol is a glucocorticoid hormone mainly involved in metabolism, stress responses, and immune regulation. It is not considered one of the principal hormones regulating normal calcium homeostasis. Therefore, B is correct.

[24/09/2026 8:28 AM] Imaanaa Harmee: **9. Nervous system adjusts homeostasis by release of:**

- A. Energy**
- B. Hormones**
- C. Neurotransmitter**
- D. Enzymes**

Answer: C. Neurotransmitter

Explanation:

The nervous system regulates many homeostatic functions through electrical impulses and neurotransmitters. Neurons communicate with other neurons, muscles, and glands by releasing neurotransmitters at synapses. This allows the nervous system to rapidly coordinate functions such as heart rate, blood vessel diameter, breathing, digestion, and body temperature regulation. Hormones are primarily associated with the endocrine system, although some nervous-system structures can influence or stimulate hormone release.

Energy and enzymes are not the primary signaling substances by which neurons communicate. Therefore, the best answer is C. Neurotransmitter.

10. In the loop of homeostasis, the disturbance is detected by:

- A. Stimuli**
- B. Effectors**
- C. Receptors**
- D. Control unit**

Answer: C. Receptors

Explanation:

A homeostatic control system generally consists of a stimulus, receptor, control center, and effector. The receptor or sensor is the component that detects a change in the regulated variable. For example, thermoreceptors detect changes in temperature, while baroreceptors detect changes in blood pressure. After detecting the disturbance, the receptor sends information to the control center, which evaluates the information and determines an appropriate response. The control center then sends instructions to the effector, such as a muscle or gland, which produces the corrective response. A stimulus is the actual change or disturbance, but it does not detect the disturbance itself. Therefore, C. Receptors is the correct answer.

[24/09/2026 8:28 AM] Imaanaa Harmee: 11. The atom that forms the most of the mass of the human body is:

- A. Oxygen
- B. Hydrogen
- C. Carbon
- D. Sodium

Answer: A. Oxygen

Explanation:

Oxygen is the element that contributes the greatest proportion of the human body's mass. This is mainly because the human body contains a large amount of water, and water consists of hydrogen and oxygen, with oxygen contributing substantially more mass per molecule. Oxygen also occurs in many important biological molecules such as carbohydrates, proteins, lipids, and nucleic acids. By mass, oxygen accounts for roughly 65% of the human body, while carbon contributes about 18%, hydrogen about 10%, and nitrogen about 3%. Therefore, among the options, oxygen forms the greatest portion of body mass.

12. Majority of the human body is:

- A. Protein
- B. Lipid
- C. Carbohydrate
- D. Water

Answer: D. Water

Explanation:

The human body is composed primarily of water. In an average adult, water accounts for approximately 50–60% of body weight, although the exact percentage varies with age, sex, and body composition. Water is essential because it serves as the major solvent

for biochemical reactions, transports nutrients and waste products, participates in chemical reactions, helps regulate body temperature, and contributes to maintaining blood volume and cellular function. Although proteins, lipids, and carbohydrates are all essential components of the body, none constitutes as large a proportion of total body mass as water. Therefore, D. Water is correct.

13. Which one is not true about the composition of cell membrane?

- A. Proteins 55%**
- B. Lipids 42%**
- C. Carbohydrates 3%**
- D. Minerals 7%**

Answer: D. Minerals 7%

Explanation:

The cell membrane is primarily composed of lipids, proteins, and carbohydrates. A commonly cited approximate composition by mass is about 55% protein, 42% lipid, and 3% carbohydrate. Proteins perform many important functions, including transport, receptor activity, enzymatic activity, and cell signaling. Lipids, particularly phospholipids, form the fundamental bilayer structure of the membrane, while carbohydrates are attached mainly to membrane proteins and lipids and contribute to cell recognition and communication. Minerals are not considered a major structural component of the cell membrane at 7%. Therefore, D is the statement that is not true.

14. Which serves as the last source of energy and mainly used for structure and function of a cell?

- A. Carbohydrates**
- B. Proteins**
- C. Lipids**
- D. Nucleic acids**

Answer: B. Proteins

Explanation:

Proteins are primarily structural and functional molecules of cells rather than the body's preferred energy source. Carbohydrates are generally the body's readily available energy source, while lipids provide a major long-term energy store. Proteins are mainly used to build and repair cellular structures and to perform essential functions as enzymes, receptors, channels, transporters, antibodies, and contractile

proteins. However, when carbohydrate and fat stores are insufficient, proteins can be broken down and used as an energy source. For this reason, proteins are often described as a last or secondary source of energy compared with carbohydrates and fats. Therefore, B. Proteins is the intended answer.

15. Which one is not true about the human cells?

- A. There are 4 main types of cells that build tissue.**
- B. The dominant cells in number are red blood cells.**
- C. The largest cell in size is nerve cells.**
- D. The biggest cell is egg cell.**

Answer: C. The largest cell in size is nerve cells.

Explanation:

The statement that nerve cells are the largest cells in size is not generally correct. The human ovum (egg cell) is the largest human cell by diameter, measuring approximately 100–120 micrometers. Some neurons can be extremely long—certain motor neurons can extend from the spinal cord to the limbs—but being very long does not mean that a neuron is the largest cell by overall cell body size. Red blood cells are indeed extremely numerous, and the four basic tissue types are epithelial, connective, muscle, and nervous tissue.

[24/09/2026 8:28 AM] Imaanaa Harmee: **16. Which one of the following organ system has no direct contact with external environment?**

- A. Respiratory system**
- B. GIT system**
- C. Circulatory system**
- D. Renal system**

Answer: C. Circulatory system

Explanation:

The circulatory system does not have direct contact with the external environment under normal conditions. It is a closed system in which blood circulates through the heart and blood vessels. In contrast, the respiratory system has direct communication with the outside through the airways, allowing air to enter the lungs. The gastrointestinal tract is also continuous with the external environment through the mouth and anus, because substances from outside the body enter the digestive tract. The renal/urinary system also has an external opening through which urine is eliminated. The circulatory system therefore remains enclosed within the body and transports oxygen, nutrients, hormones, and waste products between tissues. Hence, C is correct.

17. Which one of the following is not the function of smooth endoplasmic reticulum?

- A. Protein synthesis**
- B. Lipid synthesis**
- C. Glycogen storage**
- D. Ca storage**

Answer: A. Protein synthesis

Explanation:

The smooth endoplasmic reticulum (SER) does not have ribosomes attached to its surface, so it is not the major site of protein synthesis. Protein synthesis occurs primarily on ribosomes, particularly those attached to the rough endoplasmic reticulum (RER). The SER has several other important functions, including lipid synthesis, detoxification of certain substances, carbohydrate metabolism, and storage and release of calcium ions, especially in muscle cells where it forms the sarcoplasmic reticulum. Therefore, protein synthesis is the function that does not characterize the smooth ER, making A the correct answer.

18. Which is a set of stacked membranes with flattened membranous saccules and cisterns that communicates with the different organelles?

- A. Ribosome**
- B. Endoplasmic reticulum**
- C. Golgi apparatus**
- D. Mitochondria**

Answer: C. Golgi apparatus

Explanation:

The Golgi apparatus consists of stacks of flattened membrane-bound sacs called cisternae. It is an important component of the cell's processing and trafficking system. Proteins and lipids arriving from the endoplasmic reticulum are modified, sorted, and packaged by the Golgi apparatus before being transported to different destinations, such as the plasma membrane, lysosomes, or secretory vesicles. The Golgi therefore communicates functionally with the endoplasmic reticulum and other organelles through transport vesicles. Ribosomes are structures responsible primarily for protein synthesis, while mitochondria are specialized for ATP production. Thus, C. Golgi apparatus is correct.

19. The part of Golgi apparatus that serves as output region in secreting substances as vesicle is:

- A. Cis Golgi networks**
- B. Trans Golgi networks**
- C. Medial Golgi networks**
- D. Golgi lumen**

Answer: B. Trans Golgi networks

Explanation:

The trans-Golgi network (TGN) is the output or shipping side of the Golgi apparatus. Proteins and lipids enter the Golgi from the cis side, pass through the medial regions where they undergo further modification, and eventually reach the trans-Golgi network. The TGN sorts and packages these processed molecules into vesicles that transport them to their appropriate destinations, including the plasma membrane, secretory pathways, and lysosomes. The cis-Golgi network is mainly associated with receiving material from the endoplasmic reticulum, while the medial Golgi is involved in processing. Therefore, B. Trans Golgi networks is correct.

[24/09/2026 8:28 AM] Imaanaa Harmee: **20. The different substances that make up the cell collectively are:**

- A. Organelles**
- B. Cytosol**
- C. Protoplasm**
- D. Cytoplasm**

Answer: C. Protoplasm

Explanation:

Protoplasm refers broadly to the living material of the cell and includes the cytoplasm and nucleus, encompassing the cell's living contents. The term cytoplasm refers to the material between the plasma membrane and the nucleus and includes the cytosol, organelles, and various inclusions. Cytosol specifically refers to the fluid portion of the cytoplasm, excluding the membrane-bound organelles. Organelles are specialized structures within cells, such as mitochondria, Golgi apparatus, and endoplasmic reticulum, rather than the collective substance making up the entire living cell. Therefore, for this question, C. Protoplasm is the best answer.

[24/09/2026 8:28 AM] Imaanaa Harmee: **21. Which one is the pyrimidine base in RNA?**

- A. Adenine**
- B. Guanine**
- C. Uracil**
- D. Thymine**

Answer: C. Uracil

Explanation:

The nitrogenous bases are divided into two groups: purines and pyrimidines. Purines are adenine (A) and guanine (G), while pyrimidines include cytosine (C), thymine (T), and uracil (U). In RNA, uracil replaces thymine, so the pyrimidine bases found in RNA are cytosine and uracil. DNA contains cytosine and thymine instead. Therefore, among the choices, uracil (C) is the correct answer.

22. The order of central dogma of life is:

- A. DNA → mRNA → tRNA → rRNA → protein**
- B. tRNA → rRNA → mRNA → protein → DNA**
- C. Protein → DNA → rRNA → mRNA → tRNA**
- D. DNA → protein → mRNA → rRNA → tRNA**

Answer: A. DNA → mRNA → tRNA → rRNA → protein

Explanation:

The central dogma of molecular biology describes the flow of genetic information from DNA to RNA to protein. First, a segment of DNA is transcribed to produce messenger RNA (mRNA). The mRNA then carries the genetic information to the ribosome, where it is translated into a protein. During translation, tRNA brings specific amino acids according to the codons present on the mRNA, while rRNA is an important structural and catalytic component of the ribosome. Therefore, although the fundamental central-dogma summary is DNA → RNA → protein, the only option that follows the appropriate information flow is A.

23. Rigidity of the cell membrane is determined by:

- A. Protein**
- B. Glycoprotein**
- C. Cholesterol**
- D. Phospholipid**

Answer: C. Cholesterol

Explanation:

Cholesterol is an important component of the plasma membrane and helps regulate its fluidity, stability, and mechanical properties. It is located between the phospholipids in the lipid bilayer. At higher temperatures, cholesterol helps restrain excessive movement

of phospholipids, making the membrane less fluid; at lower temperatures, it prevents phospholipids from packing too tightly and therefore helps maintain membrane fluidity. In this sense, cholesterol acts as a buffer for membrane fluidity and contributes to membrane stability or rigidity. Therefore, C. Cholesterol is the best answer.

24. The part of lipid bilayer plasma membrane facing to each other to the center is:

- A. Hydrophobic head**
- B. Hydrophilic tail**
- C. Lipophobic head**
- D. Lipophilic tail**

Answer: D. Lipophilic tail

Explanation:

The plasma membrane is composed primarily of a phospholipid bilayer. Each phospholipid has a hydrophilic, water-attracting head and hydrophobic, water-repelling fatty-acid tails. The hydrophilic heads face the watery environments on the extracellular and intracellular sides of the membrane, while the hydrophobic/lipophilic tails face inward toward each other, forming the central portion of the bilayer. This hydrophobic interior is important because it prevents many water-soluble substances and charged ions from freely crossing the membrane. Therefore, D. Lipophilic tail is the correct choice.

25. Peripheral proteins that bind to the intracellular surface contribute to:

- A. Cytoskeleton**
- B. Extracellular matrix**
- C. Glycocalyx**
- D. Enzymes**

Answer: A. Cytoskeleton

Explanation:

Peripheral membrane proteins can attach to the intracellular surface of the plasma membrane and interact with components of the cytoskeleton. The cytoskeleton provides structural support, maintains cell shape, participates in cell movement, and helps organize membrane proteins. Peripheral proteins can act as connecting elements between the plasma membrane and cytoskeletal structures. In contrast, the extracellular matrix is outside the cell, while the glycocalyx is composed largely of carbohydrate-containing molecules on the extracellular surface of the plasma

membrane. Some peripheral proteins can function as enzymes, but the question specifically asks what intracellular-surface peripheral proteins contribute to structurally, making A. Cytoskeleton the best answer.

[24/09/2026 8:28 AM] Imaanaa Harmee: **26. The movement of large molecules across cell membrane takes place by:**

- A. Passive transport**
- B. Active transport**
- C. Vesicular transport**
- D. Diffusion**

Answer: C. Vesicular transport

Explanation:

Large molecules and particles generally cannot cross the plasma membrane through ordinary channels or by simple diffusion. Instead, cells use vesicular transport, in which portions of the plasma membrane form vesicles that carry material into or out of the cell. Endocytosis brings material into the cell, while exocytosis releases material outside the cell. Phagocytosis is used for large particles such as microorganisms, whereas pinocytosis and receptor-mediated endocytosis handle extracellular fluid and specific molecules. Vesicular transport requires cellular energy, particularly for active processes such as endocytosis and exocytosis. Therefore, C is correct.

27. Which one of the following is not true about transport of substances via cell membrane?

- A. O₂, CO₂, N₂, steroids, urea, glycerol, ethanol alcohol, lipid soluble drugs and ions are transported by simple diffusion, towards their concentration gradient.**
- B. The rate of facilitated diffusion is directly proportional to the concentration of substances.**
- C. Water is transported by osmosis towards its concentration through aquaporins in semipermeable membrane.**
- D. Substances are transported against concentration/electrochemical gradient, uphill direction by expenditure of energy.**

Answer: A. O₂, CO₂, N₂, steroids, urea, glycerol, ethanol alcohol, lipid soluble drugs and ions are transported by simple diffusion, towards their concentration gradient.

Explanation:

Option A is incorrect because ions do not normally cross the lipid bilayer by simple diffusion. Charged ions such as Na⁺, K⁺, Ca²⁺, and Cl⁻ require membrane proteins such as ion channels or transporters because the hydrophobic interior of the phospholipid bilayer is highly resistant to charged particles. Small nonpolar molecules such as O₂ and CO₂ can cross readily by simple diffusion, and many lipid-soluble substances can also cross the membrane in this way. Facilitated diffusion moves substances down their

concentration or electrochemical gradient using membrane proteins, whereas active transport moves substances against their gradient and requires energy. Water moves by osmosis, often through aquaporins. Therefore, A is the incorrect statement.

28. Which one of the following factors promotes the rate of diffusion?

- A. Thick membrane**
- B. Large weight molecule**
- C. Water soluble molecule**
- D. Lipid soluble molecule**

Answer: D. Lipid soluble molecule

Explanation:

The rate at which a substance diffuses across a cell membrane depends partly on its lipid solubility. The plasma membrane has a hydrophobic lipid interior, so substances that are highly lipid-soluble can dissolve in the membrane and cross it relatively easily. Small nonpolar molecules such as oxygen and carbon dioxide therefore diffuse rapidly. A thick membrane increases the distance that molecules must travel and slows diffusion, while larger/heavier molecules generally diffuse more slowly. Water-soluble substances often require membrane channels or transport proteins because they do not readily dissolve in the lipid bilayer. Thus, D. Lipid soluble molecule promotes diffusion across the membrane.

[24/09/2026 8:28 AM] Imaanaa Harmee: **26. The movement of large molecules across cell membrane takes place by:**

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The rate at which a substance diffuses across a cell membrane depends partly on its lipid solubility. The plasma membrane has a hydrophobic lipid interior, so substances that are highly lipid-soluble can dissolve in the membrane and cross it relatively easily. Small nonpolar molecules such as oxygen and carbon dioxide therefore diffuse rapidly. A thick membrane increases the distance that molecules must travel and slows diffusion, while larger/heavier molecules generally diffuse more slowly. Water-soluble substances often require membrane channels or transport proteins because they do not readily

dissolve in the lipid bilayer. Thus, D. Lipid soluble molecule promotes diffusion across the membrane.

[24/09/2026 8:28 AM] Imaanaa Harmee: **31. The lung volume and capacity that cannot be determined by simple spirometer is:**

- A. Tidal volume, residual volume, functional residual capacity**
- B. Residual volume, functional residual capacity, total lung capacity**
- C. Total lung capacity, residual volume, tidal volume**
- D. Tidal volume, inspiratory reserve volume**
- E. Vital capacity and residual volume**

Answer: B. Residual volume, functional residual capacity, total lung capacity

Explanation:

A simple spirometer can measure volumes of air that can be moved into or out of the lungs, such as tidal volume, inspiratory reserve volume, expiratory reserve volume, and vital capacity. However, it cannot directly measure residual volume (RV) because residual volume is the air remaining in the lungs after maximal expiration and cannot be voluntarily expelled into the spirometer. Since functional residual capacity (FRC) = residual volume + expiratory reserve volume, it also cannot be measured directly by simple spirometry. Likewise, total lung capacity (TLC) = vital capacity + residual volume, so TLC cannot be directly determined either. Therefore, the three capacities/volumes that cannot be determined by simple spirometry are RV, FRC, and TLC, making B correct.

32. Complete obstruction of a major bronchus usually results in:

- A. Collapse of the alveoli supplied by the bronchus.**
- B. A rise in local intrapleural pressure**
- C. An increase in physiological dead space**
- D. An increase in blood flow to the lung tissue supplied by the bronchus**
- E. Hypoxemia**

Answer: A. Collapse of the alveoli supplied by the bronchus.

Explanation:

When a major bronchus becomes completely obstructed, fresh air can no longer enter the alveoli supplied by that bronchus. The air already present in those alveoli is gradually absorbed into the pulmonary blood, causing the alveoli to lose volume and eventually collapse, a condition called atelectasis. The local intrapleural pressure does not rise as a direct consequence of the obstruction. Blood flow to the poorly ventilated region is generally reduced through hypoxic pulmonary vasoconstriction, rather than increased. The obstructed region may contribute to shunt-like physiology rather than

increasing physiological dead space. The resulting mismatch between ventilation and perfusion can cause hypoxemia, but the most direct consequence of complete bronchial obstruction is alveolar collapse. Thus, A is the best answer.

33. Lack of surfactant substances as occurs in infants with respiratory distress syndrome (RDS) results in:

- A. An increased lung compliance**
- B. Stabilization of alveolar volume**
- C. An increased retroactive force of the lungs by increasing alveolar surface tension**
- D. A reduced alveolar-arterial O₂ tension difference**
- E. A decrease in the filtration forces in the pulmonary capillaries**

Answer: C. An increased retroactive force of the lungs by increasing alveolar surface tension

Explanation:

Pulmonary surfactant is produced primarily by type II alveolar cells and reduces the surface tension at the air-liquid interface within the alveoli. This reduces the tendency of alveoli to collapse and makes the lungs easier to expand, thereby increasing lung compliance. When surfactant is deficient, as occurs in premature infants with respiratory distress syndrome, alveolar surface tension increases substantially. According to the principle described by the Law of Laplace, increased surface tension increases the collapsing pressure within the alveolus. Consequently, the lungs become harder to expand, alveolar collapse occurs, and the work of breathing increases. Gas exchange is impaired, which can increase the alveolar-arterial oxygen difference. Therefore, C is correct.

[24/09/2026 8:28 AM] Imaanaa Harmee: **34. The total lung capacity consists of:**

- A. Vital capacity & tidal volume**
- B. Residual volume & vital capacity**
- C. Residual volume & vital capacity**
- D. Tidal volume & functional residual capacity**
- E. Tidal volume & residual volume**

Answer: B. Residual volume & vital capacity

Explanation:

Total lung capacity (TLC) is the maximum amount of air contained in the lungs after a person performs a maximal inspiration. It is composed of all four basic lung volumes: tidal volume, inspiratory reserve volume, expiratory reserve volume, and residual

volume. Mathematically, $TLC = \text{vital capacity} + \text{residual volume}$. Vital capacity represents the maximum amount of air that can be voluntarily exhaled after maximal inspiration, while residual volume is the air that remains in the lungs even after maximal expiration. Therefore, B is correct. Note that option C is identical to B, so if both are exactly as written, both B and C represent the correct relationship.

35. Not the function of surfactant is:

- A. Increases surface area**
- B. Decreases pressure in the alveoli**
- C. Prevents lung collapse**
- D. Increases work of breathing**
- E. Facilitates gas exchange**

Answer: D. Increases work of breathing

Explanation:

Pulmonary surfactant reduces alveolar surface tension, which makes the lungs easier to expand and therefore decreases the work of breathing. It also helps stabilize alveoli and prevents them from collapsing, particularly during expiration. By maintaining functional alveolar stability and improving lung compliance, surfactant indirectly facilitates effective gas exchange. Therefore, saying that surfactant increases the work of breathing is incorrect. Surfactant deficiency, rather than normal surfactant activity, increases the work required to inflate the lungs. Thus, D is the correct answer.

36. Which structure of the following does not belong to the respiratory portion of the breathing organ?

- A. Terminal bronchioles**
- B. Respiratory bronchioles**
- C. Alveolar ducts**
- D. Alveolar sacs**
- E. None of the above**

Answer: A. Terminal bronchioles

Explanation:

The respiratory portion of the respiratory system is the region where gas exchange can occur. It begins with the respiratory bronchioles and continues through the alveolar ducts, alveolar sacs, and alveoli. Terminal bronchioles, in contrast, belong to the conducting zone. Their main function is to conduct and distribute air; they do not

normally participate directly in gas exchange because they lack alveoli in their walls. Respiratory bronchioles have some alveoli and therefore mark the beginning of the respiratory zone. Consequently, A. Terminal bronchioles is the correct answer.

37. Air-tight chest wall is important in that it:

- A. Increases lung compliance**
- B. Decreases the work of breathing**
- C. Creates negative pressure in the chest**
- D. Prevents lung collapse**
- E. Causes the lung to become "dry"**

Answer: C. Creates negative pressure in the chest

Explanation:

The airtight nature of the chest wall is essential for maintaining the negative intrapleural pressure that keeps the lungs expanded. The lungs naturally have an elastic tendency to recoil inward, while the chest wall tends to recoil outward. Because the pleural cavity is sealed and contains only a thin film of fluid, these opposing forces create a subatmospheric or negative intrapleural pressure. This pressure helps keep the visceral and parietal pleural surfaces closely apposed and keeps the lungs expanded against the chest wall. If the chest wall is opened, as in a penetrating injury, air can enter the pleural space and the negative pressure is lost, potentially causing lung collapse (pneumothorax). Therefore, C is correct.

[24/09/2026 8:28 AM] Imaanaa Harmee: **38. Of the following, the respiratory function of the respiratory system is:**

- A. Defense**
- B. Phonation**
- C. Gas conduction**
- D. Olfaction**
- E. Blood pressure regulation**

Answer: C. Gas conduction

Explanation:

One of the primary functions of the respiratory system is to conduct air to and from the respiratory surfaces, allowing oxygen to reach the alveoli and carbon dioxide to be removed. This is directly related to its respiratory function.

The respiratory system also performs several other functions, including defense against inhaled particles and pathogens, phonation through airflow across the vocal cords, and olfaction through receptors in the nasal cavity. Regulation of blood pressure is not considered a primary respiratory function, although the respiratory system can

indirectly influence cardiovascular function and blood gases. Among the choices, C. Gas conduction is the best answer.

39. Which one of the following is true about tidal volume?

- A. The amount of air that normally moves into (or out of) the lung with each respiration.**
- B. The amount of air that enters the lung but does not participate in gas exchange.**
- C. The largest amount of air expired after maximal expiratory effort.**
- D. The largest amount of gas that can be moved into and out of the lungs in one minute.**
- E. The volume of air one breathes in with exertion.**

Answer: A. The amount of air that normally moves into (or out of) the lung with each respiration.

Explanation:

Tidal volume (TV) is the volume of air that enters or leaves the lungs during a normal, quiet breath. In a healthy resting adult, it is approximately 500 mL per breath, although the exact value varies with body size and other factors. Option B describes dead-space volume, which is air that does not participate directly in gas exchange. Option C describes vital capacity or a component of maximal expiratory volume, depending on the exact context. Option D describes minute ventilation, which is the total volume of air moved per minute and is calculated as tidal volume $\times$ respiratory rate. Therefore, A is the correct definition of tidal volume.

40. Decreasing the surface area of the respiratory membrane would affect:

- A. Inhalation**
- B. Cellular respiration**
- C. External respiration**
- D. Mucus production**
- E. Internal respiration**

Answer: C. External respiration

Explanation:

External respiration is the exchange of oxygen and carbon dioxide between the alveolar air and pulmonary capillary blood. The efficiency of this process depends on several factors, including the surface area available for diffusion, the thickness of the respiratory membrane, and the relevant pressure gradients. A decrease in respiratory membrane surface area means there is less area available for gas diffusion, reducing the transfer of oxygen into the blood and carbon dioxide out of the blood. This is seen in

conditions that destroy or reduce alveolar surface area, such as emphysema. Internal respiration, by contrast, refers to gas exchange between systemic blood and tissue cells. Therefore, decreased respiratory membrane surface area primarily affects external respiration, making C correct.

[24/09/2026 8:28 AM] Imaanaa Harmee: **41. In the lungs, surfactant substance is produced by:**

- A. Pneumocyte type I**
- B. Pneumocyte type II**
- C. Pneumocyte type III**
- D. Alveolar macrophages**
- E. White blood cell**

Answer: B. Pneumocyte type II

Explanation:

Pulmonary surfactant is produced primarily by type II pneumocytes, also called type II alveolar cells. These specialized epithelial cells synthesize and secrete surfactant into the alveolar fluid. Surfactant reduces surface tension at the air-liquid interface of the alveoli, thereby preventing alveolar collapse, particularly during expiration, and increasing lung compliance. Type I pneumocytes are extremely thin cells that cover most of the alveolar surface and are specialized mainly for gas exchange. Alveolar macrophages, on the other hand, provide immune defense by removing particles and microorganisms. Therefore, B is correct.

42. The extent to which the lung expands is its:

- A. Resistance**
- B. Compliance**
- C. Viscosity**
- D. Recoiling ability**
- E. Dryness**

Answer: B. Compliance

Explanation:

Lung compliance refers to the ease with which the lungs can expand when a pressure difference is applied. It is defined as the change in lung volume divided by the change in transpulmonary pressure: $C = \Delta V / \Delta P$. A lung with high compliance expands relatively easily, whereas a lung with low compliance requires greater pressure to achieve the same change in volume. Surfactant increases lung compliance by reducing alveolar surface tension. Lung compliance is therefore different from elastic recoil, which represents the tendency of the lungs to return toward their resting size after being stretched. Thus, B. Compliance is correct.

43. Gastric secretion is stimulated by all except one. Identify that:

- A. Thinking about a delicious food**
- B. Smelling a nice food**
- C. The presence of a protein meal in the stomach**
- D. The presence of large amount of acidic chyme in the duodenum**
- E. Drinking coffee and alcohol**

Answer: D. The presence of large amount of acidic chyme in the duodenum

Explanation:

Gastric acid secretion is stimulated during the cephalic and gastric phases of digestion. Thinking about or smelling appealing food can activate vagal pathways and stimulate gastric secretion even before food enters the stomach. Protein in the stomach stimulates gastrin release, which promotes gastric acid secretion. Coffee and alcohol can also stimulate gastric secretory activity. In contrast, when a large amount of acidic chyme enters the duodenum, it triggers inhibitory mechanisms, including the release of secretin and other intestinal signals, which reduce gastric acid secretion and slow gastric emptying. This protects the duodenum from excessive acidity. Therefore, D is the exception.

44. A hormone that protects the duodenum by causing neutralization of HCl is:

- A. Gastrin**
- B. Secretin**
- C. Somatostatin**
- D. Insulin**
- E. GIP**

Answer: B. Secretin

Explanation:

Secretin is released by S cells in the duodenal mucosa when acidic chyme containing HCl enters the duodenum. Its major protective role is to stimulate the pancreas and biliary system to provide bicarbonate-rich secretions, which neutralize gastric acid and raise the pH of the intestinal contents. Secretin also inhibits gastric acid secretion and slows gastric emptying, giving the duodenum time to neutralize the incoming acid. Gastrin primarily stimulates gastric acid secretion, while somatostatin generally inhibits gastrointestinal secretions. Therefore, B. Secretin is correct. Protein digestion continues extensively in the small intestine through pancreatic proteases such as trypsin, chymotrypsin, and carboxypeptidases, followed by intestinal

peptidases. However, the initial chemical digestion of proteins occurs in the stomach, making B correct.

[24/09/2026 8:28 AM] Imaanaa Harmee: **45. Chemical digestion of proteins begins in the:**

- A. Mouth**
- B. Stomach**
- C. Pancreas**
- D. Small intestine**
- E. Large intestine**

Answer: B. Stomach

Explanation:

Although mechanical processing of food begins in the mouth, significant chemical digestion of proteins begins in the stomach. Gastric glands secrete hydrochloric acid, which denatures proteins and converts pepsinogen into pepsin, the major gastric proteolytic enzyme. Pepsin begins breaking proteins into smaller peptides. Protein digestion continues extensively in the small intestine through pancreatic proteases such as trypsin, chymotrypsin, and carboxypeptidases, followed by intestinal peptidases. However, the initial chemical digestion of proteins occurs in the stomach, making B correct.

46. Identify a correct statement from the following options with respect to salivary secretion:

- A. Parotid glands produce much of the salivary secretions.**
- B. The pH of saliva is more of alkaline.**
- C. Center for salivary secretions is located in the inferior portion of the pons.**
- D. Reflexes originating in the upper GIT can inhibit salivary secretions.**
- E. Thinking about a delicious food doesn't affect salivary secretion.**

Answer: D. Reflexes originating in the upper GIT can inhibit salivary secretions.

Explanation:

Salivary secretion is controlled by the salivatory nuclei in the brainstem, located primarily in the pons and medulla, and is strongly influenced by autonomic nervous activity. Salivation can be stimulated by seeing, smelling, tasting, or even thinking about appetizing food, so E is false. The pH of saliva is generally around neutral and can vary rather than being simply described as strongly alkaline, making B inaccurate. The major salivary glands are the parotid, submandibular, and sublingual glands, with the submandibular glands contributing a large proportion of resting saliva, so A is not the best statement. Reflexes associated with gastrointestinal irritation or nausea can inhibit salivation, making D the correct choice.

47. Which of the following is true regarding absorptive function of the GIT?

- A. Proteins are absorbed only in the form of amino acids.**
- B. None of the lipid end products are directly absorbed into the bloodstream.**
- C. CHOs are absorbed mainly in the form of disaccharides.**
- D. Bicarbonates are absorbed by anion exchange in the large intestine.**
- E. None of the above**

Answer: D. Bicarbonates are absorbed by anion exchange in the large intestine.

Explanation:

The large intestine participates in the absorption of electrolytes and water, and bicarbonate handling occurs through intestinal ion-exchange mechanisms, including exchange with chloride. Option A is too absolute because although most dietary proteins are ultimately absorbed as amino acids and small peptides, dipeptides and tripeptides are also absorbed and then broken down within enterocytes. Carbohydrates are absorbed mainly as monosaccharides, not disaccharides. Lipid digestion products are absorbed into enterocytes and most long-chain fatty acids are packaged into chylomicrons and enter the lymphatic system before reaching the bloodstream, while some shorter-chain fatty acids can enter blood directly. Therefore, D is the intended correct answer.

48. Lamina propria of the GI mucosae contains:

- A. ENS**
- B. Entero-endocrine cells**
- C. Contains lymph nodes**
- D. Myenteric plexus**
- E. None of the above**

Answer: C. Contains lymph nodes

Explanation:

The lamina propria is a layer of loose connective tissue located beneath the epithelium of the gastrointestinal mucosa. It contains blood vessels, lymphatic vessels, immune cells, and often lymphoid aggregates or lymphatic nodules, which contribute to mucosal immune defense. Enteroendocrine cells are primarily epithelial cells rather than components of the lamina propria. The myenteric (Auerbach) plexus is located between

the circular and longitudinal layers of the muscularis externa, while the submucosal (Meissner) plexus is located in the submucosa. Therefore, C is the best answer.

[24/09/2026 8:28 AM] Imaanaa Harmee: 49. Which of the following is true with respect to GI innervation?

- A. Parasympathetic nerves provide extensive innervation to the small intestine
- B. The sacral parasympathetic innervates proximal part half of the LI
- C. The rectal and anal regions are better supplied with PaNS
- D. Sympathetic fibers innervate all portions of the GIT
- E. Sympathetic innervation is more extensive in the oral

Answer: A. Parasympathetic nerves provide extensive innervation to the small intestine

Explanation:

The gastrointestinal tract receives both parasympathetic and sympathetic innervation, in addition to extensive intrinsic control through the enteric nervous system. Parasympathetic stimulation generally promotes gastrointestinal activity, including motility and secretion. The vagus nerve supplies much of the gastrointestinal tract from the esophagus through the proximal large intestine, while the sacral parasympathetic nerves primarily supply the distal colon, rectum, and anus. Therefore, the statement that the sacral parasympathetic supplies the proximal half of the large intestine is incorrect. Sympathetic fibers do extend throughout the gastrointestinal tract, but A is the clearest correct statement because parasympathetic nerves provide extensive functional innervation to the small intestine.

50. Which of the following is true regarding regulation of food intake and energy balance?

- A. Destruction of the lateral nuclei of the hypothalamus causes aphagia.
- B. Stimulation of the ventromedial nuclei of the hypothalamus leads to hyperphagia.
- C. Stimulation of lateral nuclei of the hypothalamus causes anorexia.
- D. Regulation of food intake is exclusively the function of the hypothalamus.
- E. Amygdala and cerebral cortex are not involved in the regulation of feeding.

Answer: A. Destruction of the lateral nuclei of the hypothalamus causes aphagia.

Explanation:

The hypothalamus plays a major role in regulating food intake and energy balance. The lateral hypothalamic area has traditionally been associated with hunger and feeding behavior, so destruction of this area can result in markedly reduced food intake or aphagia. Conversely, stimulation of the lateral hypothalamic area promotes feeding. The ventromedial hypothalamus is traditionally associated with satiety; stimulation tends to reduce food intake, while lesions can produce hyperphagia and obesity.

Therefore, B and C reverse the expected effects. Regulation of feeding is not exclusively controlled by the hypothalamus because the brainstem, limbic system, cerebral cortex, gastrointestinal tract, and numerous hormones and metabolic signals also participate. Hence, A is the correct answer.

[24/09/2026 8:28 AM] Imaanaa Harmee: **51. True about GI reflex is:**

- A. The gastro-colic reflex is inhibitory to defecation**
- B. The entero-gastric reflex is facilitatory to gastric emptying**
- C. Mixing contractions are controlled by reflexes occurring within the ENS**
- D. The colono-ileal reflex facilitates ileal emptying into the colon**
- E. None of the above**

Answer: C. Mixing contractions are controlled by reflexes occurring within the ENS.

Explanation:

The gastrointestinal tract contains an extensive enteric nervous system (ENS) that coordinates many local digestive activities, including secretion, blood flow, and motility. Mixing contractions, particularly segmentation contractions in the small intestine, are largely coordinated by local enteric reflexes and help mix intestinal contents with digestive secretions while bringing the contents into contact with the mucosal surface for absorption. The other statements describe reflexes incorrectly. The gastrocolic reflex generally promotes colonic motility following a meal rather than inhibiting defecation. The enterogastric reflex is inhibitory to gastric motility and emptying when chyme enters the small intestine. The colonoileal reflex generally inhibits ileal emptying when the colon is distended. Therefore, C is correct.

52. The total number of premolar teeth in babies is:

- A. 8**
- B. 0**
- C. 12**
- D. 16**
- E. 20**

Answer: B. 0

Explanation:

Babies initially develop 20 primary (deciduous) teeth, consisting of 8 incisors, 4 canines, and 8 molars. Importantly, primary teeth have no premolars. Premolars appear only in the permanent dentition, which normally contains 8 premolars—two in each quadrant. Therefore, although children eventually develop permanent premolars as their primary molars are replaced, the number of premolars in the primary dentition is zero. Hence, B. 0 is correct.

53. Enterochromaffin cells produce:

- A. Mucous**
- B. Electrolytes**
- C. Digestive enzymes**

D. Hormones

E. Serotonin

Answer: E. Serotonin

Explanation:

Enterochromaffin (EC) cells are specialized enteroendocrine cells located primarily in the gastrointestinal mucosa. They are particularly important for producing and releasing serotonin (5-hydroxytryptamine, 5-HT). Serotonin participates in the regulation of gastrointestinal motility, secretion, and sensory signaling and can be released in response to mechanical or chemical stimulation of the intestinal mucosa. Although enteroendocrine cells as a broad group produce numerous gastrointestinal hormones and signaling molecules, the characteristic product associated with enterochromaffin cells is serotonin. Therefore, E is correct.

54. Center for chewing is located in the:

A. Pons

B. Medulla

C. Hypothalamus

D. Midbrain

E. Spinal cord

Answer: A. Pons

Explanation:

The central neural mechanisms controlling mastication (chewing) are located mainly in the pons, particularly in association with the trigeminal motor system. Chewing requires coordinated activity of the muscles of mastication, which are primarily supplied by the mandibular division of the trigeminal nerve (cranial nerve V). Sensory information from the oral cavity and teeth helps coordinate the rhythmic chewing movements. Although the medulla and other brainstem structures participate in broader orofacial functions, the classical chewing center is associated with the pons. Therefore, A. Pons is the expected answer.

55. A blood vessel that takes plasma to the kidney is:

A. Efferent arteriole

B. Afferent arteriole

C. Vasa recta

D. Peritubular capillary

E. Renal vein

Answer: B. Afferent arteriole

Explanation:

The afferent arteriole carries blood from the renal circulation toward the glomerulus. It enters the renal corpuscle and supplies the glomerular capillary network, where plasma

is filtered to form the initial filtrate. The efferent arteriole carries blood away from the glomerulus and subsequently forms the peritubular capillaries and, in juxtamedullary nephrons, the vasa recta. The renal vein carries blood away from the kidney toward the systemic circulation.

[24/09/2026 8:28 AM] Imaanaa Harmee: **56. The busiest segment of renal tubule is:**

- A. Collecting duct**
- B. Distal convoluted tubule**
- C. Proximal convoluted tubule**
- D. Descending loop of Henle**
- E. Ascending loop of Henle**

Answer: C. Proximal convoluted tubule

Explanation:

The proximal convoluted tubule (PCT) is the most active and metabolically demanding segment of the renal tubule because it performs the majority of tubular reabsorption. Approximately 65–70% of filtered sodium and water is normally reabsorbed here, along with large amounts of bicarbonate, chloride, potassium, phosphate, and other substances. Under normal conditions, virtually all filtered glucose and amino acids are also reabsorbed in the proximal tubule. Because of its enormous transport activity, the PCT contains abundant mitochondria and a highly developed brush border that increases its surface area. Therefore, C. Proximal convoluted tubule is correct.

57. Of the options given below, identify the one in favour of GFR:

- A. Activation of sympathetic nervous system**
- B. Glomerular capillary oncotic pressure**
- C. Bowman's capsular hydrostatic pressure**
- D. Glomerular capillary hydrostatic pressure**
- E. Endothelin**

Answer: D. Glomerular capillary hydrostatic pressure

Explanation:

The glomerular capillary hydrostatic pressure is the major force that favors filtration across the glomerular filtration barrier. It pushes water and small dissolved substances from the glomerular capillaries into Bowman's space. In contrast, glomerular capillary oncotic pressure opposes filtration because plasma proteins draw water back toward the capillaries. Bowman's capsular hydrostatic pressure also opposes filtration because pressure in Bowman's space pushes fluid back toward the glomerular capillaries. Sympathetic activation can constrict renal vessels and reduce renal blood flow and GFR, while endothelin is a potent vasoconstrictor. Therefore, D is the force that favors GFR.

58. Identify that is not normally found in the glomerular filtrate:

- A. Glucose
- B. Amino acid
- C. Fatty acid
- D. Sodium
- E. Blood cells

Answer: E. Blood cells

Explanation:

The glomerular filtration barrier permits water and many small molecules to pass from the blood into Bowman's space while normally preventing blood cells and most plasma proteins from crossing. Therefore, the normal glomerular filtrate contains substances such as glucose, amino acids, sodium, urea, and other small solutes at concentrations related to their plasma concentrations. Blood cells, including red blood cells and white blood cells, are normally retained within the circulation. The presence of significant blood cells in urine can indicate damage to the filtration barrier or another pathological process. Therefore, E. Blood cells is correct.

59. Renal blood flow is principally regulated by:

- A. Tubuloglomerular autoregulation
- B. Myogenic contraction
- C. Sympathetic nervous system
- D. Nitric oxide
- E. Parasympathetic nervous system

Answer: B. Myogenic contraction

Explanation:

Renal blood flow and glomerular filtration are maintained relatively constant over a range of arterial pressures through renal autoregulation. One of the two major intrinsic mechanisms responsible is the myogenic mechanism. When arterial pressure rises, the increased stretch of afferent arteriolar smooth muscle causes it to contract, thereby limiting the increase in renal blood flow and protecting the glomerulus. When pressure falls, the afferent arteriole relaxes and allows greater blood flow. The other major mechanism is tubuloglomerular feedback, which involves the macula densa and juxtaglomerular apparatus. Although sympathetic activity, nitric oxide, and other factors influence renal circulation, the myogenic mechanism is a principal intrinsic regulator. Therefore, B is the best answer.

[24/09/2026 8:35 AM] Imaanaa Harmee: **60. In the process of urine formation, which of the following is wrong?**

- A. Filtration is facilitated by the filtration slits in the podocytes
- B. Kidneys reabsorb 100% of essential substances
- C. Kidneys reabsorb hydrogen ions when the pH is high
- D. Kidneys secrete substances that are excess
- E. Kidneys reabsorb things that are deficient

Answer: C. Kidneys reabsorb hydrogen ions when the pH is high

Explanation:

The kidneys maintain acid-base balance through a combination of hydrogen ion

secretion and bicarbonate reabsorption. When the body is acidotic or blood pH is low, the kidneys increase hydrogen ion secretion and conserve bicarbonate. They do not normally respond to a high pH by reabsorbing hydrogen ions as a primary mechanism; instead, renal mechanisms can reduce hydrogen ion secretion and increase bicarbonate excretion when appropriate. The other statements describe important aspects of renal function: podocyte filtration slits contribute to the glomerular filtration barrier, essential substances are extensively reabsorbed, excess substances can be secreted into the tubular fluid, and substances that the body needs or is deficient in can be conserved through reabsorption. Thus, C is the incorrect statement.

[24/09/2026 8:35 AM] Imaanaa Harmee: **61. The number of capillaries in the kidneys is:**

- A. One**
 - B. Two**
 - C. Three**
 - D. Four**
 - E. Five**
- Answer: B. Two**

Explanation:

The kidney has two major capillary beds arranged in series: the glomerular capillaries and the peritubular capillaries. Blood enters the glomerulus through the afferent arteriole, where filtration occurs. It then leaves through the efferent arteriole and enters the peritubular capillary network, which surrounds the renal tubules and is important for reabsorption and secretion. In juxtamedullary nephrons, the peritubular capillary system includes the vasa recta, which extend into the renal medulla and participate in the countercurrent exchange mechanism. Thus, the classical answer is two capillary beds, making B correct.

62. The structure of nephron that receives processed urine and passes to the ureter is:

- A. Renal papilla**
 - B. Renal pyramid**
 - C. Renal calyx**
 - D. Renal pelvis**
 - E. Loop of Henle**
- Answer: D. Renal pelvis**

Explanation:

After urine has been formed and collected by the collecting ducts, it drains through the renal papillae into the minor calyces. The minor calyces join to form major calyces, which empty into the renal pelvis. The renal pelvis acts as a funnel-shaped collecting region and continues directly with the ureter, which carries urine from the kidney to the urinary bladder. Therefore, although the renal papilla and calyces participate in urine drainage, the structure that receives urine and then passes it into the ureter is the renal pelvis, making D correct.

63. Mechanism by which glucose and acids are reabsorbed from the kidneys is:

- A. Primary active transport**
- B. Sodium-dependent secondary active transport**
- C. Hydrogen-dependent secondary active transport**
- D. Osmosis**

E. Facilitated diffusion

Answer: B. Sodium-dependent secondary active transport

Explanation:

Glucose and many amino acids are reabsorbed primarily in the proximal convoluted tubule through sodium-dependent secondary active transport. The sodium-potassium ATPase on the basolateral membrane maintains a low intracellular sodium concentration by actively pumping sodium out of the tubular cell. This sodium gradient provides the energy for sodium to move from the tubular fluid into the cell together with glucose or amino acids through sodium-linked cotransporters on the apical membrane. The glucose can then leave the tubular cell through facilitated diffusion into the blood. Therefore, the initial reabsorption mechanism for glucose is sodium-dependent secondary active transport, making B correct.

64. True for Klinefelter Syndrome:

- A. Defective SRY**
- B. 5-alpha reductase deficiency**
- C. Small fragment of Y is translocated to an X**
- D. Extra X chromosome**
- E. Failure of expressing TDF and MIF**

Answer: D. Extra X chromosome

Explanation:

Klinefelter syndrome is a sex-chromosome disorder most commonly characterized by the karyotype 47,XXY, meaning that a male has an additional X chromosome. The extra X chromosome is usually caused by nondisjunction during meiosis. Individuals commonly have small, firm testes, impaired spermatogenesis, reduced testosterone production, and may have gynecomastia and a relatively tall stature. The presence of the Y chromosome means that testicular development occurs, but the additional X chromosome interferes with normal testicular function. Defective SRY and failure of testis-determining factors are associated with other disorders of sexual development. Therefore, D is correct.

[24/09/2026 8:35 AM] Imaanaa Harmee: **65. Which one of the following statements about spermatogenesis is not correct?**

- A. One matured sperm cell is formed from one primary spermatocyte**
- B. Sertoli cells are required for sperm nutrition**
- C. Spermatogenesis requires gonadotropin-releasing hormone**
- D. Leydig cell secretion of testosterone is in demand**
- E. Sub-basal temperature is required**

Answer: A. One matured sperm cell is formed from one primary spermatocyte

Explanation:

A single primary spermatocyte undergoes meiosis I to produce two secondary spermatocytes, and each secondary spermatocyte undergoes meiosis II to produce two spermatids. Thus, one primary spermatocyte ultimately gives rise to four spermatids, which differentiate through spermiogenesis into four spermatozoa. Therefore, it is incorrect to say that one mature sperm cell is formed from one primary spermatocyte. Sertoli cells provide structural and nutritional support for developing sperm cells, while testosterone from Leydig cells is essential for normal spermatogenesis. GnRH stimulates pituitary release of LH and FSH, which regulate testicular function. Spermatogenesis also requires a testicular temperature somewhat lower than core body temperature. Hence, A is the incorrect statement.

66. Functions of the Sertoli cells in the seminiferous tubules include:

- A. Secretion of FSH into the tubular lumen**
- B. Secretion of testosterone into the tubular lumen**
- C. Maintenance of the blood-testis barrier**
- D. Synthesis of estrogen after puberty**

Answer: C. Maintenance of the blood-testis barrier

Explanation:

Sertoli cells are essential supporting cells within the seminiferous tubules. One of their major functions is forming and maintaining the blood-testis barrier through tight junctions between adjacent Sertoli cells. This barrier separates developing germ cells from the systemic immune system and creates a specialized environment for spermatogenesis. Sertoli cells also provide nutrients and structural support to developing sperm cells and secrete substances such as androgen-binding protein and inhibin B. FSH is secreted by the anterior pituitary, while testosterone is produced by Leydig cells under the influence of LH. Therefore, C is correct.

67. From the structures listed below, identify that develops from Mullerian duct:

- A. Ovaries**
- B. Female genital tract**
- C. Male accessory glands**
- D. Male genital tract**
- E. Testis**

Answer: B. Female genital tract

Explanation:

The Müllerian ducts (paramesonephric ducts) are embryonic structures that give rise to much of the female internal reproductive tract. They develop into the fallopian tubes, uterus, and upper portion of the vagina. The ovaries do not develop from the Müllerian ducts; they arise from the gonadal ridges. In males, the Müllerian ducts normally regress under the influence of anti-Müllerian hormone (AMH) produced by Sertoli cells. Male internal reproductive structures such as the epididymis, vas deferens, and seminal vesicles arise largely from the Wolffian (mesonephric) ducts. Therefore, B is correct.

68. The Leydig cells that secrete testosterone are under influence of which hormone?

- A. Follicle-stimulating hormone**

- B. Growth hormone**
 - C. Luteinizing hormone**
 - D. T3 and T4**
 - E. Aldosterone**
- Answer: C. Luteinizing hormone**

Explanation:

Leydig cells, also called interstitial cells, are located between the seminiferous tubules in the testes and are responsible for producing testosterone. Their activity is stimulated primarily by luteinizing hormone (LH), which is secreted by the anterior pituitary. LH binds to receptors on Leydig cells and stimulates steroidogenesis, resulting in testosterone production. FSH primarily acts on Sertoli cells, where it supports spermatogenesis and stimulates production of factors such as androgen-binding protein and inhibin B. Therefore, C. Luteinizing hormone is correct.

[24/09/2026 8:35 AM] Imaanaa Harmee: **69. The phases of the female sexual cycle that occur simultaneously are:**

- A. Secretory phase and menstrual phase**
 - B. Secretory phase and luteal phase**
 - C. Menstrual phase and luteal phase**
 - D. Secretory phase and proliferative phase**
 - E. Ovulatory phase and secretory phase**
- Answer: B. Secretory phase and luteal phase**

Explanation:

The female reproductive cycle can be described using two related systems: the ovarian cycle and the endometrial (uterine) cycle. The ovarian cycle consists of the follicular phase, ovulation, and luteal phase. The endometrial cycle consists mainly of the menstrual, proliferative, and secretory phases. The luteal phase of the ovarian cycle occurs at the same time as the secretory phase of the endometrial cycle. During the luteal phase, the corpus luteum produces substantial progesterone, which causes the endometrium to become thick, glandular, and secretory, preparing it for possible implantation. Therefore, B is correct.

70. A hormone that is responsible for proliferative phase of endometrial cycle is:

- A. Progesterone**
 - B. FSH**
 - C. LH**
 - D. Estrogen**
 - E. Testosterone**
- Answer: D. Estrogen**

Explanation:

The proliferative phase of the endometrial cycle occurs after menstruation and corresponds approximately to the follicular phase of the ovarian cycle. During this period, developing ovarian follicles produce increasing amounts of estrogen, particularly estradiol. Estrogen stimulates proliferation of the endometrial epithelial

and stromal cells, causing the endometrium to become thicker and preparing it for possible implantation. After ovulation, progesterone becomes the dominant hormone and converts the proliferative endometrium into a secretory endometrium. Therefore, D. Estrogen is the correct answer.

[24/09/2026 8:35 AM] Imaanaa Harmee: 71. The neurotransmitter that is most commonly released from postganglionic sympathetic neurons is:

- A. Acetylcholine
- B. Norepinephrine
- C. Epinephrine
- D. Dopamine

Answer: B. Norepinephrine

Explanation:

Most postganglionic sympathetic neurons release norepinephrine (noradrenaline) at their target organs. Norepinephrine binds mainly to α - and β -adrenergic receptors and produces effects such as increased heart rate, vasoconstriction, and pupil dilation. There are important exceptions: postganglionic sympathetic fibers supplying eccrine sweat glands release acetylcholine rather than norepinephrine. Also, the adrenal medulla releases mainly epinephrine and some norepinephrine directly into the bloodstream, but this is not the usual neurotransmitter released by a postganglionic sympathetic neuron. Therefore, B is correct.

72. Acetylcholine released by autonomic nervous system binds to what type of receptors on skeletal muscles?

- A. Muscarinic receptors
- B. Beta-adrenergic receptors
- C. Nicotinic receptors
- D. None

Answer: C. Nicotinic receptors

Explanation:

At the neuromuscular junction, somatic motor neurons release acetylcholine (ACh), which binds to nicotinic acetylcholine receptors (Nm) on the skeletal muscle membrane. These receptors are ligand-gated ion channels. When ACh binds, the channels open, allowing mainly sodium ions to enter the muscle fiber, producing depolarization that initiates the action potential and ultimately causes muscle contraction. Muscarinic receptors are found on many parasympathetic target organs, whereas beta-adrenergic receptors respond to norepinephrine or epinephrine. Therefore, C. Nicotinic receptors is correct.

73. Which one is false?

- A. Sympathetic preganglionic neuron is short
- B. Parasympathetic ganglia located near to organ to be innervated
- C. Parasympathetic nervous system contains adrenergic receptor on effector organ
- D. Adrenal gland has role in sympathetic

Answer: C. Parasympathetic nervous system contains adrenergic receptor on effector organ

Explanation:

The parasympathetic nervous system generally uses acetylcholine as its neurotransmitter at the effector organ, where it acts primarily on muscarinic receptors. Therefore, saying that parasympathetic activity uses adrenergic receptors on its effector organs is false. Sympathetic preganglionic neurons are generally short because sympathetic ganglia are located relatively close to the spinal cord. Parasympathetic ganglia are usually located near or within the target organs, so parasympathetic preganglionic fibers are generally long and postganglionic fibers short. The adrenal medulla is also functionally part of the sympathetic system because sympathetic preganglionic fibers stimulate it to release catecholamines into the blood. Thus, C is correct.

74. A woman presents to the clinic with increased heart rate, mydriatic pupils, and dry mouth. This condition results from the activation of which system function?

- A. Sympathetic**
- B. Parasympathetic**
- C. Both**

Answer: A. Sympathetic

Explanation:

The combination of increased heart rate, mydriasis (pupil dilation), and dry mouth is characteristic of increased sympathetic nervous system activity. Sympathetic activation prepares the body for the "fight-or-flight" response. It increases heart rate and cardiac output, dilates the pupils to improve visual awareness, and generally inhibits salivary secretion, producing a dry mouth. Parasympathetic activation generally produces the opposite pattern: it slows the heart and promotes salivation and digestion. Therefore, A. Sympathetic is correct.

[24/09/2026 8:35 AM] Imaanaa Harmee: **75. A middle-aged man went to a foreign country because he had been offered a job there. However, he had been unemployed for quite a long time. What endocrine glands were exhausted most of all in this man?**

- A. Thymus gland**
- B. Thyroid gland**
- C. Adrenal glands**
- D. Parathyroid glands**

Answer: C. Adrenal glands

Explanation:

This question is referring to prolonged stress, which activates the hypothalamic-pituitary-adrenal (HPA) axis. During stress, the hypothalamus releases CRH, which stimulates the anterior pituitary to release ACTH. ACTH then stimulates the adrenal cortex to produce cortisol. The adrenal glands also participate in the body's stress response through catecholamines released from the adrenal medulla. Prolonged psychological or physical stress therefore places significant functional demands on the

adrenal system. The thyroid and parathyroid glands are primarily involved in metabolism and calcium regulation, respectively, while the thymus is mainly associated with immune development. Thus, C. Adrenal glands is the intended answer.

76. It was identified that the conduction velocity in the nerve fibers was equal to 120 m/sec. Specify these fibers:

- A. Preganglionic sympathetic**
 - B. Preganglionic parasympathetic**
 - C. Postganglionic sympathetic**
 - D. Somatic motor neuron axons**
- Answer: D. Somatic motor neuron axons**

Explanation:

A conduction velocity of approximately 120 m/s is characteristic of large, heavily myelinated A α nerve fibers, including somatic motor axons. These fibers have a large diameter and thick myelin sheath, allowing rapid saltatory conduction from the central nervous system to skeletal muscles. Autonomic preganglionic fibers are generally smaller, myelinated B fibers, and postganglionic sympathetic fibers are mostly small, unmyelinated C fibers, so their conduction velocities are much lower. Therefore, a velocity of 120 m/s strongly indicates a somatic motor neuron axon, making D correct.

77. What type of sympathetic receptor is found on the iris circular muscle?

- A. Nicotinic receptor**
 - B. Alpha-adrenergic receptor**
 - C. Beta-adrenergic receptor**
 - D. None**
- Answer: B. Alpha-adrenergic receptor**

Explanation:

The circular (sphincter) muscle of the iris is responsible for constricting the pupil. Sympathetic stimulation primarily acts on the radial/dilator muscle of the iris through α_1 -adrenergic receptors, producing pupil dilation (mydriasis). The question's wording is anatomically problematic because the circular sphincter muscle is primarily controlled by the parasympathetic system through muscarinic M3 receptors, not a sympathetic receptor. If the question intends the sympathetic receptor associated with the iris muscle responsible for dilation, the answer is α_1 -adrenergic receptor (B). Thus, B is the likely intended exam answer, but strictly speaking, the circular muscle itself is parasympathetically controlled.

78. What is the target cell for vasopressin?

- A. Posterior pituitary**
- B. Renal tubule**

C. Anterior pituitary

D. Hypothalamus

Answer: B. Renal tubule

Explanation:

Vasopressin, also called antidiuretic hormone (ADH), acts importantly on the kidney, particularly the collecting ducts and late distal nephron. It binds primarily to V_2 receptors on principal cells, activating a signaling pathway that promotes insertion of aquaporin-2 water channels into the apical membrane. This increases water reabsorption and reduces urine volume, helping maintain body water and plasma osmolality. Vasopressin also has V_1 -mediated effects on vascular smooth muscle, where it can cause vasoconstriction. The posterior pituitary stores and releases ADH but is not its target cell. Therefore, B is correct.

[24/09/2026 8:35 AM] Imaanaa Harmee: **79. A target cell for parathyroid hormone can be all of the following, except:**

A. Brain

B. Bone

C. Kidneys

D. Intestine

Answer: A. Brain

Explanation:

The major physiological target tissues of parathyroid hormone (PTH) are the bone and kidneys. PTH increases blood calcium by promoting processes that increase calcium availability from bone and by increasing calcium reabsorption in the kidney. It also stimulates renal activation of vitamin D to calcitriol, which subsequently increases intestinal calcium absorption. Importantly, PTH does not act directly on the intestine as its principal target; its intestinal effect is indirect through calcitriol.

Nevertheless, among the choices, the clearest tissue that is not a target of PTH is the brain. Therefore, A. Brain is the intended answer.

80. Which of the following could not be a characteristic of graded potential?

A. It can be summated

B. It has refractory period

C. It has varied amplitude

D. It does not have overshoot

Answer: B. It has refractory period

Explanation:

Graded potentials are local changes in membrane potential whose amplitude varies according to the strength of the stimulus. They can undergo summation, meaning that multiple graded potentials can combine to produce a larger change in membrane

potential. Unlike action potentials, graded potentials do not have a true absolute or relative refractory period because they are not all-or-none events generated by the voltage-gated sodium channel mechanism characteristic of action potentials. Their amplitude decreases with distance from the site of origin. They also generally do not exhibit the characteristic overshoot of an action potential, in which the membrane potential becomes positive. Therefore, B. It has refractory period is the statement that could not characterize a graded potential.

[24/09/2026 8:35 AM] Imaanaa Harmee: **85. The endocrine system:**

- A. Releases chemicals into the bloodstream for distribution throughout the body**
- B. Releases hormones that can alter the metabolic activities of many different tissues and organs**
- C. Produces effects that can last for hours, days, or even longer**
- D. Can alter the gene activity of cells**
- E. All of the above**

Answer: E. All of the above

Explanation:

All of the statements describe important characteristics of the endocrine system. Endocrine glands release hormones into the bloodstream, allowing these chemical messengers to travel to distant target cells. Hormones can modify cellular metabolism, growth, development, reproduction, and many other physiological processes. Compared with many nervous-system effects, endocrine effects can be relatively long-lasting, sometimes persisting for hours, days, or even longer. Steroid and certain other hormones can enter cells and influence gene transcription, thereby producing long-term changes in cellular function. Since A, B, C, and D are all correct, the best answer is E. All of the above.

86. The action potential of a neuron is initiated by:

- A. Efflux of Na^+**
 - B. Influx of K^+**
 - C. Influx of Na^+**
 - D. Efflux of K^+**
 - E. Influx of Cl^-**
- Answer: C. Influx of Na^+**

Explanation:

When a neuron reaches its threshold membrane potential, voltage-gated sodium channels rapidly open. Sodium ions move from the extracellular fluid into the neuron because of both their concentration gradient and electrical gradient. This influx of Na^+ causes rapid depolarization of the membrane and initiates the rising phase of the action potential. Subsequently, sodium channels inactivate and potassium channels open, allowing K^+ to leave the cell and produce repolarization. Therefore, the event responsible for initiating the action potential is C. Influx of Na^+ .

87. Concerning glial cells one is wrong:

- A. Oligodendrocytes work in PNS
- B. Satellite cells serve as blood-brain barrier
- C. Microglia function in sensory & autonomic ganglia
- D. Ependymal cells regulate the external chemical environment
- E. All

Answer: E. All

Explanation:

This question contains multiple incorrect statements, so the options are poorly constructed. Oligodendrocytes are CNS glial cells, not PNS cells; in the PNS, Schwann cells form myelin. Satellite cells are located around neuronal cell bodies in peripheral sensory and autonomic ganglia and help regulate their local chemical environment, but they do not form the blood-brain barrier. Microglia are the resident immune/phagocytic cells of the CNS, not the principal glial cells of peripheral sensory and autonomic ganglia. Ependymal cells line the ventricles and central canal and are involved in CSF-related functions and regulation of the environment around CNS neurons. Therefore, A, B, and C contain errors, meaning the question as written is flawed. If forced to choose the option representing the overall problem, E. All is the closest, but it is not a well-written single-best-answer question.

[24/09/2026 8:35 AM] Imaanaa Harmee: **88. Arrange the following in the correct order for skeletal muscle fiber contraction.**

1. Sarcoplasmic reticulum releases Ca^{2+} .
2. Ca^{2+} combines with troponin, uncovering myosin-binding sites.
3. Acetylcholine is released from the axon terminal and binds to receptors in the motor end plate.
4. Action potential travels along the sarcolemma and into transverse tubules.
5. Energized myosin heads (cross-bridges) attach to actin.
6. Thin filaments slide toward the center of the sarcomere.

- A. 3, 4, 1, 2, 5, 6
- B. 4, 3, 2, 1, 5, 6
- C. 1, 2, 3, 4, 5, 6
- D. 3, 1, 3, 5, 2, 6
- E. 3, 1, 4, 5, 2, 6

Answer: A. 3, 4, 1, 2, 5, 6

Explanation:

Skeletal muscle contraction begins when a motor neuron releases acetylcholine (ACh) into the neuromuscular junction. ACh binds to nicotinic receptors on the motor end plate and generates an action potential in the muscle fiber (3). The action potential then spreads along the sarcolemma and down the T-tubules (4). This activates the excitation-contraction coupling mechanism, causing the sarcoplasmic reticulum to release Ca^{2+} (1). Calcium binds to troponin C, causing tropomyosin to move away from the actin-binding sites (2). Energized myosin heads can then attach to exposed actin sites and form cross-bridges (5). Repeated cross-bridge cycling pulls the thin filaments toward the center of the sarcomere, producing contraction (6). Thus, the correct sequence is 3 → 4 → 1 → 2 → 5 → 6, or A.

89. Which of the following statements about skeletal muscle tissue is NOT true?

- A. Skeletal muscle requires a large blood supply.**
- B. Skeletal muscle fibers have many mitochondria.**
- C. The arrangement of thick and thin filaments produces the striations in skeletal muscle tissue.**
- D. Skeletal muscle fibers contain gap junctions that help conduct action potentials from one fiber to another.**
- E. A skeletal muscle fiber has many nuclei.**

Answer: D. Skeletal muscle fibers contain gap junctions that help conduct action potentials from one fiber to another.

Explanation:

Skeletal muscle fibers are multinucleated, contain numerous mitochondria, and require a substantial blood supply because muscle contraction has a high metabolic demand. Their characteristic striated appearance results from the highly organized arrangement of actin and myosin into repeating sarcomeres. However, skeletal muscle fibers are not connected by gap junctions for the purpose of spreading action potentials from one fiber to another. Each skeletal muscle fiber is individually innervated by a somatic motor neuron, and its action potential propagates along that individual fiber. Gap junctions are especially important in cardiac and some smooth muscle, where electrical activity can spread directly between cells. Therefore, D is the incorrect statement.

90. The release of neurotransmitters from the synaptic vesicles is dependent on the presence of ____ in the synaptic end bulb.

- A. Glucose**
- B. Calcium**
- C. Proteins**
- D. Sodium**
- E. Potassium**

Answer: B. Calcium

Explanation:

Neurotransmitter release at a chemical synapse is triggered by the arrival of an action potential at the presynaptic terminal. Depolarization opens voltage-gated Ca^{2+} channels, allowing calcium ions to enter the synaptic end bulb. The increased intracellular calcium concentration activates proteins involved in synaptic vesicle docking and fusion, particularly the SNARE complex, causing the vesicles to fuse with the presynaptic membrane and release neurotransmitter into the synaptic cleft by exocytosis. Without this calcium influx, normal rapid neurotransmitter release does not occur. Therefore, B. Calcium is correct.

[24/09/2026 8:35 AM] Imaanaa Harmee: **91. The speed of nerve impulse conduction is increased by:**

- A. Cold**
- B. A very strong stimulus**
- C. Small diameter of the axon**
- D. Myelination**
- E. Astrocytes**

Answer: D. Myelination

Explanation:

The speed of nerve impulse conduction depends strongly on the diameter of the axon and the presence of myelin. Myelin forms an insulating sheath around the axon and allows the action potential to "jump" between the nodes of Ranvier, a process called saltatory conduction. This makes conduction much faster than continuous conduction along an unmyelinated axon. Larger axon diameter also increases conduction velocity, so a small diameter would actually slow conduction. Cold generally decreases nerve conduction velocity, while the strength of a stimulus does not determine the speed of an already generated action potential. Astrocytes support neurons but do not directly increase axonal conduction velocity. Therefore, D is correct.

92. Which of the following is NOT true concerning hormones?

- A. Responses to hormones are generally slower and longer lasting than the responses stimulated by the nervous system.**
- B. Hormones are generally controlled by negative feedback systems.**
- C. The hypothalamus inhibits the release of some hormones.**
- D. Most hormones are released steadily throughout the day.**
- E. Hormone secretion is determined by the body's need to maintain homeostasis.**

Answer: D. Most hormones are released steadily throughout the day.

Explanation:

Hormone secretion is not generally steady throughout the day. Many hormones are released in pulses or rhythmic patterns, and some follow circadian rhythms. For example, cortisol secretion varies substantially over the 24-hour period, while growth hormone is released in pulses, especially during sleep. Hormonal responses are generally slower but longer-lasting than nervous responses. Most endocrine systems are regulated by negative feedback, which helps maintain homeostasis. The hypothalamus can either stimulate or inhibit endocrine secretion depending on the hormonal axis. Therefore, the statement that most hormones are released steadily throughout the day is incorrect, making D the answer.

93. Which of the following statements concerning hormone action is NOT true?

- A. Hormones bring about changes in metabolic activities of cells.**
- B. Target cells must have receptors for a hormone.**
- C. Lipid-soluble hormones may directly enter target cells and affect specific genes.**
- D. When a hormone attaches to a membrane receptor, it is termed the first messenger.**
- E. ATP is a common second messenger in target cells.**

Answer: E. ATP is a common second messenger in target cells.

Explanation:

Hormones produce their effects by binding to specific receptors on or inside target cells. Lipid-soluble hormones such as steroid hormones can cross the plasma membrane and bind intracellular receptors, and the resulting hormone-receptor complex can influence gene transcription. Water-soluble hormones generally bind membrane receptors and

act as first messengers, activating intracellular signaling molecules known as second messengers. Common second messengers include cAMP, IP₃, DAG, and Ca²⁺. ATP itself is not generally classified as a common second messenger. Therefore, E is the incorrect statement.

94. Which nerve carries most of the parasympathetic output from the brain?

- A. Spinal**
- B. Vagus**
- C. Oculomotor**
- D. Facial**
- E. Glossopharyngeal**

Answer: B. Vagus

Explanation:

The vagus nerve (cranial nerve X) carries the majority of the body's cranial parasympathetic outflow. It supplies parasympathetic fibers to major thoracic and abdominal organs, including the heart, lungs, and much of the gastrointestinal tract. Other cranial nerves also carry parasympathetic fibers: the oculomotor nerve controls pupil constriction and accommodation, while the facial and glossopharyngeal nerves contribute to salivary and lacrimal gland regulation. However, the vagus nerve carries the largest and most extensive portion of parasympathetic output from the brain. Therefore, B is correct.

[24/09/2026 8:35 AM] Imaanaa Harmee: **95. Which of the following would NOT be affected by the autonomic nervous system?**

- A. Heart**
- B. Intestines**
- C. Urinary bladder**
- D. Skeletal muscle**
- E. Reproductive organs**

Answer: D. Skeletal muscle

Explanation:

The autonomic nervous system (ANS) regulates involuntary functions of cardiac muscle, smooth muscle, and glands. It therefore affects the heart, gastrointestinal tract, urinary bladder, and many aspects of reproductive function. Skeletal muscle, however, is primarily controlled by the somatic nervous system, which allows voluntary control of skeletal movement. Although autonomic activity can indirectly influence skeletal muscle function through changes in blood flow, metabolism, and other physiological processes, the skeletal muscle fibers themselves are controlled by somatic motor neurons rather than autonomic neurons. Thus, D is correct.

96. All of the following hormones are amines EXCEPT:

- A. Epinephrine**
- B. Melatonin**
- C. Norepinephrine**
- D. Thyroid hormone**
- E. None**

Answer: E. None

Explanation:

All four listed hormones are derived from amino acids, particularly tyrosine or tryptophan, and are classified as amine hormones. Epinephrine and norepinephrine are catecholamines derived from tyrosine. Thyroid hormones (T_3 and T_4) are also derived from tyrosine, although they behave physiologically much more like lipid-soluble hormones. Melatonin is derived from tryptophan through serotonin. Therefore, none of A–D is an exception, making E. None the correct answer.

97. Steroid hormones are synthesized in:

- A. Nuclei**
- B. Rough endoplasmic reticulum**
- C. Ribosomes**
- D. Smooth endoplasmic reticulum**
- E. None**

Answer: D. Smooth endoplasmic reticulum

Explanation:

Steroid hormones are synthesized from cholesterol and are produced primarily in endocrine cells that contain abundant smooth endoplasmic reticulum (SER) and mitochondria. Examples include cortisol, aldosterone, estrogen, progesterone, and testosterone. The smooth ER contains many of the enzymes involved in steroid biosynthesis, while mitochondria also perform important early and late steps in the pathway. In contrast, rough ER and ribosomes are primarily involved in the synthesis of peptide and protein hormones. Therefore, among the options, D. Smooth endoplasmic reticulum is correct.

98. Hormone produced from parafollicular cells of thyroid and important for regulation of blood calcium level is:

- A. Calcitonin**
- B. Calcitriol**
- C. Melatonin**
- D. Parathyroid**

Answer: A. Calcitonin

Explanation:

Parafollicular cells, also called C cells, of the thyroid gland secrete calcitonin. Calcitonin contributes to calcium homeostasis by tending to lower plasma calcium concentration, particularly by inhibiting osteoclast-mediated bone resorption. In contrast, parathyroid hormone (PTH) is produced by the parathyroid glands and generally increases blood calcium. Calcitriol, the active form of vitamin D, increases intestinal absorption of calcium and phosphate. Therefore, the thyroid hormone produced by parafollicular cells that participates in calcium regulation is calcitonin, making A correct.

99. All of the following hormones can be freely transported in blood EXCEPT:

- A. ADH**
- B. Aldosterone**
- C. Epinephrine**
- D. Insulin**
- E. Oxytocin**

Answer: B. Aldosterone

Explanation:

The wording here requires an important distinction. Water-soluble hormones such as ADH, epinephrine, insulin, and oxytocin can circulate in the blood largely dissolved in plasma because they are not dependent on carrier proteins for transport. Aldosterone, being a steroid hormone and therefore lipid-soluble, is transported in the circulation partly bound to plasma proteins, including albumin and corticosteroid-binding proteins. Thus, it is not freely dissolved to the same extent as the listed peptide and catecholamine hormones. Therefore, B. Aldosterone is the intended answer.

[24/09/2026 8:35 AM] Imaanaa Harmee: **100. Receptor for all of the following hormones is found in cytoplasm or in the nucleus EXCEPT:**

- A. Aldosterone**
 - B. Epinephrine**
 - C. Testosterone**
 - D. Thyroid hormone**
- Answer: B. Epinephrine**

Explanation:

Hormones that are lipid-soluble can cross the plasma membrane and bind to intracellular receptors located in the cytoplasm or nucleus. Aldosterone is a steroid hormone and acts through intracellular mineralocorticoid receptors. Testosterone is also a steroid hormone and acts through an intracellular androgen receptor. Thyroid hormones, although chemically derived from amino acids, are lipid-soluble and act primarily through nuclear thyroid hormone receptors that regulate gene transcription. Epinephrine, however, is a water-soluble catecholamine and cannot freely cross the lipid bilayer of the cell membrane. It therefore acts through cell-surface adrenergic receptors and intracellular second-messenger systems. Hence, B. Epinephrine is correct.

Exam:-Physiology

Year:-2015 E.C

Dr. Doc

[24/09/2026 8:35 AM] Imaanaa Harmee: **1. General properties of negative feedback include all of the following EXCEPT:**

- A. It is unwavering of the set point**
- B. Ideal value of a controlled variable is defined as its "set point".**
- C. It is a closed loop control system in which the output accelerates the control system.**
- D. The term negative is used because the effector response opposes the departure from the set point.**
- E. It often works towards homeostasis.**

Answer: C. It is a closed loop control system in which the output accelerates the control system.

Explanation:

Negative feedback is a closed-loop regulatory mechanism in which the response of the system opposes or reduces the original disturbance. The purpose is to maintain a physiological variable near its desired set point and therefore promote homeostasis. The word "negative" refers to the fact that the response counteracts the initial change rather than amplifying it. For example, when blood glucose rises after a meal, insulin secretion promotes glucose uptake and storage, bringing blood glucose back toward its normal range. In contrast, a system in which the output accelerates or amplifies the original disturbance is characteristic of positive feedback. Therefore, C is the exception.

2. Which of the following is not the function of smooth endoplasmic reticulum?

- A. Protein synthesis**
- B. Calcium storage in muscle cells**
- C. Glycogen storage in liver cells**
- D. Drug metabolism in hepatocytes**
- E. Lipid synthesis**

Answer: A. Protein synthesis

Explanation:

The smooth endoplasmic reticulum (SER) is involved mainly in lipid synthesis, steroid synthesis, detoxification of drugs and toxins, carbohydrate metabolism, and calcium storage. In muscle cells, the specialized SER is called the sarcoplasmic reticulum, which stores and releases calcium during muscle contraction. In hepatocytes, the SER participates in drug metabolism and detoxification. Protein synthesis, however, is primarily a function of the rough endoplasmic reticulum (RER) because ribosomes are attached to its cytoplasmic surface. Therefore, A is correct.

3. The cell membrane is composed of:

- A. 45% proteins and 55% carbohydrates**

- B. 55% proteins and 42% lipids**
- C. 55% lipids and 13% cholesterol**
- D. 45% proteins and 3% carbohydrates**
- E. 45% phospholipids and 13% neutral fats**

Answer: B. 55% proteins and 42% lipids

Explanation:

The plasma membrane is primarily composed of proteins, lipids, and carbohydrates. A commonly taught approximate composition by mass is about 55% protein, 42% lipid, and 3% carbohydrate. Membrane proteins perform many essential functions, including transport, receptor activity, enzymatic activity, and cell adhesion. Lipids, particularly phospholipids and cholesterol, form the structural bilayer and contribute to membrane fluidity and stability. Carbohydrates are attached mainly to proteins and lipids on the extracellular surface, forming part of the glycocalyx. Therefore, B gives the expected approximate composition.

4. Which one of the following does not belong to the cell membrane proteins?

- A. Ion channels**
- B. Na⁺-K⁺ ATPase**
- C. Carrier proteins**
- D. Membrane receptor**
- E. None of the above**

Answer: E. None of the above

Explanation:

All of options A–D are examples of membrane proteins. Ion channels provide pathways through which ions can cross the membrane. The Na⁺-K⁺ ATPase is an integral membrane protein that actively transports sodium and potassium. Carrier proteins transport specific substances across the membrane, while membrane receptors detect extracellular signals such as hormones and neurotransmitters. Since all four listed structures belong to membrane proteins, none of A–D is the exception. Therefore, E. None of the above is correct.

[24/09/2026 8:35 AM] Imaanaa Harmee: **5. Which of the following statements is correct about the structure of cell membrane?**

- A. Portions of the lipid bilayer that face the ECF and ICF are hydrophobic**
- B. Polar molecules can easily pass through the lipid bilayer**
- C. Protein channels within the lipid bilayer allow the passage of ions**
- D. The polar heads of the lipid bilayer face each other in the interior of the cell**
- E. None of the above**

Answer: C. Protein channels within the lipid bilayer allow the passage of ions

Explanation:

The plasma membrane consists primarily of a phospholipid bilayer. The hydrophilic (polar) heads face the aqueous extracellular and intracellular fluids, while the hydrophobic tails face inward toward one another. Because the interior of the membrane is hydrophobic, ions and most polar molecules cannot readily cross it

directly. Specialized membrane proteins, including ion channels and carrier proteins, provide pathways for these substances. Therefore, A, B, and D are incorrect, while C correctly describes an important function of membrane proteins.

6. Which of the following is true about the $\text{Na}^+\text{-K}^+$ ATPase?

- A. It is made up of five subunits**
- B. Its alpha subunit is facing the ECF and has binding site for Na^+ ions**
- C. Its ECF domain contains four binding sites**
- D. The ICF domain has K^+ ion binding sites**
- E. ATPase binding site is on its ECF domain**

Answer: B. Its alpha subunit is facing the ECF and has binding site for Na^+ ions.

Explanation:

The $\text{Na}^+\text{-K}^+$ ATPase is an integral membrane protein that uses ATP to transport 3 Na^+ out of the cell and 2 K^+ into the cell during each cycle. Its major functional component is the alpha subunit, which contains the ion-binding sites and ATP-binding/catalytic machinery. During the pump cycle, sodium binds on the intracellular side, while potassium binds on the extracellular side before being transported inward. Therefore, the wording of option B is actually problematic: the alpha subunit spans the membrane and has binding sites on both sides, but Na^+ binds on the intracellular side, not the extracellular side. Likewise, K^+ binds extracellularly. Thus, none of the options is perfectly correct as written. If your provided key says B, it likely reflects an error in the question/key.

7. Which one of the following is false about facilitated diffusion?

- A. It requires carrier proteins to transport substances**
- B. It needs ATP to transport substances**
- C. It transports substances down their concentration gradient**
- D. Carrier proteins are involved**
- E. It is a saturable process**

Answer: B. It needs ATP to transport substances

Explanation:

Facilitated diffusion is a passive transport mechanism, meaning that it does not require direct expenditure of ATP. Instead, substances move across the membrane down their concentration or electrochemical gradient with the assistance of membrane proteins. These proteins may be carrier proteins or channels. Because the number of available carrier proteins is limited, facilitated diffusion can become saturated when all available transport proteins are occupied. This distinguishes it from simple diffusion, which does not require a carrier and is not saturable in the same manner. Therefore, B is false.

8. A mechanism by which water molecules are transported through the cell membrane is:

- A. Osmosis**
- B. Simple diffusion**
- C. Facilitated diffusion**
- D. Primary active transport**
- E. Secondary active transport**

Answer: A. Osmosis

Explanation:

Osmosis is the net movement of water across a selectively permeable membrane from an area of relatively lower solute concentration toward an area of relatively higher effective solute concentration. In biological membranes, much of the rapid water movement occurs through specialized membrane proteins called aquaporins, although water can also cross the lipid bilayer to a limited extent. Osmosis does not require ATP because water moves according to its water potential/osmotic gradient. Therefore, A. Osmosis is the expected answer.

[24/09/2026 8:35 AM] Imaanaa Harmee: **9. Which of the following substances is lipid-soluble?**

- A. O₂ and CO₂**
- B. Water**
- C. Sodium and potassium**
- D. Glucose**
- E. All**

Answer: A. O₂ and CO₂

Explanation:

Oxygen and carbon dioxide are small, nonpolar molecules that are readily soluble in the lipid portion of the plasma membrane. Consequently, they can cross the lipid bilayer by simple diffusion. Water is polar, while sodium and potassium are charged ions and therefore cannot freely cross the hydrophobic membrane core. Glucose is also a relatively large polar molecule and requires specific transport proteins to cross cell membranes. Therefore, A. O₂ and CO₂ is correct.

10. Glucose and amino acids are absorbed from the GIT and reabsorbed from the renal tubules by:

- A. Simple diffusion**
- B. Osmosis**
- C. Secondary active transport**
- D. Primary active transport**
- E. None of the above**

Answer: C. Secondary active transport

Explanation:

Glucose and many amino acids are transported across intestinal and renal tubular epithelial cells using sodium-dependent secondary active transport. The $\text{Na}^+\text{-K}^+$ ATPase on the basolateral membrane uses ATP to maintain a low intracellular sodium concentration. This sodium gradient then provides the energy that drives sodium and glucose or amino acids together across the apical membrane through specific cotransporters. The transport of glucose itself does not directly hydrolyze ATP at the cotransporter; instead, it uses the sodium gradient established by primary active transport. This is why the mechanism is classified as secondary active transport, making C correct.

[24/09/2026 8:35 AM] Imaanaa Harmee: **15. Identify the correct statement from the following.**

- A. Motor neurons transmit impulses from periphery to the CNS**
- B. Microglia provides nutrients to neurons in the CNS**
- C. Neuroglia are excitable cells**
- D. Nerve insulators for central and peripheral NS are of the same structure**
- E. Satellite cells are located in CNS**

Answer: D. Nerve insulators for central and peripheral NS are of the same structure

Explanation:

The intended concept here is the myelin sheath, which electrically insulates axons and greatly increases the speed of action potential conduction. However, there is an important distinction: the cells that produce myelin are not the same in the CNS and PNS. Oligodendrocytes produce myelin in the CNS, whereas Schwann cells produce myelin in the PNS.

Therefore, if option D literally means that the structures/cells are the same, it is not scientifically correct. Motor neurons carry impulses from the CNS to effectors, while sensory neurons carry information toward the CNS. Microglia are immune/phagocytic cells rather than the primary nutrient-providing glia, and satellite cells are located in peripheral ganglia. The supplied answer D appears to depend on imprecise wording.

16. During the rising phase of action potential:

- A. Voltage-gated Na^+ channels are opened**
- B. Voltage-gated K^+ channels opened**
- C. Voltage-gated Na^+ channels are closed**
- D. Ligand-gated K^+ channels are closed**
- E. None of the above**

Answer: A. Voltage-gated Na^+ channels are opened

Explanation:

The rising phase of a neuronal action potential is primarily caused by the rapid opening of voltage-gated sodium channels. Once the membrane reaches threshold, these channels open and Na^+ rapidly enters the neuron because of its electrochemical gradient. This produces rapid depolarization, driving the membrane potential toward a positive value. Voltage-gated potassium channels also participate in the action potential, but

they open more slowly and are primarily responsible for the subsequent repolarization phase. Therefore, A is correct.

17. Which one of the following is wrong about the action potential of the nerve fiber?

- A. It is caused by a transient membrane depolarization**
 - B. It needs a threshold stimulus to be generated**
 - C. It has a definite RMP**
 - D. It is all-or-none in its nature**
 - E. Amplitude of action potential is stimulus dependent**
- Answer: E. Amplitude of action potential is stimulus dependent**

Explanation:

A nerve action potential follows the all-or-none principle. Once the membrane reaches its threshold, a full action potential is generated. Increasing the strength of the stimulus above threshold does not increase the amplitude of the action potential. Instead, stronger stimuli may increase the frequency of action potentials or recruit additional nerve fibers. The action potential is a transient change in membrane potential and occurs from a neuron with a characteristic resting membrane potential (RMP). Therefore, the statement that its amplitude depends on stimulus strength is incorrect, making E correct.

18. Fastest rate of conduction of nerve impulse is achieved in:

- A. Type A nerve fibers**
 - B. Type B nerve fibers**
 - C. Type C nerve fibers**
 - D. Autonomic nerve fibers**
 - E. Unmyelinated somatic nerve fibers**
- Answer: A. Type A nerve fibers**

Explanation:

Type A nerve fibers are large-diameter, heavily myelinated fibers and therefore conduct action potentials at the highest velocities. Their thick myelin sheath allows saltatory conduction, in which the action potential effectively jumps from one node of Ranvier to the next. Type B fibers are smaller and generally conduct more slowly, while Type C fibers are small, unmyelinated, and have the slowest conduction. Therefore, A. Type A nerve fibers is correct.

[24/09/2026 8:35 AM] Imaanaa Harmee: **19. In the peripheral nerve fibers, myelin sheath is produced by:**

- A. Schwann cells**
 - B. Oligodendrocytes**
 - C. Microglial cells**
 - D. Ependymal cells**
 - E. Astrocytes**
- Answer: A. Schwann cells**

Explanation:

In the peripheral nervous system (PNS), myelin is produced by Schwann cells. A single Schwann cell can wrap its membrane repeatedly around a segment of one axon, forming a myelin sheath. The gaps between adjacent myelinated segments are called nodes of Ranvier, which allow saltatory conduction and greatly increase the speed of nerve impulse transmission. In contrast, oligodendrocytes produce myelin in the central nervous system (CNS). Microglia provide immune functions, ependymal cells line the ventricles, and astrocytes provide various support functions. Therefore, A is correct.

20. Which one of the following is wrong about the neuromuscular junction of skeletal muscle?

- A. The neurotransmitter is always acetylcholine**
- B. The postsynaptic receptor is always nicotinic receptor**
- C. Excitatory postsynaptic potential is always produced**
- D. EPSP causes contraction**
- E. None of the above**

Answer: D. EPSP causes contraction

Explanation:

At the skeletal neuromuscular junction (NMJ), the motor neuron releases acetylcholine, which binds to nicotinic acetylcholine receptors on the motor end plate. This produces an excitatory end-plate potential and initiates an action potential in the muscle fiber. The action potential then travels along the sarcolemma and T-tubules, leading to calcium release from the sarcoplasmic reticulum. Calcium binds to troponin, allowing actin-myosin interaction and ultimately producing contraction. Therefore, strictly speaking, the EPSP/end-plate potential does not directly cause contraction; it initiates the electrical events that lead to calcium release and contraction. Thus, D is the intended answer.

[24/09/2026 8:35 AM] Imaanaa Harmee: **21. An action potential arriving at the neuromuscular junction causes release of:**

- A. Acetylcholine**
- B. Na⁺ ions**
- C. Ca²⁺ ions**
- D. Noradrenalin**
- E. None of the above**

Answer: A. Acetylcholine

Explanation:

When an action potential reaches the presynaptic terminal of a motor neuron at the neuromuscular junction, it opens voltage-gated calcium channels. Calcium enters the nerve terminal and triggers fusion of synaptic vesicles with the presynaptic membrane. The neurotransmitter released into the synaptic cleft is acetylcholine (ACh). ACh then binds to nicotinic receptors on the motor end plate, producing an end-plate potential and initiating an action potential in the skeletal muscle fiber. Therefore, A. Acetylcholine is correct.

22. If a person has to continue his activities after 15 seconds, the source of energy to continue the activity for the next 45–60 seconds is obtained from:

- A. Stored ATP**
- B. Phosphagen system**
- C. Anaerobic metabolism**
- D. Aerobic metabolism**
- E. Myoglobin**

Answer: C. Anaerobic metabolism

Explanation:

Energy production during exercise changes according to the duration and intensity of activity. Stored ATP provides energy for only a few seconds, while the phosphagen system (mainly ATP and creatine phosphate) provides rapid energy for approximately the first 10–15 seconds of intense activity. As the activity continues for the next several tens of seconds, anaerobic glycolysis becomes an important source of ATP. It breaks down glucose or glycogen without requiring oxygen and can provide ATP rapidly, although lactate and hydrogen ions accumulate. Aerobic metabolism becomes increasingly important as exercise continues for longer periods. Therefore, for the period described, C. Anaerobic metabolism is the intended answer.

23. Which drug of the following has a muscle relaxant effect?

- A. D-tubocurarine**
- B. Methacholine**
- C. Nicotine**
- D. Carbacholine**
- E. Acetylcholine**

Answer: A. D-tubocurarine

Explanation:

D-tubocurarine is a classic competitive antagonist of nicotinic acetylcholine receptors at the skeletal neuromuscular junction. By preventing acetylcholine from activating these receptors, it prevents effective depolarization of the muscle end plate and produces skeletal muscle paralysis/relaxation. It is therefore classified as a non-depolarizing neuromuscular blocker. The other listed substances are cholinergic agonists or nicotine-related agonists and generally stimulate cholinergic receptors rather than producing the classic neuromuscular-blocking effect. Therefore, A is correct.

24. The function of T-tubules in skeletal muscle fibers is:

- A. Storage of calcium ion**
- B. Conduction of depolarization waves to the sarcoplasmic reticulum**
- C. Contain postsynaptic receptors**
- D. Storage of neurotransmitters**
- E. None of the above**

Answer: B. Conduction of depolarization waves to the sarcoplasmic reticulum

Explanation:

Transverse tubules (T-tubules) are invaginations of the skeletal muscle cell membrane (sarcolemma). Their major role is to carry the action potential deep into the muscle fiber, allowing the electrical signal to reach the vicinity of the sarcoplasmic reticulum. This triggers the excitation-contraction coupling mechanism and causes calcium release from the sarcoplasmic reticulum. The sarcoplasmic reticulum, rather than the T-tubules, is the principal site of calcium storage. Therefore, B is correct.

[24/09/2026 8:35 AM] Imaanaa Harmee: **25. From the given options below, identify that is energy independent:**

- A. Release of calcium from the sarcoplasmic reticulum**
- B. Returning of calcium to the sarcoplasmic reticulum**
- C. Detaching myosin head from the active site of actin**
- D. Moving the myosin head and actin toward the center**
- E. Muscle relaxation**

Answer: A. Release of calcium from the sarcoplasmic reticulum

Explanation:

The release of Ca^{2+} from the sarcoplasmic reticulum occurs through calcium-release channels that open in response to excitation of the muscle fiber. The release itself does not directly require ATP; it occurs largely according to the calcium electrochemical gradient.

In contrast, returning calcium to the sarcoplasmic reticulum requires the SERCA calcium pump, which uses ATP. ATP is also required for myosin to detach from actin, and the cross-bridge cycle requires ATP for continued contraction. Thus, the energy-independent process listed is A. Release of calcium from the sarcoplasmic reticulum.

26. The principal contractile proteins in skeletal muscle are:

- A. Actin and troponin**
- B. Actin and myosin**
- C. Troponin and tropomyosin**
- D. Myosin and tropomyosin**
- E. Actin and tropomyosin**

Answer: B. Actin and myosin

Explanation:

The two principal contractile proteins of skeletal muscle are actin and myosin. Actin forms the major component of the thin filament, while myosin forms the thick filament. During contraction, the myosin heads attach to exposed binding sites on actin and undergo a power stroke, pulling the thin filaments toward the center of the sarcomere. Troponin and tropomyosin are regulatory proteins that control access to the actin-binding sites rather than being the primary contractile proteins. Therefore, B is correct.

27. The contractile component of a skeletal muscle that is not shortened during contraction is:

- A. The sarcomeres**

- B. The A-bands**
 - C. The I-bands**
 - D. The Z-lines**
 - E. The H-bands**
- Answer: B. The A-bands**

Explanation:

According to the sliding-filament theory, muscle contraction occurs because the thin actin filaments slide toward the center of the sarcomere over the thick myosin filaments. The filaments themselves do not become shorter. The A-band, which corresponds primarily to the length of the thick filaments, therefore remains essentially constant in length during contraction. The I-band and H-zone become shorter as actin moves inward, while the Z-lines move closer together. Thus, **B. The A-bands** is correct.

28. Not true of the single-unit smooth muscle is:

- A. Innervated in motor units comparable to those of skeletal muscles**
- B. Has gap junctions**
- C. Responsive to neural and hormonal controls**
- D. No pacemaker cells**
- E. Less common**

Answer: A. Innervated in motor units comparable to those of skeletal muscles

Explanation:

Single-unit (unitary) smooth muscle is characterized by large groups of smooth muscle cells that function together because they are connected by gap junctions. It is found in organs such as the gastrointestinal tract, uterus, and some blood vessels. These muscles can exhibit spontaneous electrical activity and may contain pacemaker regions, and their activity can be modified by autonomic nerves and hormones. Unlike skeletal muscle, smooth muscle is not organized into motor units in the same way. Therefore, the statement that single-unit smooth muscle is innervated in motor units comparable to skeletal muscle is incorrect. **A** is the best answer.

[24/09/2026 8:35 AM] Imaanaa Harmee: **29. Parasympathetic stimulation produces:**

- A. Bronchoconstriction**
- B. Increased gut motility**
- C. Decreased heart activities**
- D. Relaxes GIT sphincters**
- E. All are correct**

Answer: E. All are correct

Explanation:

Parasympathetic activity generally promotes rest-and-digest functions. In the respiratory system, it promotes bronchoconstriction and increased airway secretions. In the gastrointestinal tract, it increases motility and secretory activity and generally relaxes gastrointestinal sphincters, facilitating movement of contents through the digestive tract. In the cardiovascular system, parasympathetic stimulation, particularly through the vagus nerve, decreases heart rate and certain aspects of cardiac activity. Since **A**, **B**, **C**, and **D** are all consistent with parasympathetic activity, **E. All are correct** is the best answer.

30. Sympathetic stimulation on the cardiovascular system produces:

- A. Increased heart rate**
- B. Increased force of contraction of the cardiac muscle**
- C. Vasoconstriction of the systemic blood vessels**
- D. Increases cardiac output**
- E. All are correct**

Answer: E. All are correct

Explanation:

Sympathetic stimulation produces several coordinated cardiovascular effects that support the fight-or-flight response. Activation of β_1 -adrenergic receptors in the heart increases heart rate (positive chronotropy) and increases the force of contraction (positive inotropy). Sympathetic stimulation also produces vasoconstriction in many systemic blood vessels, particularly through α_1 -adrenergic receptors. The combination of increased heart rate and contractility increases stroke volume and therefore generally increases cardiac output. These changes help maintain or increase blood flow and blood pressure during stress or physical activity. Therefore, E. All are correct.

[24/09/2026 8:40 AM] Imaanaa Harmee: **31. Autonomic nervous system differs from the somatic nervous system in that:**

- A. The latter is made up of one long thick neuron**
- B. The former controls involuntary activities**
- C. The former is made up of two consecutive neurons**
- D. The latter controls voluntary activities**
- E. All of the above**

Answer: E. All of the above

Explanation:

The somatic nervous system and autonomic nervous system (ANS) differ in their organization and functions. The somatic motor pathway generally uses one motor neuron extending from the CNS to skeletal muscle, whereas the autonomic pathway uses two consecutive neurons: a preganglionic neuron and a postganglionic neuron. The ANS regulates largely involuntary functions, including heart activity, smooth muscle, glands, and many visceral functions. The somatic system is primarily responsible for voluntary control of skeletal muscles. Therefore, the statements in A–D describe the major differences, making E. All of the above the intended answer.

32. Of the given options, which is not innervated by the oculomotor nerve?

- A. Superior rectus**
- B. Medial rectus**
- C. Inferior rectus**

D. Inferior oblique

E. Superior oblique

Answer: E. Superior oblique

Explanation:

The oculomotor nerve (cranial nerve III) supplies most of the extraocular muscles: superior rectus, inferior rectus, medial rectus, and inferior oblique. It also supplies the levator palpebrae superioris and carries parasympathetic fibers responsible for pupil constriction and accommodation. The important exception among the extraocular muscles is the superior oblique, which is supplied by the trochlear nerve (cranial nerve IV). A useful rule is LR6 SO4, all others III: lateral rectus is CN VI, superior oblique is CN IV, and the remaining extraocular muscles are CN III. Therefore, E is correct.

33. A neurotransmitter produced by a postganglionic sympathetic fiber to the sweat glands is:

A. Acetylcholine

B. Adrenaline

C. Noradrenaline

D. Dopamine

E. All

Answer: A. Acetylcholine

Explanation:

Although most postganglionic sympathetic neurons release norepinephrine (noradrenaline), there is an important exception: sympathetic fibers supplying eccrine sweat glands release acetylcholine. The acetylcholine binds primarily to muscarinic receptors on the sweat glands and stimulates sweating. This is an important example of a sympathetic pathway that is cholinergic rather than adrenergic. Therefore, A. Acetylcholine is correct.

34. Which one of the following is wrong about neuronal activities of the ANS?

A. All preganglionic fibers are cholinergic

B. Postganglionic parasympathetic fibers are cholinergic

C. Preganglionic parasympathetic fibers are cholinergic

D. Most postganglionic sympathetic fibers are adrenergic

E. None of the above

Answer: E. None of the above

Explanation:

All statements A–D are essentially correct. All autonomic preganglionic neurons release acetylcholine, so both sympathetic and parasympathetic preganglionic fibers are cholinergic. Postganglionic parasympathetic neurons also release acetylcholine, usually acting on muscarinic receptors. Most postganglionic sympathetic neurons release norepinephrine, making them adrenergic. The major exception is sympathetic innervation of eccrine sweat glands, where acetylcholine is released. Therefore, none of A–D is wrong, so E. None of the above is correct.

[24/09/2026 8:40 AM] Imaanaa Harmee: **35. A signaling mechanism that causes muscle contraction is:**

- A. Endocrine**
- B. Paracrine**
- C. Synaptic**
- D. Contact**
- E. Autocrine**

Answer: C. Synaptic

Explanation:

Synaptic signaling occurs when a neuron releases a neurotransmitter across a synaptic junction to affect another cell. At the skeletal neuromuscular junction, a motor neuron releases acetylcholine, which binds to receptors on the skeletal muscle fiber and initiates an action potential leading to contraction. Endocrine signaling involves hormones traveling through the bloodstream, while paracrine signaling affects nearby cells and autocrine signaling affects the cell that released the signal. Contact-dependent signaling requires direct cell-to-cell interaction. Therefore, C. Synaptic is the best answer.

[24/09/2026 8:40 AM] Imaanaa Harmee: **36. Which of the following is not a sympathomimetic drug?**

- A. Epinephrine**
- B. Norepinephrine**
- C. Methoxamine**
- D. Pilocarpine**
- E. Epinephrine**

Answer: D. Pilocarpine

Explanation:

A sympathomimetic drug mimics the effects of sympathetic stimulation, usually by activating adrenergic receptors or increasing catecholaminergic activity. Epinephrine and norepinephrine are direct adrenergic agonists, while methoxamine is primarily an α_1 -adrenergic agonist. Pilocarpine, however, is a muscarinic cholinergic agonist and therefore mimics parasympathetic rather than sympathetic activity. It increases salivation and lacrimation and causes pupil constriction. Thus, D. Pilocarpine is correct.

37. Which of the following cranial nerves is responsible for corneal sensitivity?

- A. Glossopharyngeal nerve**
- B. Trochlear nerve**
- C. Trigeminal nerve**
- D. Abducens nerve**
- E. Oculomotor nerve**

Answer: C. Trigeminal nerve

Explanation:

The sensory innervation of the cornea is provided mainly by the ophthalmic division

(V1) of the trigeminal nerve (cranial nerve V). Specifically, sensory fibers travel through the nasociliary branch of V1. These fibers detect touch, pain, and temperature on the cornea and form the afferent limb of the corneal reflex. When the cornea is touched, sensory information travels through CN V1 to the brainstem, while the facial nerve (CN VII) provides the motor limb that causes blinking. Therefore, C. Trigeminal nerve is correct.

38. The work of breathing decreases when:

- A. Lung compliance increases**
 - B. The subject exercises**
 - C. The rate of breathing increases with constant minute volume**
 - D. The person lies down**
 - E. Functional residual capacity increases**
- Answer: A. Lung compliance increases**

Explanation:

Lung compliance describes how easily the lungs expand in response to a change in transpulmonary pressure. When compliance increases, less pressure and therefore less muscular effort are required to produce a given change in lung volume, so the work of breathing decreases. Conditions such as pulmonary fibrosis reduce compliance and make the lungs stiff, increasing the work required for inspiration. Although increased compliance can make expansion easier, excessively high compliance, as in emphysema, can impair elastic recoil and expiration. For the question as stated, however, A is the correct choice.

39. The volumes and capacities that cannot be measured by direct spirometry include:

- A. Residual volume, expiratory reserve volume and inspiratory capacity**
 - B. Residual volume, functional residual capacity and total lung capacity**
 - C. Inspiratory reserve volume, residual volume, vital capacity**
 - D. Total lung capacity, residual volume and expiratory reserve volume**
 - E. Tidal volume, inspiratory reserve volume and inspiratory capacity**
- Answer: B. Residual volume, functional residual capacity and total lung capacity**

Explanation:

Simple spirometry measures air that can be inhaled or exhaled, but it cannot directly measure the air remaining in the lungs after maximal expiration—the residual volume (RV). Because functional residual capacity (FRC) includes residual volume ($FRC = ERV + RV$), it also cannot be measured directly by simple spirometry. Similarly, total lung capacity ($TLC = VC + RV$) cannot be measured directly because it contains the unmeasurable residual volume. Other volumes such as tidal volume and inspiratory reserve volume can be measured with spirometry. Therefore, B is correct.

[24/09/2026 8:40 AM] Imaanaa Harmee: **40. The total lung capacity consists of:**

- A. Vital capacity & tidal volume**
- B. Residual volume & vital capacity**

C. Functional residual capacity & vital capacity

D. Tidal volume & functional residual capacity

E. Tidal volume & residual volume

Answer: B. Residual volume & vital capacity

Explanation:

Total lung capacity (TLC) is the maximum amount of air contained in the lungs after a maximal inspiration. It consists of all four basic lung volumes: tidal volume (TV), inspiratory reserve volume (IRV), expiratory reserve volume (ERV), and residual volume (RV). Vital capacity consists of IRV + TV + ERV, while residual volume is the air remaining after maximal expiration.

Therefore:

TLC = Vital Capacity + Residual Volume

Thus, B. Residual volume & vital capacity is correct.

[24/09/2026 8:40 AM] Imaanaa Harmee: **41. All of the following events occur during expiration except:**

A. Inspiratory muscles relax and the rib cage descends due to gravity

B. Thoracic cavity volume decreases

C. Intrapulmonary volume decreases

D. Intrapulmonary pressure rises above atmospheric pressure (+1 mm Hg)

E. Gases flow out of the lungs against the pressure gradient

Answer: E. Gases flow out of the lungs against the pressure gradient

Explanation:

During normal expiration, the inspiratory muscles relax, allowing the elastic recoil of the lungs and chest wall to reduce the volume of the thoracic cavity. As lung volume decreases, intrapulmonary (alveolar) pressure increases slightly above atmospheric pressure, typically by about 1 mmHg. Air then moves from an area of higher pressure inside the lungs to lower atmospheric pressure outside. Therefore, gases flow down, not against, the pressure gradient. The statement in E is therefore incorrect, making E the answer.

42. Not the function of surfactant is:

A. Increases surface area

B. Decreases pressure in the alveoli

C. Prevents lung collapse

D. Increases work of breathing

E. Facilitates gas exchange

Answer: D. Increases work of breathing

Explanation:

Pulmonary surfactant, produced mainly by type II pneumocytes, reduces the surface tension of the fluid lining the alveoli. This helps prevent alveolar collapse, particularly during expiration, and makes the lungs easier to expand. Consequently, surfactant

decreases rather than increases the work of breathing. By maintaining open and stable alveoli, it also improves the effective surface available for gas exchange. Therefore, D is the statement that is not a function of surfactant.

43. Which structure does not belong to the respiratory portion of the breathing organ?

- A. Terminal bronchioles**
- B. Respiratory bronchioles**
- C. Alveolar ducts**
- D. Alveolar sacs**
- E. None of the above**

Answer: A. Terminal bronchioles

Explanation:

The respiratory portion of the respiratory system is the region where gas exchange can occur. It begins with the respiratory bronchioles and continues through the alveolar ducts, alveolar sacs, and alveoli. Terminal bronchioles belong to the conducting portion because they conduct air but do not normally participate directly in gas exchange. Therefore, A. Terminal bronchioles is correct.

44. The respiratory function of the respiratory system is:

- A. Word production**
- B. Air conduction**
- C. Defense**
- D. Olfaction**
- E. Regulation of pH**

Answer: B. Air conduction

Explanation:

The respiratory system performs several functions, including air conduction, gas exchange, defense, phonation, olfaction, and acid-base regulation. However, among the options, the question identifies air conduction as a respiratory function. Air conduction involves transporting air through the nasal passages, pharynx, larynx, trachea, bronchi, and conducting bronchioles toward the respiratory portion of the lungs. Therefore, according to the provided key, B is the intended answer. Note that regulation of pH is also an important physiological function of respiration, because CO₂ removal influences blood pH.

45. Which of the following would cause an increase in glomerular filtration rate?

- A. Contraction of mesangial cells**
- B. Ureter obstruction**
- C. Release of renin**

D. Dilation of the afferent arterioles

E. Volume depletion

Answer: D. Dilation of the afferent arterioles

Explanation:

The afferent arteriole carries blood into the glomerulus. When it dilates, resistance to blood entering the glomerular capillaries decreases. This increases glomerular capillary hydrostatic pressure, which promotes filtration and therefore increases GFR. In contrast, afferent arteriolar constriction decreases renal blood flow and GFR. Ureter obstruction increases pressure within Bowman's space, opposing filtration. Volume depletion activates mechanisms that generally reduce renal perfusion and GFR. Therefore, D is correct.

[24/09/2026 8:40 AM] Imaanaa Harmee: **46. Which of the following best describes the urine of a 46-year-old man with a damaged pituitary gland?**

A. A higher flow of hypotonic urine

B. A higher flow of hypertonic urine

C. A normal flow of hypertonic urine

D. A lower flow of hypotonic urine

E. A lower flow of hypertonic urine

Answer: A. A higher flow of hypotonic urine

Explanation:

The posterior pituitary releases antidiuretic hormone (ADH), also called vasopressin. ADH increases water reabsorption in the late distal tubule and collecting ducts by promoting insertion of aquaporin-2 channels into the apical membrane of collecting duct cells. Damage to the pituitary can therefore reduce ADH availability, causing central diabetes insipidus. Without sufficient ADH, less water is reabsorbed, resulting in a large volume of dilute (hypotonic) urine. Therefore, A is correct.

47. Sympathetic stimulation decreases GFR via:

A. Constricting afferent arteriole

B. Dilating efferent arteriole

C. Dilating afferent arteriole

D. Inhibiting nitric oxide

E. Activating endothelin

Answer: A. Constricting afferent arteriole

Explanation:

Strong sympathetic stimulation causes vasoconstriction of the renal blood vessels, particularly the afferent arterioles. Afferent arteriolar constriction decreases blood flow into the glomerular capillaries and lowers glomerular capillary hydrostatic pressure, thereby reducing GFR. This mechanism is particularly important during severe stress, hemorrhage, or significant volume depletion, when the sympathetic nervous system prioritizes blood flow to vital organs such as the heart and brain. Therefore, A is correct.

48. Most filtrate of the glomerulus undergoes reabsorption in which of the following?

- A. Proximal tubule**
- B. Descending limb of the loop of Henle**
- C. Ascending limb of the loop of Henle**
- D. Distal tubule**
- E. Collecting duct**

Answer: A. Proximal tubule

Explanation:

The proximal convoluted tubule (PCT) is the major site of renal tubular reabsorption. Approximately two-thirds of the filtered water and sodium are normally reabsorbed here, along with nearly all filtered glucose and amino acids under normal conditions, plus substantial amounts of bicarbonate and other solutes. The proximal tubule therefore performs the largest fraction of the total reabsorptive work of the nephron. The loop of Henle, distal tubule, and collecting duct perform additional selective reabsorption and secretion. Thus, A. Proximal tubule is correct.

49. Decreasing the resistance of the afferent arteriole in the glomerulus will decrease which of the following?

- A. Glomerular filtration rate**
- B. Renal plasma flow**
- C. Filtration fraction**
- D. Renin release from juxtaglomerular cells**
- E. All**

Answer: D. Renin release from juxtaglomerular cells

Explanation:

Decreasing afferent arteriolar resistance means afferent arteriolar dilation. This allows more blood to enter the glomerulus, increasing renal plasma flow and glomerular capillary pressure, and therefore generally increasing GFR. At the same time, increased perfusion pressure at the juxtaglomerular apparatus and increased NaCl delivery to the macula densa suppress renin secretion. Thus, among the listed variables, renin release decreases. Therefore, D is correct.

50. Most filtrate of the glomerulus undergoes reabsorption in which of the following?

- A. Proximal tubule**
- B. Descending limb of the loop of Henle**
- C. Ascending limb of the loop of Henle**
- D. Distal tubule**
- E. Collecting duct**

Answer: A. Proximal tubule

Explanation:

The proximal tubule is responsible for the largest proportion of renal tubular reabsorption. As filtrate leaves Bowman's capsule, the proximal tubule reabsorbs approximately 65% of filtered Na^+ and water, as well as essentially all filtered glucose and amino acids under normal circumstances. It also reabsorbs large amounts of bicarbonate, chloride, potassium, phosphate, and other substances. Later portions of the nephron fine-tune the composition and concentration of urine, but they do not

reabsorb as much of the original filtrate as the proximal tubule. Therefore, A. Proximal tubule is correct.

[24/09/2026 8:40 AM] Imaanaa Harmee: **51. Response to high pressure in the renal blood flow is:**

- A. Tubuloglomerular autoregulation**
- B. Myogenic contraction**
- C. Release of ANP**
- D. Sympathetic stimulation**
- E. Parasympathetic stimulation**

Answer: B. Myogenic contraction

Explanation:

The myogenic mechanism is an important component of renal autoregulation. When arterial pressure and renal blood flow increase, the increased stretch of the smooth muscle in the walls of the afferent arteriole causes the muscle to contract. This contraction increases vascular resistance and prevents excessive increases in renal blood flow and GFR. This mechanism helps keep renal function relatively stable despite changes in systemic blood pressure. Therefore, B. Myogenic contraction is correct.

52. The filtration fraction is increased by which of the following?

- A. Increasing renal blood flow**
- B. Increasing afferent arteriolar resistance**
- C. Increasing efferent arteriolar resistance**
- D. Increasing plasma oncotic pressure**
- E. Increasing the pressure within Bowman's capsule**

Answer: C. Increasing efferent arteriolar resistance

Explanation:

The filtration fraction (FF) is the proportion of renal plasma flow that is filtered by the glomeruli:

$$\text{FF} = \text{GFR} \div \text{Renal Plasma Flow}$$

Constriction of the efferent arteriole increases resistance to blood leaving the glomerulus. Moderate efferent constriction raises glomerular capillary hydrostatic pressure, which increases GFR while renal plasma flow decreases. Because GFR rises relative to renal plasma flow, the filtration fraction increases. Severe efferent constriction can eventually reduce GFR because of excessive increases in glomerular capillary oncotic pressure and reduced renal perfusion. Therefore, C is the best answer.

53. Structure in the renal medulla that drains urine to the ureter is:

- A. Calyx**
- B. Pyramid**
- C. Renal pelvis**

D. Glomeruli

E. Hilum

Answer: C. Renal pelvis

Explanation:

Urine formed by the nephrons passes through the collecting ducts and drains through the renal papillae into the minor calyces. Several minor calyces join to form major calyces, which empty into the renal pelvis. The renal pelvis then narrows and continues as the ureter, which transports urine to the urinary bladder. The renal pyramid is a structural region of the medulla, not the final collecting structure before the ureter. Therefore, C. Renal pelvis is correct.

54. Determinant factor for the fetus to become a male individual is:

A. The presence of the hypothalamus

B. The presence of penis

C. The presence of the SRY-protein

D. The absence of the X-chromosome

E. All of the above

Answer: C. The presence of the SRY-protein

Explanation:

The major genetic determinant of typical male sex development is the presence of the SRY gene on the Y chromosome. SRY encodes a transcription factor that initiates the development of the undifferentiated embryonic gonad into a testis. The developing testes then produce hormones such as testosterone and anti-Müllerian hormone, which direct subsequent differentiation of the male reproductive tract and genitalia. A male does not develop because of the absence of an X chromosome; typically, males have one X and one Y chromosome. Therefore, C is the intended answer. More precisely, it is the SRY gene and its functional product, rather than simply the protein alone, that initiates testicular differentiation.

55. Müllerian duct develops into the:

A. Ovaries

B. Female genital tract

C. Male accessory glands

D. Female external genitalia

E. Ovaries

Answer: B. Female genital tract

Explanation:

The Müllerian duct, also called the paramesonephric duct, gives rise to major portions of the female internal reproductive tract. It develops into the uterine tubes (fallopian tubes), uterus, cervix, and upper portion of the vagina. The ovaries themselves develop from the gonadal ridges and are not derivatives of the Müllerian duct. The male reproductive ducts, in contrast, largely arise from the Wolffian (mesonephric) ducts under the influence of testosterone. Therefore, B. Female genital tract is correct.

[24/09/2026 8:40 AM] Imaanaa Harmee: **56. The urogenital slit remains open and**

develops into:

- A. The female external genitalia**
- B. The male external genitalia**
- C. The female internal genitalia**
- D. The ovaries**
- E. Fallopian tube**

Answer: A. The female external genitalia

Explanation:

During embryonic development, the external genital structures initially have a common, undifferentiated appearance. In the absence of the androgen-driven pathway responsible for typical male external genital development, the urogenital region remains open, and the genital folds develop into structures of the female external genitalia, including the labia minora. The genital tubercle and labioscrotal swellings also undergo characteristic female differentiation. Thus, according to the developmental terminology used in this question, A is correct.

57. The Wolffian duct develops into the:

- A. Testes**
- B. Male accessory organs**
- C. Male external genitalia**
- D. Female external genitalia**
- E. Female ductile system**

Answer: B. Male accessory organs

Explanation:

The Wolffian duct, or mesonephric duct, develops into important portions of the male internal reproductive duct system under the influence of testosterone from the fetal testes. Its derivatives include the epididymis, ductus deferens, ejaculatory ducts, and seminal vesicles. The testes themselves develop from the gonadal ridge, while the male external genitalia develop from the genital tubercle, genital folds, and labioscrotal swellings under androgen influence. Therefore, B. Male accessory organs is correct.

58. Which one of the following is known to be the hormone of ovulation?

- A. FSH**
- B. Estradiol**
- C. Progesterone**
- D. hCG**
- E. LH**

Answer: E. LH

Explanation:

Ovulation is triggered primarily by the mid-cycle luteinizing hormone (LH) surge. During the late follicular phase, sustained high estrogen levels from the dominant follicle switch from negative to positive feedback on the hypothalamus and pituitary. This produces a large surge in LH, which promotes final maturation of the oocyte and

follicular rupture, releasing the secondary oocyte from the ovary. FSH also rises somewhat, but LH is the principal hormonal trigger for ovulation. Therefore, E. LH is correct.

59. Which of the following is not a haploid cell?

- A. Spermatogonium**
- B. Secondary oocyte**
- C. Spermatids**
- D. Oocyte**
- E. Ovum**

Answer: A. Spermatogonium

Explanation:

A haploid cell contains one set of chromosomes (n), whereas a diploid cell contains two sets ($2n$). A spermatogonium is a germ cell that remains diploid ($2n$) and undergoes mitosis before entering meiosis. After meiosis I, secondary spermatocytes are haploid, and after meiosis II, spermatids are haploid. The secondary oocyte and ovum are also haploid. Therefore, A. Spermatogonium is the cell that is not haploid.

60. Usual site of fertilization in the female reproductive system is:

- A. Ovary**
- B. Fallopian tubes**
- C. Uterus**
- D. Vagina**
- E. Clitoris**

Answer: B. Fallopian tubes

Explanation:

Fertilization normally occurs in the ampulla of the uterine (fallopian) tube, which is the wider portion of the tube. After ovulation, the secondary oocyte enters the uterine tube and travels toward the uterus. Sperm deposited in the vagina travel through the cervix and uterus into the uterine tube, where one sperm may fuse with the oocyte. The resulting zygote then undergoes early development while traveling toward the uterus, where implantation normally occurs. Therefore, B. Fallopian tubes is correct.

[24/09/2026 8:40 AM] Imaanaa Harmee: **61. A chromosomal abnormality with the genotype of XXY and a phenotype of infertile male is called:**

- A. Turner's syndrome**
 - B. Klinefelter's syndrome**
 - C. XXXYY syndrome**
 - D. Trisomy**
 - E. Cushing's syndrome**
- Answer: B. Klinefelter's syndrome**

Explanation:

Klinefelter syndrome is a sex-chromosome abnormality characterized most commonly

by a 47,XXY karyotype. Individuals typically have a male phenotype but have impaired testicular development, resulting in reduced testosterone production and infertility, usually because of severe impairment of sperm production. Common features may include small testes, gynecomastia, reduced body hair, and relatively tall stature. Turner syndrome is typically 45,X and affects phenotypic females. Therefore, B is correct.

62. Ptyalin (amylase) is produced by:

- A. The parotid gland**
- B. Parietal cells**
- C. Chief cells**
- D. Enteroendocrine cells**
- E. Goblet cells**

Answer: A. The parotid gland

Explanation:

Ptyalin, or salivary α -amylase, is an enzyme secreted by the salivary glands and begins the digestion of carbohydrates in the mouth. The parotid glands produce a large amount of watery, enzyme-rich saliva containing significant quantities of amylase. Parietal cells secrete hydrochloric acid and intrinsic factor, while chief cells secrete pepsinogen. Goblet cells produce mucus, and enteroendocrine cells produce gastrointestinal hormones. Therefore, A. The parotid gland is correct.

63. The major stimulus for the release of secretin hormone is:

- A. Protein digestion products**
- B. Histamine**
- C. Somatostatin**
- D. Acidic chyme in the duodenum**
- E. Fat digestion products**

Answer: D. Acidic chyme in the duodenum

Explanation:

Secretin is produced by S cells of the duodenal mucosa. Its major stimulus is the arrival of acidic chyme from the stomach into the duodenum, which lowers duodenal pH. Secretin stimulates the pancreas and biliary system to release bicarbonate-rich fluid, helping neutralize the gastric acid and create a suitable pH for intestinal and pancreatic enzymes. Fat and protein digestion products can also influence intestinal hormone secretion, but acid in the duodenum is the major stimulus for secretin. Therefore, D is correct.

64. GI layer that contains much of the glands is:

- A. Serosa**
- B. Muscularis**
- C. Submucosa**
- D. Mucosa**
- E. All**

Answer: C. Submucosa

Explanation:

The gastrointestinal tract is organized into four major layers: mucosa, submucosa, muscularis externa, and serosa/adventitia. The submucosa contains blood vessels, lymphatic vessels, nerves, and certain important glands. For example, the duodenum contains Brunner glands in the submucosa, which secrete alkaline mucus that helps protect the duodenal mucosa from gastric acid. The mucosa also contains glands, so the wording of this question is somewhat broad, but based on the provided answer key, C. Submucosa is intended.

65. Center for chewing is located in the:

- A. Medulla**
- B. Pons**
- C. Midbrain**
- D. Spinal cord**
- E. Cerebrum**

Answer: B. Pons

Explanation:

The neural control of chewing, or mastication, is coordinated by a chewing center in the brainstem, particularly involving the pons and associated trigeminal motor nuclei. The trigeminal nerve (CN V) supplies the major muscles of mastication. The chewing center coordinates rhythmic activation and relaxation of these muscles, allowing the jaw to open and close and food to be mechanically broken down. Therefore, B. Pons is the intended answer.

[24/09/2026 8:40 AM] Imaanaa Harmee: **66. Which of the following is correct regarding GIT regulation?**

- A. Parasympathetic nervous system has excitatory effect on GI function.**
- B. Salivary secretion is mainly regulated by gastric hormone.**
- C. Sympathetic nerves have excitatory effect on the GI functions.**
- D. Secretin hormone inhibits pancreatic secretions.**
- E. None of the above.**

Answer: A. Parasympathetic nervous system has excitatory effect on GI function.

Explanation:

The parasympathetic nervous system generally promotes gastrointestinal activity. Parasympathetic stimulation increases gastrointestinal motility, secretion, and blood flow and generally facilitates digestion. The sympathetic nervous system usually has an inhibitory effect on many gastrointestinal functions, decreasing motility and secretions and causing vasoconstriction. Secretin does not simply inhibit pancreatic secretion; it strongly stimulates the pancreas to release bicarbonate-rich fluid. Salivary secretion is primarily controlled by neural mechanisms involving the salivatory nuclei rather than gastric hormones. Therefore, A is correct.

67. An endocrine function of the stomach is the fact that it secretes:

- A. HCl**
- B. Pepsin**
- C. Intrinsic factor**
- D. Mucus**
- E. Gastrin**

Answer: E. Gastrin

Explanation:

An endocrine function involves releasing a hormone into the bloodstream. The stomach contains G cells, particularly in the gastric antrum, which secrete the hormone gastrin into the circulation. Gastrin stimulates gastric acid secretion, promotes gastric mucosal growth, and supports gastric motility. HCl, pepsinogen, mucus, and intrinsic factor are secretory products of gastric cells but are not hormones released into the bloodstream as endocrine signals. Therefore, E. Gastrin is correct.

68. The hardest part of the teeth is:

- A. Crown**
- B. Dentin**
- C. Enamel**
- D. Root**
- E. None**

Answer: C. Enamel

Explanation:

Enamel is the hardest substance in the human body. It is the highly mineralized outer covering of the crown of the tooth and consists predominantly of hydroxyapatite crystals. Its high mineral content gives it exceptional hardness and allows it to withstand the mechanical forces involved in chewing. Dentin lies beneath enamel and is softer, while the root is the portion of the tooth rather than a specific tissue type. Therefore, C. Enamel is correct.

69. An essential regulation of bile release into the duodenum is:

- A. Secretin**
- B. CCK**
- C. GIP**
- D. Gastrin**
- E. Sympathetic nervous system**

Answer: B. CCK

Explanation:

Cholecystokinin (CCK) is released from I cells in the duodenum and jejunum primarily in response to fatty acids and amino acids in the intestinal lumen. One of its major actions is to cause contraction of the gallbladder, resulting in the release of stored bile into the duodenum. CCK also promotes pancreatic enzyme secretion and contributes to slowing gastric emptying. Secretin mainly stimulates bicarbonate secretion from the pancreas and bile ducts. Therefore, B. CCK is correct.

70. The hormone that causes release of bicarbonate from the pancreas is:

- A. CCK**
 - B. Secretin**
 - C. Gastrin**
 - D. Somatostatin**
 - E. Insulin**
- Answer: B. Secretin**

Explanation:

Secretin is the principal hormone responsible for stimulating the pancreas to release bicarbonate-rich aqueous fluid. When acidic chyme enters the duodenum, S cells release secretin into the bloodstream. Secretin acts primarily on pancreatic ductal cells, stimulating bicarbonate secretion. The bicarbonate neutralizes gastric acid and raises the duodenal pH, creating a favorable environment for pancreatic and intestinal digestive enzymes. CCK, in contrast, primarily stimulates pancreatic enzyme secretion and gallbladder contraction. Therefore, B. Secretin is correct.

Exam:-Physiology

Year:-2016 E.C

Dr. Doc

[24/09/2026 9:01 AM] Imaanaa Harmee: **1. General properties of negative feedback include all of the following EXCEPT:**

- A. It controls deviation from normal value**
- B. It is the most commonly occurring**
- C. It is a closed loop control system in which the output accelerates the control system.**
- D. The term negative is used because the effector response opposes the departure from the set point**
- E. It often works towards homeostasis.**

Answer: C. It is a closed loop control system in which the output accelerates the control system.

Explanation:

Negative feedback is the major regulatory mechanism used by the human body to maintain homeostasis. In a negative-feedback system, a deviation from the desired or normal value is detected, and the resulting response opposes or reduces the original disturbance. The system therefore tends to bring a controlled variable back toward its set point. It is called "negative" because the response acts in the opposite direction to the initial change. For example, when blood glucose rises, insulin secretion increases glucose uptake and storage, causing blood glucose to fall toward normal. Negative feedback is therefore a stabilizing mechanism. In contrast, a system in which the output accelerates or reinforces the original response is characteristic of positive feedback, not negative feedback. Thus, C is the exception.

2. Which of the following is not the function of smooth endoplasmic reticulum?

- A. Protein synthesis**
- B. Calcium storage in muscle cells**
- C. Glycogen storage in liver cells**
- D. Drug metabolism in hepatocytes**
- E. Lipid synthesis**

Answer: A. Protein synthesis

Explanation:

The smooth endoplasmic reticulum (SER) performs several important functions, including lipid synthesis, steroid synthesis, detoxification of drugs and toxins, and storage and regulation of calcium ions. In muscle cells, the specialized smooth ER is called the sarcoplasmic reticulum, which stores and releases Ca^{2+} during muscle contraction. The smooth ER is also involved in carbohydrate metabolism, including processes associated with hepatic glycogen metabolism. However, protein synthesis is

primarily a function of the rough endoplasmic reticulum (RER) because ribosomes attached to the RER synthesize proteins destined for secretion, membranes, or certain organelles. Therefore, A is correct.

3. The cell membrane is composed of:

- A. 45% proteins and 55% carbohydrates**
- B. 55% proteins and 43% lipids**
- C. 55% lipids and 13% cholesterol**
- D. 45% proteins and 3% carbohydrates**
- E. 45% phospholipids and 13% neutral fats**

Answer: B. 55% proteins and 43% lipids

Explanation:

The plasma membrane is composed mainly of proteins and lipids, with a smaller amount of carbohydrate. A commonly cited approximate composition by mass is about 55% protein, 42% lipid, and 3% carbohydrate. The proteins perform many functions, including transport, receptors, enzymes, and cell adhesion. The lipid component forms the basic bilayer, with phospholipids and cholesterol contributing to membrane structure and fluidity. Carbohydrates are attached mainly to membrane proteins and lipids on the extracellular surface. Therefore, among the choices, B. 55% proteins and 43% lipids is the intended answer.

4. Which one of the following does not belong to the cell membrane proteins?

- A. Ion channels**
 - B. Na⁺-K⁺ ATPase**
 - C. Carrier proteins**
 - D. Membrane receptor**
 - E. None of the above**
- Answer: E. None of the above**

Explanation:

All of options A–D are types or examples of membrane proteins. Ion channels allow specific ions to move across the membrane. The Na⁺-K⁺ ATPase is an integral membrane protein that actively transports Na⁺ and K⁺ using ATP. Carrier proteins transport specific substances across the membrane, while membrane receptors recognize extracellular signaling molecules such as hormones and neurotransmitters. Therefore, none of A–D is excluded from membrane proteins, making E. None of the above correct.

[24/09/2026 9:01 AM] Imaanaa Harmee: **5. Which of the following statements is correct about the structure of the cell membrane?**

- A. Portions of the lipid bilayer that face the ECF and ICF are hydrophobic**
- B. Polar molecules can easily pass through the lipid bilayer**
- C. Protein channels within the lipid bilayer allow the passage of ions**
- D. The polar heads of the lipid bilayer face each other in the interior of the cell**
- E. None of the above**

Answer: C. Protein channels within the lipid bilayer allow the passage of ions

Explanation:

The plasma membrane is a phospholipid bilayer. The hydrophilic (polar) heads face the aqueous extracellular and intracellular environments, while the hydrophobic fatty-acid tails face inward toward each other. Because the interior of the membrane is hydrophobic, many ions and polar molecules cannot readily cross the lipid bilayer by simple diffusion. Instead, specific membrane proteins, such as ion channels and transporters, provide pathways for these substances. Therefore, statement C is correct. Statements A, B, and D incorrectly describe the orientation or permeability of the lipid bilayer.

6. Which of the following is true regarding the $\text{Na}^+\text{-K}^+$ ATPase?

- A. It is made up of five subunits**
 - B. Its alpha subunit is facing the ECF and has binding site for Na^+ ions**
 - C. Its ECF domain contains four binding sites**
 - D. The ICF domain has K^+ ion binding sites**
 - E. ATPase binding site is on its ECF domain**
- Answer: B. Its alpha subunit is facing the ECF and has binding site for Na^+ ions.**

Explanation:

The $\text{Na}^+\text{-K}^+$ ATPase is an integral membrane protein that maintains the concentration gradients of sodium and potassium across the plasma membrane. It uses ATP to transport 3 Na^+ ions out of the cell and 2 K^+ ions into the cell during each cycle. The α -subunit contains the major functional components, including ion-binding sites and the ATPase activity. In the classic description, the Na^+ -binding sites are accessible from the intracellular side, while K^+ binds from the extracellular side. Therefore, the wording of option B is actually reversed: the α -subunit spans the membrane, but its Na^+ binding sites are exposed to the intracellular side during the relevant transport state. Thus, the supplied answer/key appears physiologically inaccurate.

If forced to select from the provided options, B is the intended key, but remember the correct transport principle: 3 Na^+ move out and 2 K^+ move in, using ATP on the intracellular side.

7. Which one of the following is false about facilitated diffusion?

- A. It requires carrier proteins to transport substances**
- B. It needs ATP to transport substances**
- C. It transports substances down their concentration gradient**
- D. Carrier proteins are involved**
- E. It is a saturable process**

Answer: B. It needs ATP to transport substances

Explanation:

Facilitated diffusion is a form of passive transport. It uses membrane proteins, particularly carrier proteins or channels, to help substances cross the membrane, but it does not directly require ATP. The transported substance moves down its

electrochemical or concentration gradient, from an area of higher concentration toward an area of lower concentration. Because carrier proteins have a finite number of binding sites, facilitated diffusion can become saturated when all available carriers are occupied. This distinguishes it from simple diffusion. Therefore, B is false and is the correct answer.

8. A mechanism by which water molecules are transported through the cell membrane is:

- A. Osmosis
- B. Simple diffusion
- C. Facilitated diffusion
- D. Primary active transport
- E. Secondary active transport

Answer: A. Osmosis

Explanation:

Osmosis is the net movement of water across a selectively permeable membrane from a region of relatively lower solute concentration toward a region of relatively higher effective solute concentration. In many cell membranes, water moves rapidly through specialized channels called aquaporins, although the underlying process is still classified as osmosis. Importantly, osmosis does not directly consume ATP. Therefore, A. Osmosis is correct.

[24/09/2026 9:01 AM] Imaanaa Harmee: 9. Which of the following substances is lipid-soluble?

- A. O₂ and CO₂
- B. Water
- C. Sodium and potassium
- D. Glucose
- E. All

Answer: A. O₂ and CO₂

Explanation:

The interior of the phospholipid bilayer is hydrophobic, so small nonpolar or lipid-soluble molecules can cross it relatively easily. Oxygen (O₂) and carbon dioxide (CO₂) are small, nonpolar gases and can move across the membrane by simple diffusion. Water is polar, while sodium and potassium are charged ions and cannot freely cross the lipid core. Glucose is also relatively large and polar and requires specific transport proteins. Therefore, A. O₂ and CO₂ is correct.

10. Glucose and amino acids are absorbed from the GIT and reabsorbed from the renal tubules by:

- A. Simple diffusion
- B. Osmosis
- C. Secondary active transport
- D. Primary active transport
- E. None of the above

Answer: C. Secondary active transport

Explanation:

Glucose and many amino acids are transported using secondary active transport, particularly through sodium-dependent cotransport mechanisms. In the small intestine and proximal renal tubule, the Na^+ gradient provides the energy that drives glucose and amino acids into cells against their own concentration gradients. This Na^+ gradient is maintained by the Na^+-K^+ ATPase, which uses ATP directly; therefore, the cotransport mechanism itself is called secondary active transport. The important distinction is that secondary active transport does not directly hydrolyze ATP at the cotransporter but depends indirectly on the ATP-generated sodium gradient. Hence, C is correct.

[24/09/2026 9:01 AM] Imaanaa Harmee: **11. Which one of the following is correct about total body water volume?**

- A. Higher in females than in males**
- B. Higher in adults than in infants**
- C. Higher in obese people than in lean people**
- D. Higher in males than in females in adults**
- E. Higher in old people than in adults**

Answer: D. Higher in males than in females in adults

Explanation:

Total body water (TBW) is generally greater in adult males than in adult females because males tend to have a greater proportion of lean body mass, especially skeletal muscle, and muscle contains considerably more water than adipose tissue. Infants have a relatively high percentage of body water, while the proportion decreases with aging. Similarly, obesity is associated with a lower percentage of body water because fat contains less water than lean tissue. Therefore, D is the correct option.

12. Which electrolytes are present at a higher concentration in the ECF than in the ICF?

- A. Na^+ , K^+ , Cl^- and HCO_3^-**
- B. K^+ , Cl^- , SO_4^{2-} and HCO_3^-**
- C. Na^+ , Cl^- , HCO_3^- and Ca^{2+}**
- D. K^+ , SO_4^{2-} , Mg^{2+} and PO_4^{3-}**
- E. Ca^{2+} , K^+ and Na^+**

Answer: C. Na^+ , Cl^- , HCO_3^- and Ca^{2+}

Explanation:

The ECF and ICF have distinctly different electrolyte compositions. Na^+ is the major cation of the ECF, whereas K^+ is the major cation inside cells. Chloride and bicarbonate are important extracellular anions, and extracellular calcium concentration is much higher than free intracellular calcium concentration. Conversely, the ICF contains high

concentrations of K^+ , Mg^{2+} , phosphate, and intracellular proteins. Therefore, the combination in C represents electrolytes that are characteristically higher in the ECF.

13. Which one of the following represents an antiport carrier?

- A. Transport of glucose by facilitated diffusion**
- B. Na^+-K^+ pump**
- C. Na^+ -glucose co-transporter**
- D. Transport of O_2 and CO_2**
- E. Osmosis**

Answer: B. Na^+-K^+ pump

Explanation:

An antiport is a membrane transport mechanism in which two substances move in opposite directions. The Na^+-K^+ ATPase is an example: it transports 3 Na^+ out of the cell while transporting 2 K^+ into the cell during each cycle. It uses energy from ATP, making it a form of primary active transport. The Na^+ -glucose transporter is a symport, because Na^+ and glucose move in the same direction. Therefore, B is correct.

14. Bipolar neurons are located in the:

- A. Anterior horn of the gray matter of spinal cord**
- B. Retina and inner ear**
- C. Lateral horn of the gray matter of the spinal cord**
- D. Posterior horn of the gray matter of the spinal cord**
- E. Peripheral effector organs**

Answer: B. Retina and inner ear

Explanation:

Bipolar neurons have two processes extending from their cell body: one dendrite and one axon. They are specialized sensory neurons and are found in particular sensory structures. Important examples include the retina, where bipolar cells participate in visual signaling, and sensory structures of the inner ear, where bipolar neurons are involved in hearing and balance. In contrast, most neurons in the spinal cord are multipolar. Therefore, B is correct.

15. Identify the correct statement from the following.

- A. Motor neurons transmit impulses from periphery to the CNS**
- B. Microglia provides nutrients to neurons in the CNS**
- C. Neuroglia are excitable cells**
- D. Nerve insulators for both central and peripheral NS are the same**
- E. None of the above**

Answer: E. None of the above

Explanation:

Option A is incorrect because motor (efferent) neurons carry impulses from the CNS toward effectors, whereas sensory neurons carry information from the periphery

toward the CNS. Option B is incorrect because microglia are the resident immune/phagocytic cells of the CNS, rather than the cells primarily responsible for providing nutrients to neurons. Option C is incorrect because neuroglial cells generally do not generate conventional action potentials like neurons. Option D is also incorrect because the myelin-producing cells differ: oligodendrocytes produce myelin in the CNS, whereas Schwann cells produce myelin in the PNS. Therefore, E is correct.

[24/09/2026 9:01 AM] Imaanaa Harmee: **16. During the depolarization phase of an action potential:**

- A. Voltage-gated Na⁺ channels are opened**
 - B. Voltage-gated K⁺ channels are opened**
 - C. Voltage-gated Na⁺ channels are closed**
 - D. Ligand-gated K⁺ channels are closed**
 - E. None of the above**
- Answer: A. Voltage-gated Na⁺ channels are opened**

Explanation:

During the rising or depolarization phase of a neuronal action potential, a threshold stimulus causes rapid opening of voltage-gated Na⁺ channels. Sodium ions then flow into the neuron down their electrochemical gradient, making the membrane potential progressively less negative and eventually positive. Voltage-gated K⁺ channels open more slowly and are mainly responsible for repolarization. Thus, the key event producing depolarization is the opening of voltage-gated Na⁺ channels, making A correct.

17. Which one of the following is wrong about the action potential of a nerve fiber?

- A. It is caused by a transient membrane depolarization**
- B. It needs a threshold stimulus to be generated**
- C. It has a definite RMP**
- D. It is all-or-none in its nature**
- E. None of the above**

Answer: E. None of the above

Explanation:

All statements A–D are correct characteristics of an action potential. An action potential is a rapid, transient change in membrane potential involving depolarization followed by repolarization and often a brief hyperpolarization. It is generated when the membrane reaches the threshold potential. The neuron has a characteristic resting membrane potential (RMP) before stimulation. Once threshold is reached, the action potential follows the all-or-none principle: its amplitude does not increase simply because the stimulus becomes stronger. Therefore, there is no incorrect statement among A–D, so E. None of the above is correct.

18. Fast rate of conduction of nerve impulse is achieved in:

- A. Type A nerve fibers**
- B. Type B nerve fibers**

- C. Type C nerve fibers**
 - D. Autonomic nerve fibers**
 - E. Unmyelinated somatic nerve fibers**
- Answer: A. Type A nerve fibers**

Explanation:

Type A nerve fibers are generally the fastest-conducting nerve fibers because they are large in diameter and heavily myelinated. Myelin allows action potentials to travel by saltatory conduction, jumping rapidly from one node of Ranvier to the next. Type B fibers are smaller and conduct more slowly, while type C fibers are small and unmyelinated and therefore have the slowest conduction velocities. Thus, among the choices, A. Type A nerve fibers provides the fastest conduction.

19. In the peripheral nerve fibers, myelin sheath is produced by:

- A. Schwann cells**
 - B. Oligodendrocytes**
 - C. Microglial cells**
 - D. Ependymal cells**
 - E. Astrocytes**
- Answer: A. Schwann cells**

Explanation:

In the peripheral nervous system (PNS), myelin is produced by Schwann cells. Each Schwann cell can wrap its membrane repeatedly around a segment of a peripheral axon, creating a myelin sheath that electrically insulates the axon and greatly increases the speed of impulse conduction. In the central nervous system (CNS), myelin is produced by oligodendrocytes. Microglia function mainly in immune defense, ependymal cells line the ventricles and contribute to CSF-related functions, and astrocytes perform numerous support and homeostatic functions. Therefore, A is correct.

[24/09/2026 9:01 AM] Imaanaa Harmee: **20. Which one of the following is wrong about the neuromuscular junction of skeletal muscle?**

- A. The neurotransmitter is always acetylcholine**
 - B. The post-synaptic receptor is always nicotinic receptor**
 - C. Excitatory post-synaptic potential is always produced**
 - D. IPSP causes contraction**
 - E. All are correct**
- Answer: D. IPSP causes contraction**

Explanation:

At the skeletal neuromuscular junction, the motor neuron releases acetylcholine (ACh), which binds to nicotinic acetylcholine receptors on the motor end plate. This produces an excitatory end-plate potential that depolarizes the muscle membrane and initiates a muscle action potential, ultimately causing skeletal muscle contraction. An IPSP (inhibitory postsynaptic potential) does not cause contraction; rather, inhibitory potentials reduce the likelihood of generating an action potential.

Therefore, D. IPSP causes contraction is the incorrect statement and is the correct answer.

[24/09/2026 9:01 AM] Imaanaa Harmee: **21. An action potential arriving at the neuromuscular junction causes opening of:**

- A. Cl^- channels**
- B. Na^+ channels**
- C. Ca^{2+} channels**
- D. K^+ channels**
- E. H^+ channels**

Answer: C. Ca^{2+} channels

Explanation:

When an action potential reaches the presynaptic terminal of a neuromuscular junction, it depolarizes the membrane and opens voltage-gated Ca^{2+} channels. Calcium ions then enter the motor neuron terminal from the extracellular fluid. The increased intracellular Ca^{2+} concentration triggers fusion of synaptic vesicles with the presynaptic membrane and causes the release of acetylcholine (ACh) into the synaptic cleft. ACh subsequently binds to nicotinic receptors on the muscle membrane, producing depolarization and initiating muscle contraction. Therefore, the immediate channel opened by the arriving action potential is the Ca^{2+} channel, making C correct.

22. A drug that blocks the neuromuscular junction of skeletal muscle is:

- A. Neostigmine**
- B. Carbachol**
- C. Physostigmine**
- D. D-tubocurarine**
- E. Acetylcholine**

Answer: D. D-tubocurarine

Explanation:

D-tubocurarine is a classic competitive antagonist of nicotinic acetylcholine receptors at the skeletal neuromuscular junction. By occupying these receptors, it prevents acetylcholine from producing the normal end-plate depolarization, resulting in skeletal muscle relaxation or paralysis. In contrast, neostigmine and physostigmine inhibit acetylcholinesterase, increasing the amount of ACh available at synapses. Carbachol and acetylcholine can stimulate cholinergic receptors. Therefore, D-tubocurarine is the drug that blocks neuromuscular transmission.

23. Which of the following is not involved in the homeostasis of glucose?

- A. Intestine**
- B. Kidneys**
- C. Brain**

D. Pancreas

E. Lung

Answer: E. Lung

Explanation:

Blood glucose homeostasis involves several organs and tissues that regulate glucose absorption, storage, utilization, and production. The intestine absorbs dietary glucose; the pancreas secretes insulin and glucagon to regulate blood glucose; the liver is a major site of glucose storage and production; the kidneys contribute to glucose reabsorption and gluconeogenesis; and the brain is a major consumer of glucose. The lungs are not a major organ involved in regulating blood glucose concentration. Therefore, E. Lung is the best answer.

24. The function of T-tubules in skeletal muscle fibers is:

A. Storage of calcium ion

B. Conduction of depolarization waves to the sarcoplasmic reticulum

C. Contain postsynaptic receptors

D. Storage of neurotransmitters

E. None of the above

Answer: B. Conduction of depolarization waves to the sarcoplasmic reticulum

Explanation:

T-tubules (transverse tubules) are deep invaginations of the skeletal muscle cell membrane, or sarcolemma. Their major function is to rapidly carry the action potential from the surface of the muscle fiber deep into its interior. This depolarization reaches the triads, where T-tubules lie next to the terminal cisternae of the sarcoplasmic reticulum. It activates voltage-sensitive proteins that trigger the release of Ca^{2+} from the sarcoplasmic reticulum, initiating contraction. The sarcoplasmic reticulum, rather than the T-tubules, is the major calcium store. Thus, B is correct.

25. One of the following is specific to fast muscle fibers EXCEPT:

A. Depends on aerobic metabolism

B. Contains large number of sarcomeres

C. Rich in capillary supply

D. Has multiple substrates

E. Contains a lot of myoglobin

Answer: A. Depends on aerobic metabolism

Explanation:

Fast muscle fibers, particularly fast glycolytic fibers, are specialized for rapid and powerful contractions. They rely heavily on anaerobic glycolysis and stored phosphagens rather than depending primarily on aerobic metabolism. They generally have fewer mitochondria, less myoglobin, and lower capillary density than slow oxidative fibers. In contrast, slow-twitch fibers are characterized by abundant mitochondria, extensive capillary supply, high myoglobin content, and strong dependence on aerobic metabolism. Therefore, A is the exception for fast fibers.

muscle are:

- A. Actin and troponin**
 - B. Actin and myosin**
 - C. Troponin and tropomyosin**
 - D. Myosin and tropomyosin**
 - E. Actin and tropomyosin**
- Answer: B. Actin and myosin**

Explanation:

The two principal contractile proteins of skeletal muscle are actin and myosin. Actin forms the thin filaments, while myosin forms the thick filaments. During contraction, myosin heads attach to active sites on actin and pull the thin filaments toward the center of the sarcomere. Troponin and tropomyosin are important regulatory proteins, because they control whether myosin can bind to actin in response to Ca^{2+} . Therefore, B. Actin and myosin is correct.

27. The contractile unit of skeletal muscle is:

- A. The sarcomere**
- B. The A-band**
- C. The I-band**
- D. The Z-lines**
- E. The H-bands**

Answer: A. The sarcomere

Explanation:

The sarcomere is the basic structural and functional contractile unit of skeletal and cardiac muscle. It extends from one Z disc to the next Z disc. Within the sarcomere are the thick myosin filaments and thin actin filaments arranged in an organized pattern. During contraction, the thin filaments slide toward the center of the sarcomere, causing the sarcomere to shorten. The A-band, I-band, and H-zone are regions within the sarcomere rather than the entire contractile unit. Therefore, A is correct.

28. Which of the following statements is wrong about muscle tissue?

- A. Cardiac and skeletal muscles have a striated appearance**
- B. Cardiac and smooth muscles are involuntary**
- C. Skeletal and cardiac muscles are innervated by the ANS**
- D. Cardiac and smooth muscles are innervated by ANS**
- E. Cardiac and smooth muscles are regulated by hormones and drugs**

Answer: C. Skeletal and cardiac muscles are innervated by the ANS

Explanation:

Skeletal muscle is primarily controlled by the somatic nervous system, not the autonomic nervous system. It can be voluntarily controlled, although many skeletal muscle activities can also occur through reflex mechanisms. Cardiac and smooth muscle, on the other hand, are involuntary and are regulated by the autonomic nervous system, hormones, local factors, and various drugs. Skeletal and cardiac muscle are both striated, whereas smooth muscle lacks visible striations. Therefore, C is the incorrect statement.

29. Which of the following events is energy independent?

- A. Release of Ca^{2+} from the sarcoplasmic reticulum**
- B. Return of Ca^{2+} back to the sarcoplasmic reticulum**
- C. Movement of myosin head to the center**
- D. Detachment of myosin head from actin**
- E. Final ionic rearrangement**

Answer: A. Release of Ca^{2+} from the sarcoplasmic reticulum

Explanation:

The release of Ca^{2+} from the sarcoplasmic reticulum occurs through Ca^{2+} release channels and does not directly require ATP. Calcium moves down its electrochemical gradient from the SR into the cytoplasm. In contrast, returning Ca^{2+} to the sarcoplasmic reticulum requires ATP through the SERCA calcium pump. ATP is also required during the cross-bridge cycle: ATP binding allows myosin to detach from actin, while ATP hydrolysis energizes the myosin head. Therefore, A is the energy-independent event.

30. Which of the following cranial nerves is stimulatory to the salivary glands?

- A. Oculomotor**
- B. Facial**
- C. Vagus**
- D. Trigeminal**
- E. Hypoglossal**

Answer: B. Facial

Explanation:

The facial nerve (CN VII) carries important parasympathetic secretomotor fibers to the submandibular and sublingual salivary glands. These fibers originate from the superior salivatory nucleus and travel through branches of the facial nerve before reaching the glands. Parasympathetic stimulation generally produces a copious, watery salivary secretion. The glossopharyngeal nerve (CN IX) also provides parasympathetic stimulation to the parotid gland, but it is not among the options. Therefore, among the choices given, B. Facial nerve is correct.

[24/09/2026 9:01 AM] Imaanaa Harmee: **31. Sympathetic stimulation on the cardiovascular system produces**

- A. Increased heart rate**
- B. Increased force of contraction of the cardiac muscle**
- C. Vasoconstriction of the systemic blood vessels**
- D. Increases cardiac output**
- E. All are correct**

Answer: E. All are correct

Explanation:

Sympathetic stimulation produces several cardiovascular effects simultaneously. It increases heart rate and the force of cardiac contraction mainly through β_1 -adrenergic

receptors in the heart. It also causes constriction of many systemic blood vessels through α_1 receptors, increasing peripheral resistance. The combination of increased heart rate and increased contractility raises cardiac output. Therefore, A, B, C, and D are all correct, making E the best answer.

32. The receptor that is found on the coronary arteries is

- A. Alpha 1
- B. Beta 1
- C. Alpha 2
- D. Beta 2
- E. Beta 3

Answer: D. Beta 2

Explanation:

β_2 -adrenergic receptors are important receptors in vascular smooth muscle, including the coronary circulation. Their activation produces vasodilation, helping increase blood flow. During sympathetic stimulation, the heart becomes more active and requires more oxygen, so coronary blood flow must increase. Although coronary blood flow is strongly influenced by local metabolic factors, β_2 -mediated vasodilation also contributes. β_1 receptors are primarily associated with the heart, where they increase heart rate and contractility. Therefore, D. Beta 2 is the expected answer.

33. Of the following, one does not receive dual autonomic innervation. Identify.

- A. The heart
 - B. The blood vessels
 - C. The liver
 - D. The stomach
 - E. Pancreas
- Answer: B. The blood vessels

Explanation:

Many visceral organs receive both sympathetic and parasympathetic autonomic innervation. The heart, stomach, liver, and pancreas receive influences from both divisions. Most systemic blood vessels, however, are predominantly controlled by the sympathetic nervous system and have little or no direct parasympathetic innervation. Sympathetic activity can cause vasoconstriction or, in certain vascular beds, contribute to vasodilation. Therefore, among these options, B. Blood vessels is the best answer.

34. A neurotransmitter produced by a postganglionic sympathetic fiber to the sweat glands is

- A. Acetylcholine
- B. Adrenaline
- C. Noradrenaline
- D. Dopamine
- E. All

Answer: A. Acetylcholine

Explanation:

The usual rule is that postganglionic sympathetic neurons release norepinephrine (noradrenaline). However, sweat glands provide an important exception. Postganglionic sympathetic fibers supplying eccrine sweat glands release acetylcholine, which acts primarily on muscarinic receptors and stimulates sweating. Thus, although these fibers are sympathetic, their neurotransmitter is acetylcholine rather than norepinephrine. Therefore, A is correct.

35. Which one of the following is wrong about neuronal activities of the ANS?

- A. All preganglionic fibers are cholinergic**
- B. Postganglionic parasympathetic fibers are cholinergic**
- C. Preganglionic parasympathetic fibers are cholinergic**
- D. Most postganglionic sympathetic fibers are adrenergic**
- E. None of the above**

Answer: E. None of the above

Explanation:

All autonomic preganglionic neurons are cholinergic, meaning they release acetylcholine at the autonomic ganglia. Parasympathetic postganglionic neurons are also generally cholinergic and release acetylcholine onto their target organs. Most sympathetic postganglionic neurons are adrenergic, releasing norepinephrine. The important exception is sympathetic innervation of sweat glands, where acetylcholine is released. Therefore, statements A–D are correct, so E. None of the above is the appropriate answer.

[24/09/2026 9:01 AM] Imaanaa Harmee: **36. Which one of the following is an acetylcholine drug?**

- A. Propranolol**
- B. Metoprolol**
- C. Atropine**
- D. Neostigmine**
- E. Pilocarpine**

Answer: E. Pilocarpine

Explanation:

Pilocarpine is a direct-acting muscarinic cholinergic agonist. It mimics some actions of acetylcholine by stimulating muscarinic receptors, particularly increasing glandular secretions such as saliva and tears. Neostigmine does not directly act as acetylcholine; instead, it inhibits acetylcholinesterase, thereby preventing the breakdown of acetylcholine and increasing its concentration. Propranolol and metoprolol are β -adrenergic blockers, while atropine is a muscarinic antagonist. Therefore, E. Pilocarpine is correct.

37. Which of the following is not a sympathomimetic drug?

- A. Epinephrine**
 - B. Norepinephrine**
 - C. Methoxamine**
 - D. Pilocarpine**
 - E. None of the above**
- Answer: D. Pilocarpine**

Explanation:

Sympathomimetic drugs mimic the effects of sympathetic stimulation, usually by activating adrenergic receptors. Epinephrine and norepinephrine are classic sympathomimetic agents, while methoxamine is primarily an α_1 -adrenergic agonist. In contrast, pilocarpine is a parasympathomimetic drug because it is a muscarinic receptor agonist and therefore mimics parasympathetic activity. It increases salivation, lacrimation, and other cholinergic effects. Thus, D. Pilocarpine is the drug that is not sympathomimetic.

38. Which cranial nerve is responsible for corneal sensitivity?

- A. Glossopharyngeal nerve**
 - B. Trochlear nerve**
 - C. Trigeminal nerve**
 - D. Abducens nerve**
 - E. Oculomotor nerve**
- Answer: C. Trigeminal nerve**

Explanation:

Corneal sensation is carried primarily by the ophthalmic division (V1) of the trigeminal nerve, cranial nerve V. Sensory fibers from the cornea travel through the nasociliary branch of V1 and ultimately reach the trigeminal sensory nuclei in the brainstem. Touching the cornea normally produces the corneal reflex, in which sensory input travels through CN V and the motor response is carried mainly by the facial nerve (CN VII), producing blinking. Therefore, C. Trigeminal nerve is correct.

39. Mechanism of action of acetylcholine when binding with M1 is via

- A. Formation of IP_3 and DAG**
 - B. Activation of adenylyl cyclase**
 - C. Hyperpolarization**
 - D. Opening Na^+ channels**
 - E. Inhibition of Ca^{2+} channels**
- Answer: A. Formation of IP_3 and DAG**

Explanation:

The M1 muscarinic receptor is a Gq-coupled receptor. When acetylcholine binds to M1, it activates the Gq protein, which stimulates phospholipase C (PLC). PLC breaks down membrane phospholipid PIP_2 into the second messengers IP_3 and DAG. IP_3 promotes

release of Ca^{2+} from intracellular stores, while DAG activates protein kinase C. This intracellular signaling produces the physiological response associated with M1 receptors. Therefore, A. Formation of IP_3 and DAG is correct.

40. If your tongue is atrophied, there is a lesion in which cranial nerve?

- A. Facial**
- B. Glossopharyngeal**
- C. Vagus**
- D. Hypoglossal**
- E. Trigeminal**

Answer: D. Hypoglossal

Explanation:

The hypoglossal nerve (CN XII) provides the major motor innervation to the muscles of the tongue. Damage to this nerve causes weakness and, when the lesion is significant or chronic, atrophy of the tongue muscles on the affected side. With a unilateral lower motor neuron lesion of CN XII, the tongue typically deviates toward the side of the lesion when protruded, because the intact side pushes the tongue toward the weak side. Facial, glossopharyngeal, vagus, and trigeminal nerves have important functions in the head and neck but do not provide the primary motor supply to the tongue muscles. Therefore, D. Hypoglossal is correct.

[24/09/2026 9:01 AM] Imaanaa Harmee: **41. The work of breathing decreases when:**

- A. Lung compliance increases.**
- B. The subject exercises.**
- C. The rate of breathing increases with constant minute volume.**
- D. The person lies down.**
- E. Functional residual capacity increases.**

Answer: A. Lung compliance increases.

Explanation:

Lung compliance is the ease with which the lungs can expand for a given change in pressure. When compliance increases, the lungs expand more easily, so the respiratory muscles require less effort to produce a given tidal volume. Consequently, the work of breathing decreases. In contrast, exercise increases ventilation and therefore generally increases respiratory work. Thus, A is correct.

42. Lack of surfactant substances as occurs in infants with respiratory distress syndrome (RDS) results in:

- A. An increased lung compliance**
- B. Stabilization of alveolar volume**
- C. Lung collapse**
- D. A reduced alveolar-arterial O_2 tension difference**
- E. A decrease in the filtration forces in the pulmonary capillaries**

Answer: C. Lung collapse

Explanation:

Pulmonary surfactant is produced mainly by type II pneumocytes and reduces the surface tension of the fluid lining the alveoli. This prevents the alveoli from collapsing, particularly at the end of expiration. In premature infants with respiratory distress syndrome, inadequate surfactant causes increased alveolar surface tension, making the lungs harder to expand and promoting alveolar collapse (atelectasis). Lung compliance therefore decreases rather than increases. The resulting ventilation-perfusion abnormalities can also produce hypoxemia. Hence, C is correct.

43. The volumes and capacities that cannot be measured by direct spirometry include:

- A. Residual volume, expiratory reserve volume and inspiratory capacity
- B. Residual volume, functional residual capacity and total lung capacity
- C. Inspiratory reserve volume, residual volume, vital capacity
- D. Total lung capacity, residual volume and expiratory reserve volume
- E. Tidal volume, inspiratory reserve volume and inspiratory capacity

Answer: B. Residual volume, functional residual capacity and total lung capacity

Explanation:

Simple spirometry can measure volumes of air that can be moved into or out of the lungs, such as tidal volume, inspiratory reserve volume, and expiratory reserve volume. However, it cannot directly measure residual volume (RV) because residual air remains in the lungs and cannot be voluntarily exhaled. Any capacity that includes residual volume also cannot be measured directly by simple spirometry. These include functional residual capacity ($FRC = ERV + RV$) and total lung capacity ($TLC = VC + RV$). Therefore, B is correct.

44. The total lung capacity consists of:

- A. Vital capacity & tidal volume
- B. Residual volume & vital capacity
- C. Functional residual capacity & vital capacity
- D. Tidal volume & functional residual capacity
- E. Tidal volume & residual volume

Answer: B. Residual volume & vital capacity

Explanation:

Total lung capacity (TLC) is the maximum amount of air contained in the lungs after a maximal inspiration. It consists of all four basic lung volumes: tidal volume, inspiratory reserve volume, expiratory reserve volume, and residual volume. Since vital capacity consists of tidal volume + inspiratory reserve volume + expiratory reserve volume, adding residual volume gives TLC: $TLC = VC + RV$. Therefore, B is correct.

45. All of the following events occur during expiration except:

- A. Inspiratory muscles relax and the rib cage descends due to gravity
- B. Thoracic cavity volume decreases
- C. Intrapulmonary volume decreases
- D. Intrapulmonary pressure rises above atmospheric pressure (+1 mm Hg)
- E. Gases flow out of the lungs against the pressure gradient

Answer: E. Gases flow out of the lungs against the pressure gradient

Explanation:

During normal expiration, the inspiratory muscles relax, and elastic recoil of the lungs and chest wall decreases thoracic volume. As lung volume decreases, intrapulmonary (alveolar) pressure rises slightly above atmospheric pressure.

[24/09/2026 9:01 AM] Imaanaa Harmee: **46. Defense function of the respiratory system is:**

- A. Gas moistening
- B. Ciliary beating
- C. Warming of the air
- D. Smell
- E. Secretion of ACE

Answer: B. Ciliary beating

Explanation:

The respiratory tract has several protective mechanisms that prevent inhaled particles and microorganisms from reaching the lower respiratory system. Ciliary beating is an important component of the mucociliary clearance system. Cilia move mucus containing trapped dust, bacteria, and other particles toward the pharynx, where it can be swallowed or expelled. Warming and moistening of air are primarily conditioning functions, while smell is an olfactory function. Therefore, B is the best answer.

47. Which structure does not belong to the respiratory portion of the breathing organ?

- A. Terminal bronchioles
- B. Respiratory bronchioles
- C. Alveolar ducts
- D. Alveolar sacs
- E. None of the above

Answer: A. Terminal bronchioles

Explanation:

The respiratory portion of the respiratory tract is the region where gas exchange can occur. It begins with the respiratory bronchioles and continues through the alveolar ducts, alveolar sacs, and alveoli. Terminal bronchioles belong to the conducting zone, not the respiratory zone. They conduct air but do not normally participate directly in gas exchange. Therefore, A. Terminal bronchioles is correct.

48. Bronchial obstruction is likely to be relieved by:

- A. Stimulation of cholinergic receptors
- B. Stimulation of beta adrenoceptors

- C. Histamine
 - D. Stimulation of mast cell membranes
 - E. Slow reactive substance of anaphylaxis
- Answer: B. Stimulation of beta adrenoceptors

Explanation:

Stimulation of β_2 -adrenergic receptors in bronchial smooth muscle causes bronchodilation. This relaxes the airway smooth muscle and increases the diameter of the bronchi, helping relieve bronchial obstruction. In contrast, cholinergic stimulation causes bronchoconstriction, while histamine and leukotrienes associated with allergic reactions also promote bronchoconstriction. Therefore, B is correct.

49. Respiratory function of the respiratory system is:

- A. Acid-base balance
 - B. Air conditioning
 - C. Phonation
 - D. Gas conduction
 - E. Olfaction
- Answer: D. Gas conduction

Explanation:

The respiratory system has several functions, including gas exchange, acid-base regulation, defense, vocalization, and olfaction. Gas conduction is a direct respiratory function because the conducting passages transport air between the atmosphere and the gas-exchange surfaces of the lungs. Air conditioning, phonation, and olfaction are specialized functions of the respiratory tract, while acid-base balance is a major regulatory function involving CO_2 elimination. Given the options, D. Gas conduction is the intended answer.

50. The amount of fluid filtered per minute in all nephrons of both kidneys is called:

- A. Glomerular filtration rate
 - B. Filtration fraction
 - C. Filtration pressure
 - D. Renal blood flow
 - E. Tubular transport maximum
- Answer: A. Glomerular filtration rate (GFR)

Explanation:

The glomerular filtration rate (GFR) is the volume of fluid filtered from the glomerular capillaries into Bowman's capsules per unit time by both kidneys. In a healthy adult, the normal GFR is approximately 125 mL/min, although it varies with age, sex, and body size. Filtration fraction is the proportion of renal plasma flow that becomes filtrate, whereas renal blood flow refers to the amount of blood supplied to the kidneys. Therefore, the definition given corresponds directly to A. GFR.

[24/09/2026 9:01 AM] Imaanaa Harmee: **51. The net pressure in the glomerular capillaries that pushes fluid outward is called:**

- A. Glomerular capillary pressure**
- B. Glomerular colloid osmotic capillary pressure**
- C. Bowman's capsular pressure**
- D. Filtration pressure**
- E. Mean systemic capillary filling pressure**

Answer: D. Filtration pressure

Explanation:

The net filtration pressure (NFP) is the net force that drives fluid from the glomerular capillaries into Bowman's capsule. It results from the balance between forces favoring filtration and forces opposing it. Glomerular capillary hydrostatic pressure favors filtration, while Bowman's capsular hydrostatic pressure and glomerular plasma colloid osmotic pressure oppose it. In simplified form, $NFP = \text{glomerular hydrostatic pressure} - (\text{Bowman's capsular hydrostatic pressure} + \text{glomerular oncotic pressure})$. Therefore, the pressure responsible for the net outward filtration force is filtration pressure, making D correct.

52. Part of the renal tubules where nutritionally important substances are reabsorbed is called:

- A. Collecting duct**
- B. Proximal tubules**
- C. Distal tubules**
- D. Loop of Henle**
- E. Urinary bladder**

Answer: B. Proximal tubules

Explanation:

The proximal tubule is the major site of reabsorption in the nephron. It reabsorbs most of the filtered water and sodium, as well as essentially all filtered glucose and amino acids under normal conditions. It also reabsorbs substantial amounts of bicarbonate and other electrolytes. This extensive reabsorptive function ensures that valuable nutrients and substances are not unnecessarily lost in urine. Therefore, B. Proximal tubules is correct.

53. The renal threshold for glucose is equal to:

- A. 340 mg/mL**
- B. 200 mg/dL**
- C. 125 mL/min**
- D. 80–120 mg/dL**
- E. 40 mg/dL**

Answer: B. 200 mg/dL

Explanation:

The renal threshold for glucose is the plasma glucose concentration at which glucose begins to appear in the urine because the filtered load starts to exceed the reabsorptive capacity of the proximal tubules. It is commonly taught as approximately 180–200 mg/dL, with 200 mg/dL being the best choice provided here. Below this level, virtually all filtered glucose is normally reabsorbed. Once the transport maximum is approached or exceeded, glucose begins to be excreted, producing glucosuria. Therefore, B is the best answer.

54. Renal tubules upon which ADH and aldosterone act to promote water-electrolyte reabsorption are:

- A. Distal tubules and collecting duct**
- B. Proximal tubules**
- C. Loop of Henle**
- D. Glomerulus**
- E. Afferent and efferent arterioles**

Answer: A. Distal tubules and collecting duct

Explanation:

ADH (vasopressin) acts primarily on the late distal tubule and collecting ducts, increasing their permeability to water by promoting insertion of aquaporin-2 channels into the apical membrane. This allows more water to be reabsorbed and produces concentrated urine. Aldosterone acts mainly on principal cells in the late distal tubule and collecting duct, increasing sodium reabsorption and potassium secretion. Thus, these portions of the nephron are major targets for hormonal regulation of water and electrolyte balance. Therefore, A is correct.

55. Which factor increases glomerular filtration rate?

- A. Afferent arteriole constriction**
- B. Efferent arteriole vasodilation**
- C. Decrease in filtration pressure**
- D. Congestive heart failure**
- E. Increase in the renal blood flow**

Answer: E. Increase in the renal blood flow

Explanation:

An increase in renal blood flow, particularly when associated with increased glomerular capillary pressure, can increase the amount of fluid available for filtration and therefore increase GFR. Afferent arteriolar constriction reduces glomerular pressure and decreases GFR. Efferent arteriolar dilation also tends to reduce glomerular capillary hydrostatic pressure and GFR. Congestive heart failure commonly reduces effective renal perfusion. Therefore, among the choices, E is the best answer.

[24/09/2026 9:01 AM] Imaanaa Harmee: **56. Which of the following is true regarding control of GFR and RBF?**

- A. Strong activation of sympathetic NS causes marked vasoconstriction of efferent arterioles**
- B. Angiotensin II causes efferent arteriolar vasoconstriction**
- C. Angiotensin II increases flow through peritubular capillaries**
- D. ANP, released by atria when stretched, decreases GFR**
- E. Endothelin increases GFR**

Answer: B. Angiotensin II causes efferent arteriolar vasoconstriction

Explanation:

Angiotensin II preferentially constricts the efferent arteriole, particularly at moderate concentrations. This helps maintain glomerular capillary hydrostatic pressure and therefore helps preserve GFR when systemic blood pressure or renal perfusion is reduced. Strong sympathetic activation primarily causes significant afferent arteriolar constriction, reducing renal blood flow and GFR. Angiotensin II also tends to reduce peritubular capillary hydrostatic pressure and increase peritubular oncotic pressure, favoring tubular reabsorption rather than increasing peritubular flow. ANP generally promotes increased GFR, while endothelin is a vasoconstrictor that tends to reduce renal blood flow and GFR. Thus, B is correct.

57. Which of the following factors is NOT in favor of GFR?

- A. Increased capillary hydrostatic pressure of the glomerulus**
- B. Decreased capsular hydrostatic pressure**
- C. Decreased glomerular oncotic pressure**
- D. Efferent arteriolar vasoconstriction**
- E. Afferent arteriolar vasoconstriction**

Answer: E. Afferent arteriolar vasoconstriction

Explanation:

Afferent arteriolar constriction reduces blood flow into the glomerulus and lowers glomerular capillary hydrostatic pressure. As a result, the filtration pressure and GFR decrease. In contrast, increased glomerular capillary hydrostatic pressure favors filtration, while decreased Bowman's capsular pressure and decreased plasma oncotic pressure reduce opposition to filtration. Moderate efferent arteriolar constriction can increase glomerular capillary pressure and initially increase GFR. Therefore, E is the factor that is not in favor of GFR.

58. Bowman's capsular hydrostatic pressure is because of:

- A. Renal blood flow**
- B. Glomerular oncotic pressure**
- C. Glomerular hydrostatic pressure**
- D. Glomerular filtrate**
- E. Sympathetic stimulation**

Answer: D. Glomerular filtrate

Explanation:

Bowman's capsule surrounds the glomerular capillaries and receives the fluid filtered from the blood. The accumulation and movement of glomerular filtrate within

Bowman's capsule generate hydrostatic pressure inside the capsule. This pressure acts against filtration, opposing the hydrostatic pressure within the glomerular capillaries. Therefore, the best answer is D. Glomerular filtrate.

59. Which factor inhibits motility of the GIT?

- A. Sympathetic stimulation**
- B. Parasympathetic stimulation**
- C. Myenteric nerve plexus**
- D. Hormone gastrin**
- E. Motilin**

Answer: A. Sympathetic stimulation

Explanation:

The sympathetic nervous system generally inhibits gastrointestinal motility and secretion. Sympathetic stimulation reduces activity of the enteric nervous system and causes contraction of many gastrointestinal sphincters, while decreasing the contractile activity of the intestinal wall. In contrast, parasympathetic stimulation generally promotes gastrointestinal motility and secretion. The myenteric plexus is a major intrinsic regulator of gut motility, while gastrin and motilin generally promote gastrointestinal motor activity. Therefore, A is correct.

[24/09/2026 9:01 AM] Imaanaa Harmee: **60. Ptyalin (amylase) is produced by:**

- A. The parotid gland**
- B. Parietal cells**
- C. Chief cells**
- D. Enteroendocrine cells**
- E. Goblet cells**

Answer: A. The parotid gland

Explanation:

Ptyalin is salivary α -amylase, an enzyme that begins the digestion of carbohydrates in the mouth by breaking down starch into smaller carbohydrates. It is secreted by the salivary glands, with the parotid gland producing a particularly large amount of watery, enzyme-rich saliva. Parietal cells secrete HCl and intrinsic factor, chief cells secrete pepsinogen, enteroendocrine cells produce hormones, and goblet cells produce mucus. Therefore, A. The parotid gland is correct.

[24/09/2026 9:01 AM] Imaanaa Harmee: **61. Secretion of HCl by parietal cells in the stomach is increased by:**

- A. Parasympathetic stimulation**
- B. Drinking coffee and alcohol**
- C. Release of histamine by mast cells**
- D. Infection with Helicobacter pylori**
- E. All are correct**

Answer: E. All are correct.

Explanation:

Parietal cells secrete hydrochloric acid (HCl), and their activity is stimulated by several mechanisms. Parasympathetic stimulation, mainly through acetylcholine, directly stimulates parietal cells. Histamine, released from enterochromaffin-like (ECL) cells in the stomach, acts on H₂ receptors on parietal cells and strongly increases HCl secretion. Coffee and alcohol can also stimulate gastric acid secretion. Helicobacter pylori infection can alter gastric physiology and, particularly in antral-predominant infection, increase gastrin release and acid secretion. Therefore, among the choices given, E is the best answer.

62. The major stimulus for the release of secretin hormone is:

- A. Protein digestion products**
- B. Histamine**
- C. Somatostatin**
- D. Acidic chyme in the duodenum**
- E. Fat digestion products**

Answer: D. Acidic chyme in the duodenum.

Explanation:

Secretin is produced by S cells in the duodenal mucosa. Its major stimulus is the arrival of acidic chyme from the stomach into the duodenum. The low pH stimulates S cells to release secretin into the blood. Secretin then acts mainly on the pancreas and biliary system to increase bicarbonate-rich secretion, which neutralizes the acid and protects the intestinal mucosa. Protein and fat digestion products are more important stimuli for CCK release, whereas histamine stimulates gastric acid secretion.

63. GIT layers that contains much of glands is:

- A. Serosa**
- B. Muscularis**
- C. Submucosa**
- D. Mucosa**
- E. All**

Answer: C. Submucosa.

Explanation:

The gastrointestinal tract has four major layers: mucosa, submucosa, muscularis externa, and serosa/adventitia. The submucosa contains important blood vessels, lymphatic vessels, nerves, and glands. A classic example is Brunner's glands in the duodenum, which are located in the submucosa and secrete alkaline mucus that helps protect the duodenal mucosa from gastric acid. Although glands can also occur in the mucosa, the question is specifically referring to the layer containing prominent submucosal glands, making C the expected answer.

64. Center for chewing is located in the:

- A. Medulla**
- B. Pons**

- C. Midbrain
- D. Spinal cord
- E. Cerebrum

Answer: B. Pons.

Explanation:

The chewing (mastication) center is located primarily in the pons, where it coordinates the rhythmic movements of the muscles of mastication. The motor nucleus of the trigeminal nerve (cranial nerve V), located in the pons, supplies the major muscles involved in chewing. Sensory information from the oral cavity helps regulate the chewing reflex. Although higher brain centers can initiate voluntary chewing, the basic rhythmic chewing pattern is coordinated by brainstem circuits.

65. Which of the following food items is chemically digested in the mouth?

- A. Protein
- B. Fat
- C. CHO
- D. Vitamin
- E. Mineral

Answer: C. CHO (carbohydrate).

Explanation:

Chemical digestion begins in the mouth primarily with carbohydrates. Salivary glands secrete salivary amylase (ptyalin), which begins breaking down starch and glycogen into smaller carbohydrates such as maltose and other oligosaccharides. Although lingual lipase is also secreted in the oral cavity, significant fat digestion occurs later, particularly in the stomach and small intestine. Vitamins and minerals do not require enzymatic chemical digestion, while substantial protein digestion begins in the stomach through pepsin.

[24/09/2026 9:01 AM] Imaanaa Harmee: **66. Which of the following is correct regarding GIT regulation?**

- A. Parasympathetic nervous system has excitatory effect on GI function.
- B. Salivary secretion is mainly regulated by gastric hormone.
- C. Sympathetic nerves have excitatory effect on the GI functions.
- D. Secretin hormone inhibits pancreatic secretions.
- E. None of the above.

Answer: A. Parasympathetic nervous system has excitatory effect on GI function.

Explanation:

The parasympathetic nervous system generally stimulates gastrointestinal activity. It increases gastrointestinal motility, secretion, and blood flow and promotes digestion. The vagus nerve is particularly important for parasympathetic regulation of the upper gastrointestinal tract. In contrast, sympathetic stimulation generally inhibits gastrointestinal motility and secretion and constricts gastrointestinal blood vessels. Salivary secretion is primarily controlled by autonomic nervous reflexes rather than gastric hormones, and secretin actually stimulates pancreatic bicarbonate secretion, so B, C, and D are incorrect.

67. Identify an enzyme secreted by the chief cells of the stomach:

- A. HCl**
 - B. Pepsin**
 - C. Mucus**
 - D. Intrinsic factor**
 - E. None of the above**
- Answer: B. Pepsin.**

Explanation:

Chief cells of the gastric glands secrete pepsinogen, the inactive precursor of pepsin. When pepsinogen enters the acidic environment of the stomach, HCl converts it into active pepsin, which begins protein digestion. Strictly speaking, chief cells secrete pepsinogen rather than active pepsin, but because the question identifies pepsin as the enzyme associated with chief-cell secretion, B is the intended answer. Parietal cells secrete HCl and intrinsic factor, while mucous cells produce mucus.

68. An endocrine function of the stomach is the fact that it secretes:

- A. HCl**
 - B. Pepsin**
 - C. Intrinsic factor**
 - D. Mucus**
 - E. Gastrin**
- Answer: E. Gastrin.**

Explanation:

The stomach has an endocrine function because its G cells secrete the hormone gastrin into the bloodstream. Gastrin stimulates gastric acid secretion, promotes gastric mucosal growth, and enhances gastric motility. The other substances listed are primarily exocrine or secretory products: HCl and intrinsic factor are produced by parietal cells, pepsinogen by chief cells, and mucus by mucous cells. Therefore, gastrin is the endocrine secretion of the stomach.

69. The hardest part of the teeth is:

- A. Crown**
 - B. Dentin**
 - C. Enamel**
 - D. Root**
 - E. None**
- Answer: C. Enamel.**

Explanation:

Enamel is the hardest substance in the human body. It forms the outer protective covering of the crown of the tooth and is composed mainly of highly mineralized hydroxyapatite crystals. Its high mineral content gives it exceptional hardness and resistance to mechanical forces during chewing. Dentin lies beneath the enamel and is

softer, while the root is covered by cementum. The crown and root describe anatomical regions rather than specific dental tissues, so enamel is the correct answer.

70. A hormone that causes release of bile from the gall bladder is:

- A. Secretin**
- B. Cholecystokinin**
- C. Gastric inhibitory peptide**
- D. Gastrin**
- E. Sympathetic nervous system**

Answer: B. Cholecystokinin (CCK).

Explanation:

Cholecystokinin (CCK) is released from I cells of the duodenum and jejunum, particularly in response to fatty acids and amino acids. One of its major actions is to cause contraction of the gallbladder, thereby releasing stored bile into the duodenum. CCK also stimulates pancreatic enzyme secretion and slows gastric emptying, allowing more time for digestion. Secretin, in contrast, mainly stimulates bicarbonate-rich pancreatic and biliary secretion rather than causing gallbladder contraction. Therefore, B is correct.

[24/09/2026 9:02 AM] Imaanaa Harmee: **71. Stimulus for the secretion of cholecystokinin is:**

- A. Protein digestion products**
- B. Histamine**
- C. Somatostatin**
- D. Acidic chyme in the duodenum**
- E. Fat digestion products**

Answer: E. Fat digestion products.

Explanation:

Cholecystokinin (CCK) is secreted mainly by I cells of the duodenum and jejunum. Its strongest stimuli are fat digestion products, especially fatty acids, and also amino acids and peptides produced during protein digestion. CCK causes contraction of the gallbladder, stimulates pancreatic enzyme secretion, and slows gastric emptying. In contrast, acidic chyme is the major stimulus for secretin release. Therefore, E is the best answer.

72. The total number of premolar teeth in adults is:

- A. 8**
- B. 4**
- C. 12**
- D. 16**
- E. 20**

Answer: A. 8.

Explanation:

An adult normally has 8 premolar teeth: two premolars in each quadrant of the mouth. There are four quadrants—upper right, upper left, lower right, and lower left—giving $2 \times 4 = 8$ premolars. Premolars are absent from the primary (baby) dentition. The complete adult dentition contains 32 teeth, including 8 incisors, 4 canines, 8 premolars, and 12 molars.

73. Which one of the following is secreted into the colon?

- A. Glucose**
- B. Glycine**
- C. Bile salts**
- D. Calcium**
- E. Bicarbonate**

Answer: E. Bicarbonate.

Explanation:

The colon secretes bicarbonate ions (HCO_3^-) into the intestinal lumen. Colonic epithelial cells secrete bicarbonate partly in exchange for chloride. This bicarbonate helps neutralize acidic products produced by bacterial metabolism and protects the colonic mucosa. The colon also absorbs water and electrolytes extensively. Bile salts are delivered into the small intestine, particularly the duodenum, and are largely reabsorbed in the terminal ileum rather than being secreted into the colon.

74. Bile salts are secreted into what region of the small intestine?

- A. Jejunum**
- B. Duodenum**
- C. Ileum**
- D. Splanchnic circulation**
- E. Enterohepatic circulation**

Answer: B. Duodenum.

Explanation:

Bile is stored and concentrated in the gallbladder and released into the duodenum through the biliary tract. The arrival of fatty food in the duodenum stimulates CCK, which causes gallbladder contraction and relaxation of the sphincter of Oddi, allowing bile to enter the intestinal lumen. Bile salts then participate in fat digestion and absorption by forming micelles. They are subsequently reabsorbed mainly in the terminal ileum and returned to the liver through the portal circulation, a process called enterohepatic circulation.

75. Center for swallowing is located in the:

- A. Hypothalamus**
- B. Medulla oblongata**
- C. Pons**
- D. Midbrain**
- E. Cortex**

Answer: B. Medulla oblongata.

Explanation:

The swallowing center is located primarily in the medulla oblongata, with contributions from the lower pons. Swallowing involves a coordinated sequence of voluntary and involuntary actions involving the mouth, pharynx, and esophagus. Once the swallowing reflex is initiated, brainstem centers coordinate the sequential contraction and relaxation of the relevant muscles while protecting the airway. Higher cortical centers can initiate swallowing voluntarily, but the main reflex coordination occurs in the medulla.

76. Identify a hormone whose action is exhibited via gene expression:

- A. Insulin**
- B. Aldosterone**
- C. Catecholamine**
- D. Calcitonin**
- E. Oxytocin**

Answer: B. Aldosterone.

Explanation:

Aldosterone is a steroid hormone, and steroid hormones are lipid-soluble and can cross the plasma membrane. They bind to intracellular receptors in the cytoplasm or nucleus. The hormone-receptor complex interacts with DNA and modifies gene transcription, resulting in the production of specific proteins. Aldosterone, for example, increases expression of proteins involved in sodium reabsorption and potassium secretion in renal tubular cells. The other listed hormones primarily act through cell-surface receptors.

[24/09/2026 9:02 AM] Imaanaa Harmee: **77. The hormone that is stored in the form of prohormone is:**

- A. Insulin**
- B. Aldosterone**
- C. Estrogen**
- D. Testosterone**
- E. Cortisol**

Answer: A. Insulin.

Explanation:

Insulin is synthesized as proinsulin, which contains a signal peptide. The signal peptide is removed to form insulin, and further processing removes the connecting C-peptide, producing mature insulin composed of A and B chains linked by disulfide bonds. Peptide hormones such as insulin are synthesized as precursor molecules because they cannot simply diffuse through cell membranes. Steroid hormones such as aldosterone, estrogen, testosterone, and cortisol are synthesized from cholesterol and are generally not stored as prohormones.

78. From the given options below, which one is an amine hormone?

- A. Growth hormone**
- B. T3**
- C. ADH**
- D. Cortisol**
- E. Oxytocin**

Answer: B. T3.

Explanation:

Triiodothyronine (T3) is an amine hormone derived from the amino acid tyrosine. Thyroid hormones are chemically derived from tyrosine residues within thyroglobulin and are lipid-soluble despite being classified as amine hormones. Growth hormone, ADH, and oxytocin are peptide hormones, while cortisol is a steroid hormone derived from cholesterol. Therefore, T3 is the correct answer.

79. Which hormone of the following has receptors inside the cell?

- A. ADH**
- B. Insulin**
- C. Growth hormone**
- D. Glucagon**
- E. Aldosterone**

Answer: E. Aldosterone.

Explanation:

Aldosterone is a steroid hormone and therefore can cross the lipid bilayer of the cell membrane. It binds to an intracellular mineralocorticoid receptor, and the hormone-receptor complex influences gene transcription. In contrast, ADH, insulin, growth hormone, and glucagon are peptide hormones that cannot readily cross the plasma membrane. They therefore bind to cell-surface receptors and activate intracellular signaling pathways.

80. A phenomenon regulated by a positive feedback mechanism is:

- A. Secretion of insulin**
- B. Milk ejection reflex**
- C. Gonadal function**
- D. Growth**
- E. Secretion of T3 and T4**

Answer: B. Milk ejection reflex.

Explanation:

The milk ejection (let-down) reflex is a classic example of positive feedback. When a baby suckles the nipple, sensory signals travel to the hypothalamus, causing the posterior pituitary to release oxytocin. Oxytocin causes contraction of myoepithelial cells around the mammary alveoli, forcing milk into the ducts and toward the nipple. Continued suckling produces additional oxytocin release, which produces further milk ejection. Thus, the response reinforces the original stimulus, which is characteristic of positive feedback.

[24/09/2026 9:07 AM] Imaanaa Harmee: **81. Being non-steroidal but having intracellular receptor is:**

- A. Epinephrine**
- B. Dopamine**
- C. T3**
- D. Somatostatin**
- E. Serotonin**

Answer: C. T3

Explanation:

T3 (triiodothyronine) is an amine hormone derived from tyrosine, so it is not a steroid hormone. However, unlike most peptide and catecholamine hormones, thyroid hormones are lipid-soluble and can cross the cell membrane. Their receptors are located inside the cell, mainly in the nucleus, where the hormone-receptor complex binds to DNA and regulates gene transcription. This makes T3 a classic exception to the usual rule that non-steroid hormones act through membrane receptors.

82. Hormones designed for the regulation of fluid and electrolyte balance are:

- A. Growth hormone and insulin**
- B. Aldosterone and insulin**
- C. Aldosterone and ADH**
- D. Oxytocin and ADH**
- E. T3 and T4**

Answer: C. Aldosterone and ADH

Explanation:

Aldosterone and antidiuretic hormone (ADH/vasopressin) are the major hormones involved in maintaining body-fluid volume and electrolyte balance. Aldosterone acts mainly on the distal nephron and collecting duct to increase sodium reabsorption and promote potassium secretion. ADH acts mainly on the collecting ducts of the kidneys, increasing their permeability to water through insertion of aquaporin-2 channels. This allows more water to be reabsorbed and helps maintain plasma osmolality and blood volume. Therefore, C is correct.

83. A hormone whose secretion is excess during the time of deep sleep is:

- A. Growth hormone**
- B. Insulin**
- C. Cortisol**
- D. Adrenalin**
- E. ADH**

Answer: A. Growth hormone

Explanation:

Growth hormone (GH) secretion is strongly associated with deep, slow-wave sleep,

particularly during the early part of the night. Pulsatile GH secretion increases substantially during deep sleep. GH promotes protein synthesis, tissue growth, and metabolic effects, partly through stimulation of insulin-like growth factor-1 (IGF-1). In contrast, cortisol generally reaches its highest levels around the early morning and is relatively lower during the early part of sleep.

84. Mullerian duct develops into the:

- A. Ovaries**
- B. Female genital tract**
- C. Male accessory glands**
- D. Female external genitalia**
- E. Ovaries**

Answer: B. Female genital tract

Explanation:

The Müllerian (paramesonephric) ducts develop into major portions of the female internal reproductive tract. They form the fallopian tubes, uterus, cervix, and upper portion of the vagina. The ovaries do not arise from the Müllerian ducts; they develop from the gonadal ridges. In males, the Müllerian ducts regress under the influence of anti-Müllerian hormone (AMH). Therefore, B is the best answer.

85. Which one of the following is known to be the hormone of ovulation?

- A. FSH**
- B. Estradiol**
- C. Progesterone**
- D. hCG**
- E. LH**

Answer: E. LH

Explanation:

Luteinizing hormone (LH) is known as the principal hormone responsible for triggering ovulation. During the late follicular phase, sustained high levels of estrogen from the dominant follicle produce positive feedback on the hypothalamus and pituitary. This causes a large LH surge, which triggers rupture of the mature ovarian follicle and release of the secondary oocyte. LH also promotes formation of the corpus luteum after ovulation.

86. Which of the following is not a haploid cell?

- A. Spermatogonium**
- B. Secondary oocyte**
- C. Spermatids**
- D. Oocyte**
- E. Ovum**

Answer: A. Spermatogonium

Explanation:

A spermatogonium is diploid (2n) because it is a germ cell that has not yet undergone meiosis. During spermatogenesis, spermatogonia develop into primary spermatocytes, which subsequently undergo meiosis to produce haploid secondary spermatocytes and spermatids. The secondary oocyte and ovum are haploid, as are spermatids. Therefore, A is the clearly correct answer.

[24/09/2026 9:07 AM] Imaanaa Harmee: **87. In the male sexual act, the erection phase is under the regulation of the:**

- A. Parasympathetic nervous system**
- B. Sympathetic nervous system**
- C. Somatic pudendal nerve**
- D. Vagus nerve from medulla**
- E. Phrenic nerve of the cervical segment**

Answer: A. Parasympathetic nervous system

Explanation:

Penile erection is primarily a parasympathetic response. Parasympathetic fibers from the sacral spinal cord stimulate the release of nitric oxide and other vasodilatory substances, producing relaxation of smooth muscle in the penile erectile tissues. This increases arterial blood flow and allows the erectile tissue to fill with blood. A useful physiological distinction is “point and shoot”: parasympathetic activity is important for erection, whereas sympathetic activity is important for ejaculation.

88. The function of the uterus is:

- A. Hormone secretion**
 - B. Essential reproductive organ**
 - C. Site for normal pregnancy**
 - D. Gametogenesis**
 - E. None of the above**
- Answer: C. Site for normal pregnancy**

Explanation:

The primary reproductive function of the uterus is to provide the site where implantation occurs and where the embryo and fetus develop during pregnancy. The endometrium undergoes cyclic changes to prepare for implantation, and during pregnancy the uterus enlarges greatly to accommodate the developing fetus. The uterus is not the site of gametogenesis; oocytes develop in the ovaries, while sperm develop in the testes. Therefore, C is the most specific answer.

89. Usual site of fertilization in the female reproductive system is:

- A. Ovary**
- B. Fallopian tubes**
- C. Uterus**

D. Vagina

E. Clitoris

Answer: B. Fallopian tubes

Explanation:

Fertilization normally occurs in the ampulla of the fallopian (uterine) tube. After ovulation, the secondary oocyte enters the fallopian tube and travels toward the uterus. Sperm travel through the female reproductive tract to reach the tube, where one sperm may fuse with the oocyte to form a zygote. The developing embryo then continues toward the uterus, where implantation normally occurs. Thus, the fallopian tube—not the uterus—is the usual site of fertilization.

90. Proliferative phase of the endometrial cycle is controlled by which hormone?

A. Progesterone

B. FSH

C. LH

D. Estrogen

E. Oxytocin

Answer: D. Estrogen

Explanation:

The proliferative phase occurs after menstruation and before ovulation and is primarily controlled by estrogen, especially estradiol, secreted by developing ovarian follicles. Estrogen stimulates proliferation of the endometrial cells, causing the endometrium to become thicker and increasing the growth of its glands and blood vessels. After ovulation, the secretory phase becomes dominant and is primarily controlled by progesterone from the corpus luteum. Therefore, D. Estrogen is correct.